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(54) **POWER TRACKING USING A SOLDIER NETWORK MIDDLEWARE SYSTEM**

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H04W 4/12 (2009.01)

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CPC **G05B 19/0428** (2013.01); **H04L 45/52**

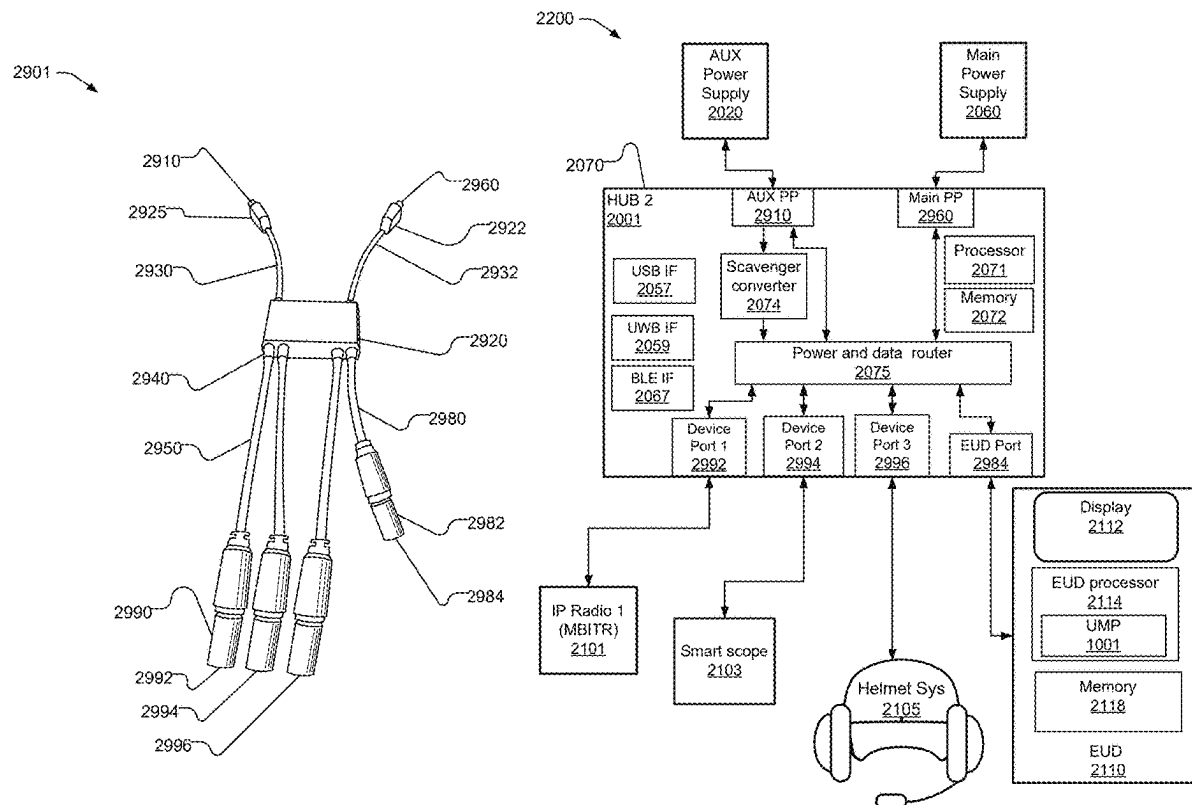
(2013.01); **H04L 51/56** (2022.05); **H04W 4/12**

(2013.01); **G05B 2219/2639** (2013.01)

(57)

ABSTRACT

A bridge system includes messaging middleware that is configured to operate on a multi-protocol soldier personal area network (PAN) and to enable sharing of data with multiple applications in an Android OS. The bridge system is implemented on smart and semi-smart multi-protocol hubs and is used to support sharing of live power data between PAN components. The data is displayed to a user on a user interface and is used by components of the system to model and predict power requirements and power usage of the PAN components.



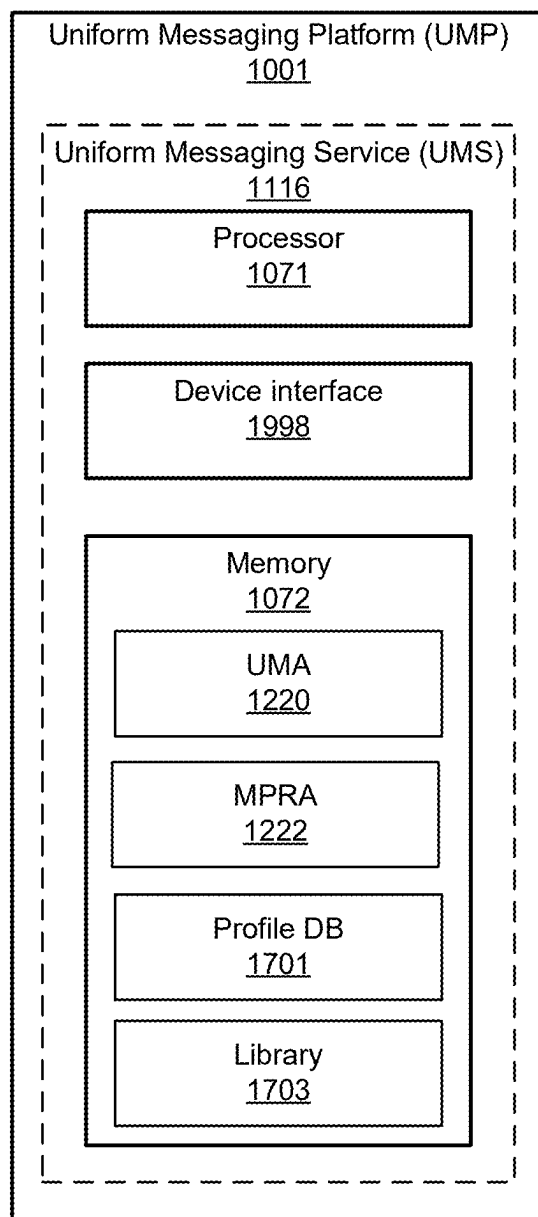


Fig. 1

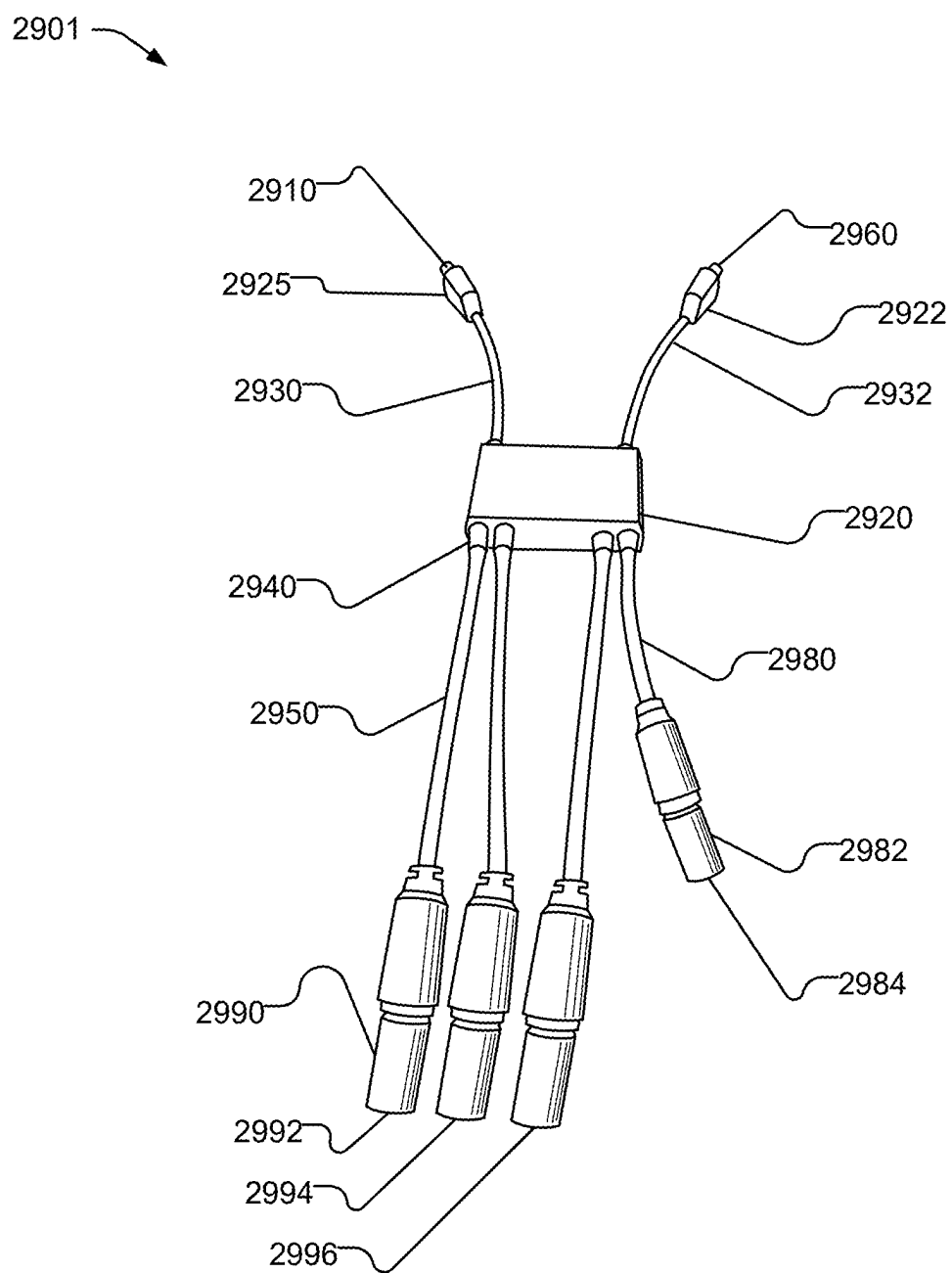


Fig. 2A

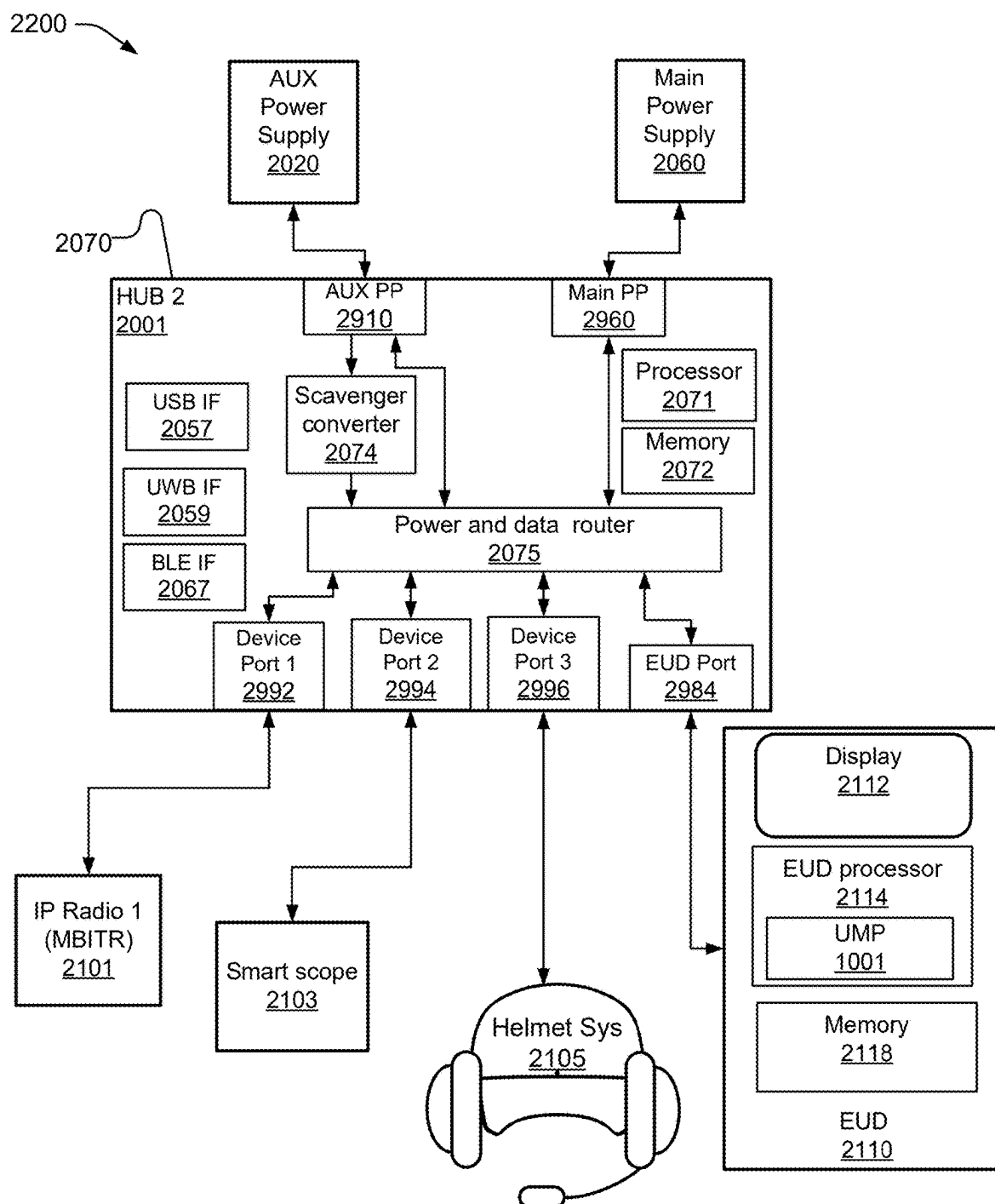


Fig. 2B

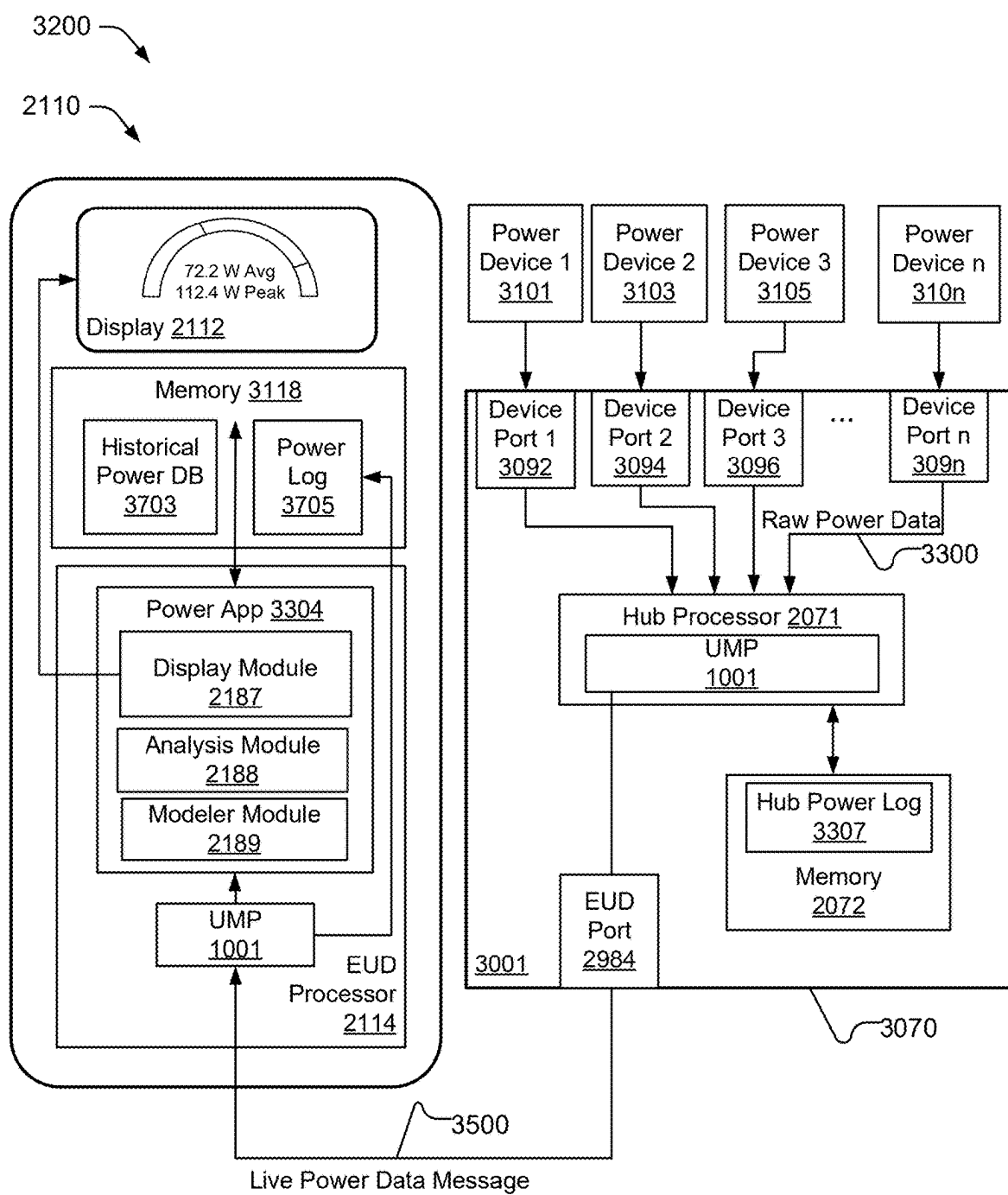


Fig. 3

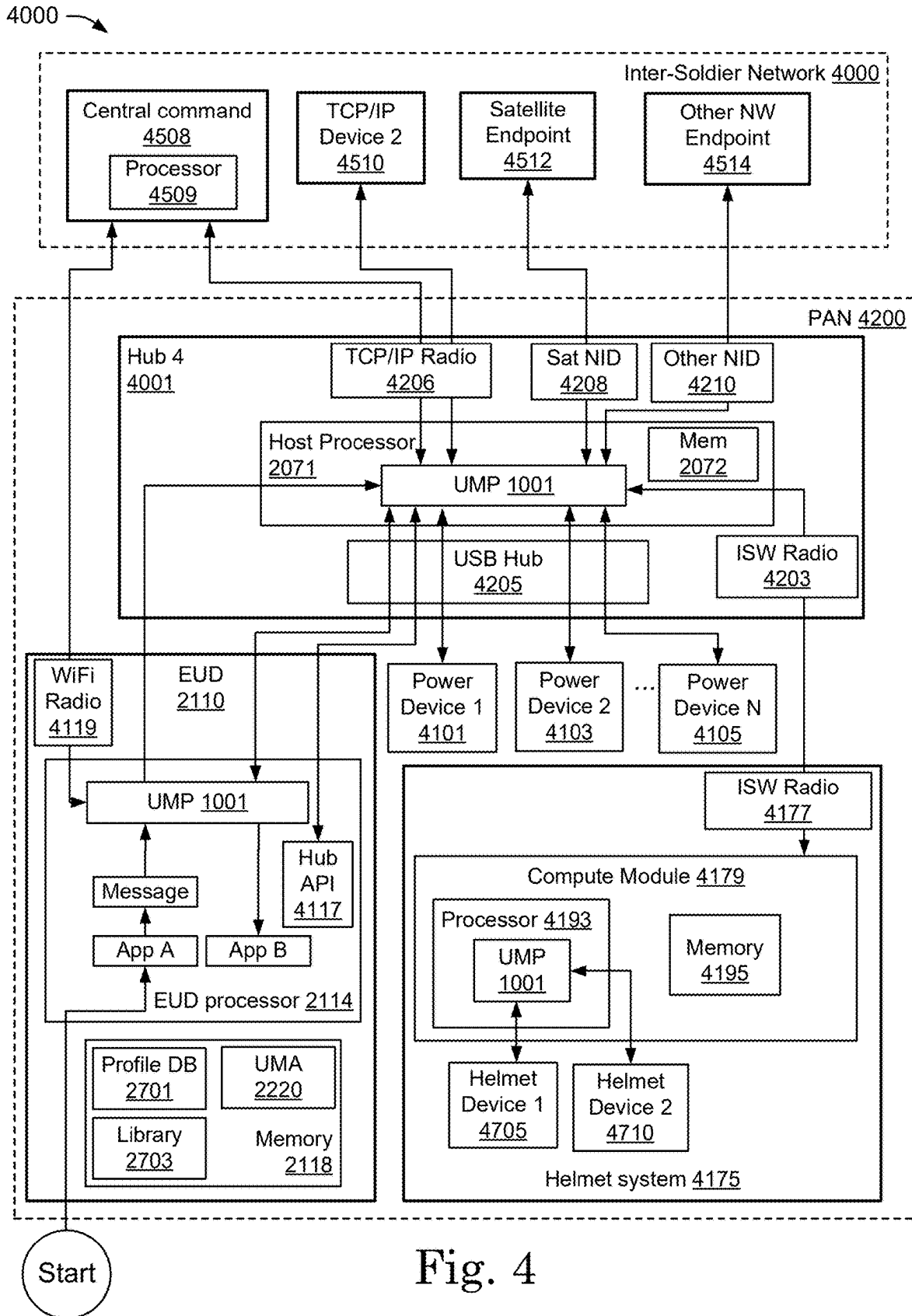


Fig. 4

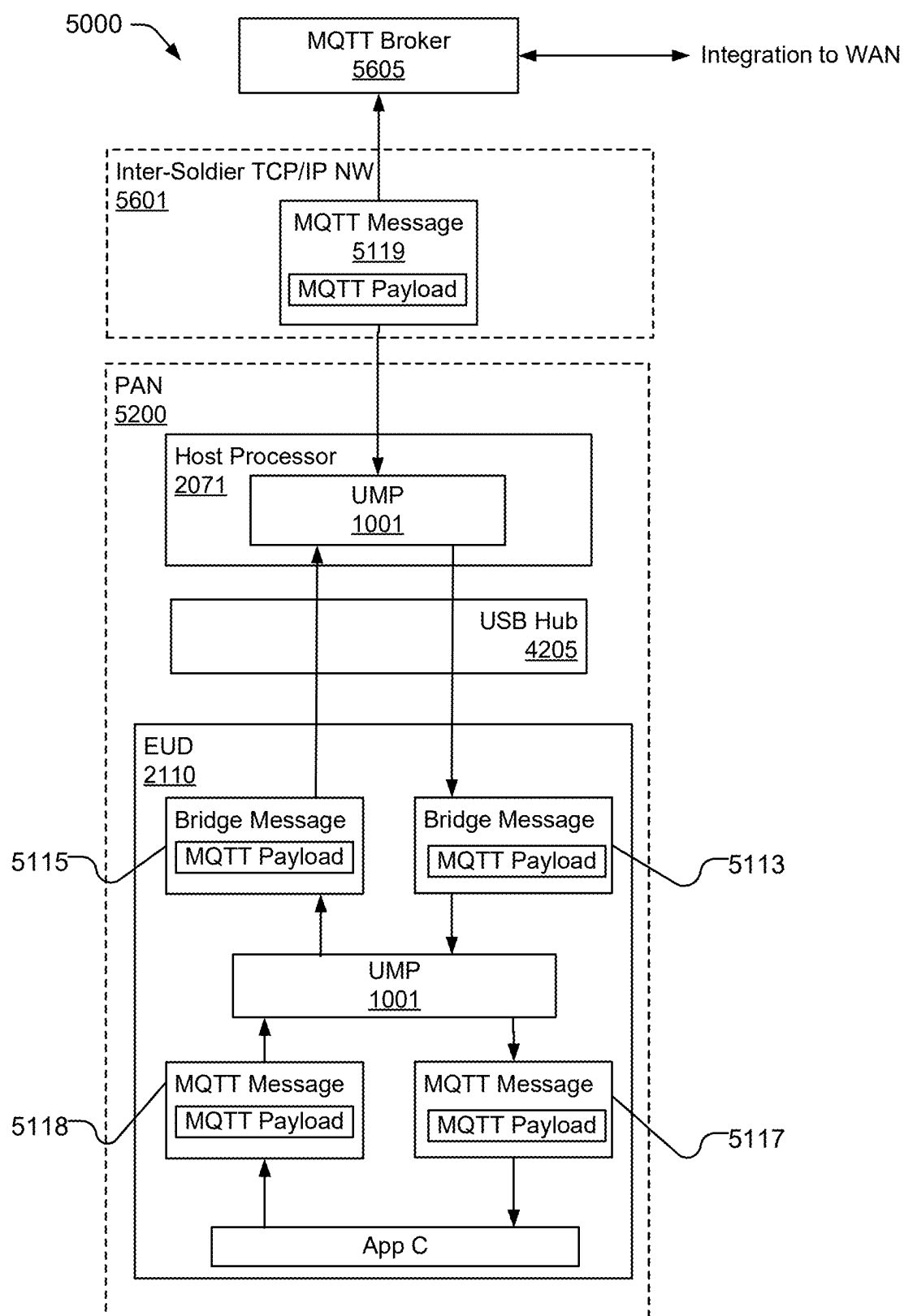


Fig. 5

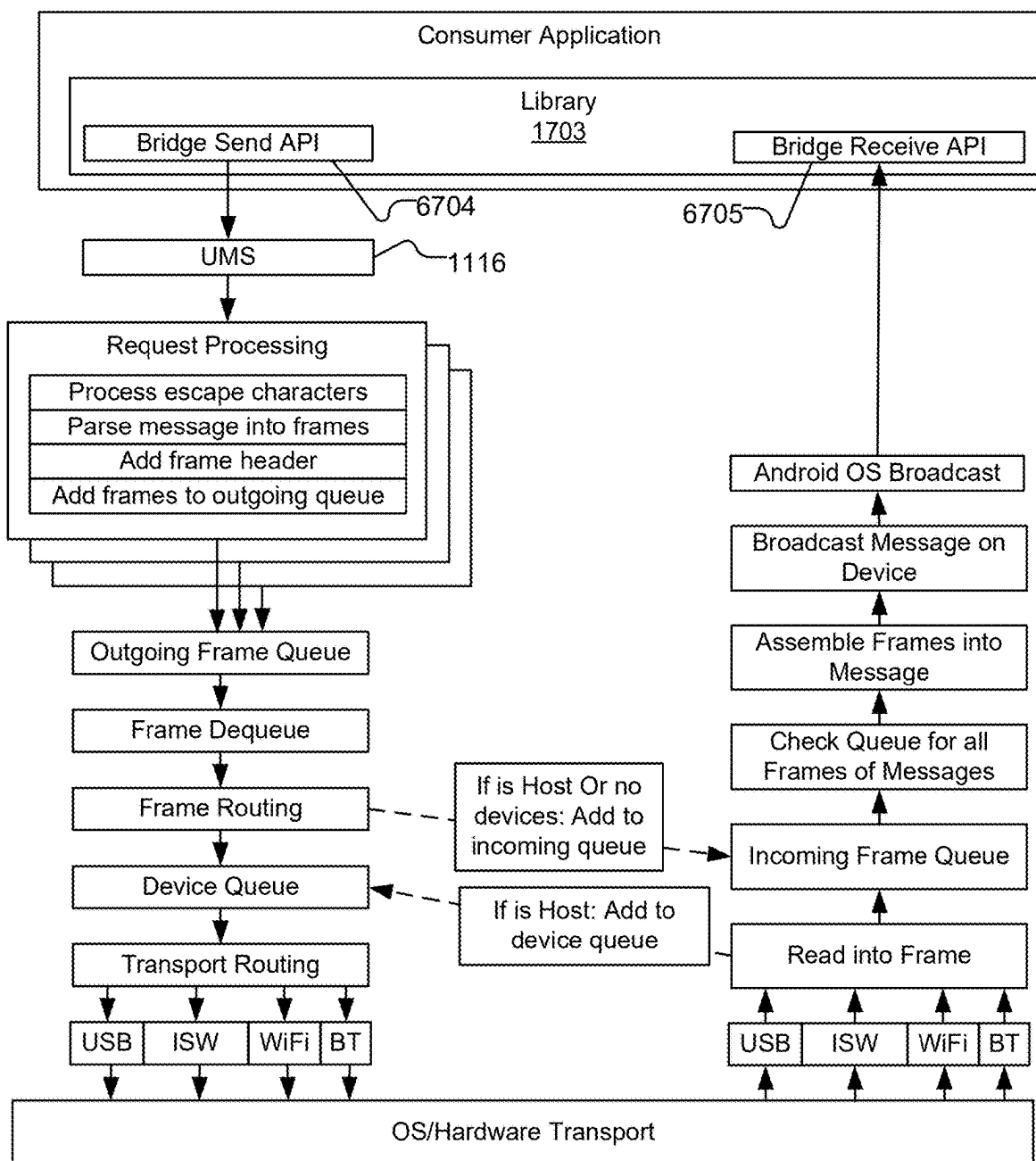


Fig. 6

7000 ↘

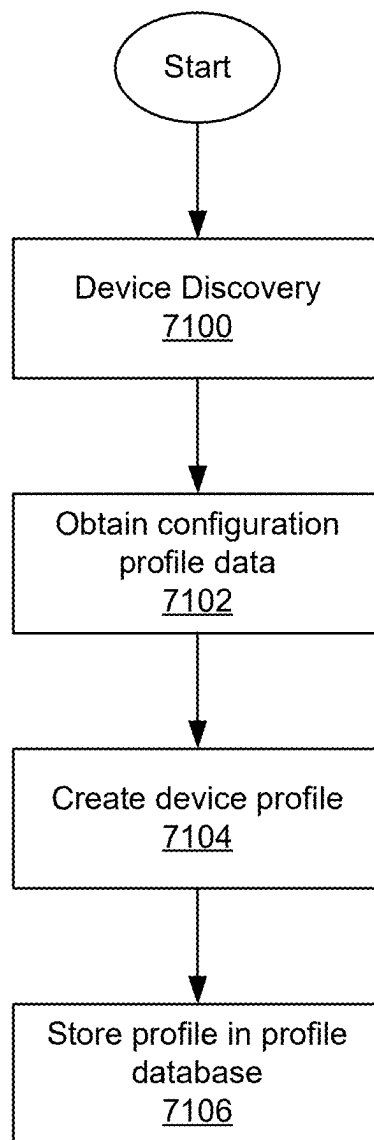


Fig. 7

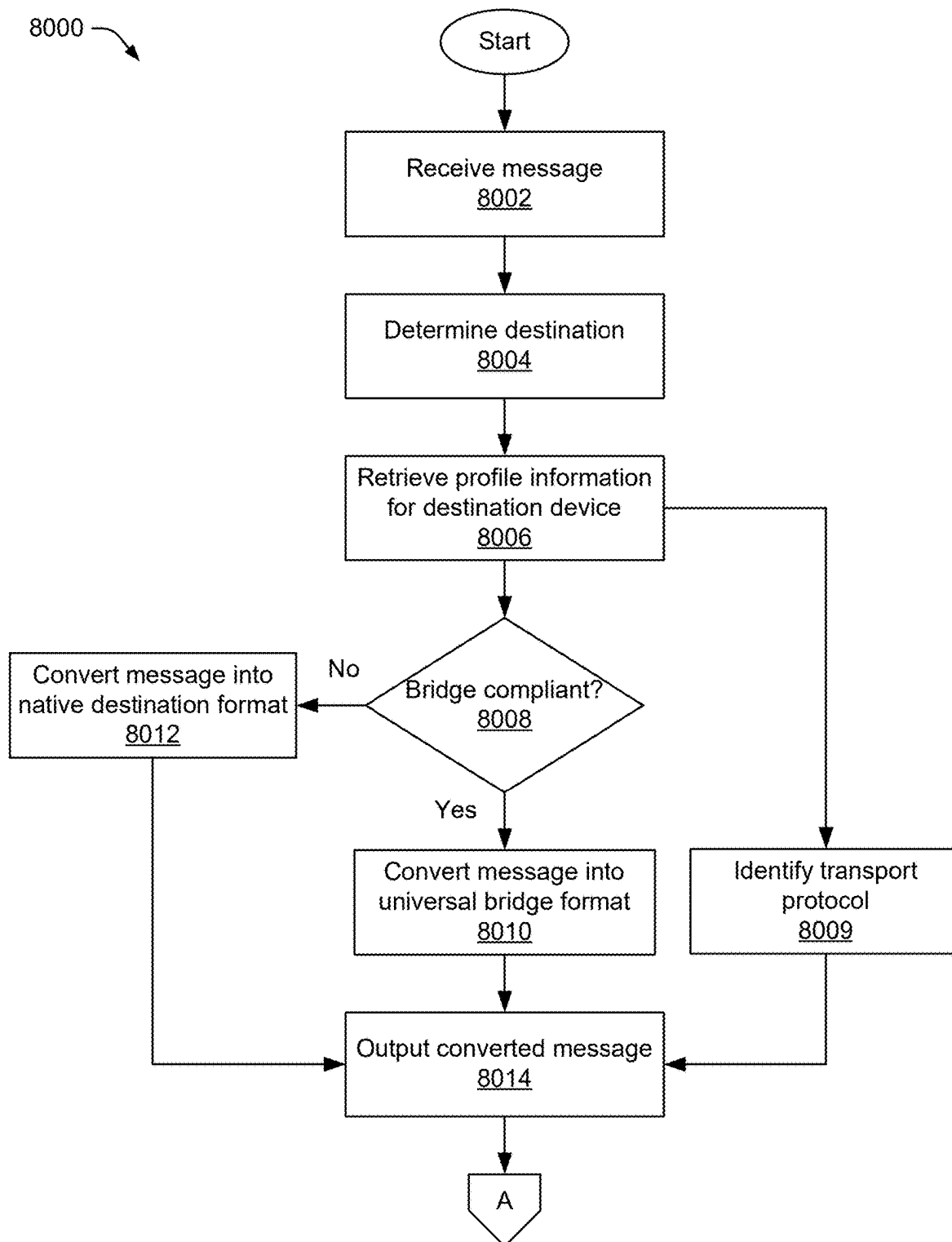


Fig. 8

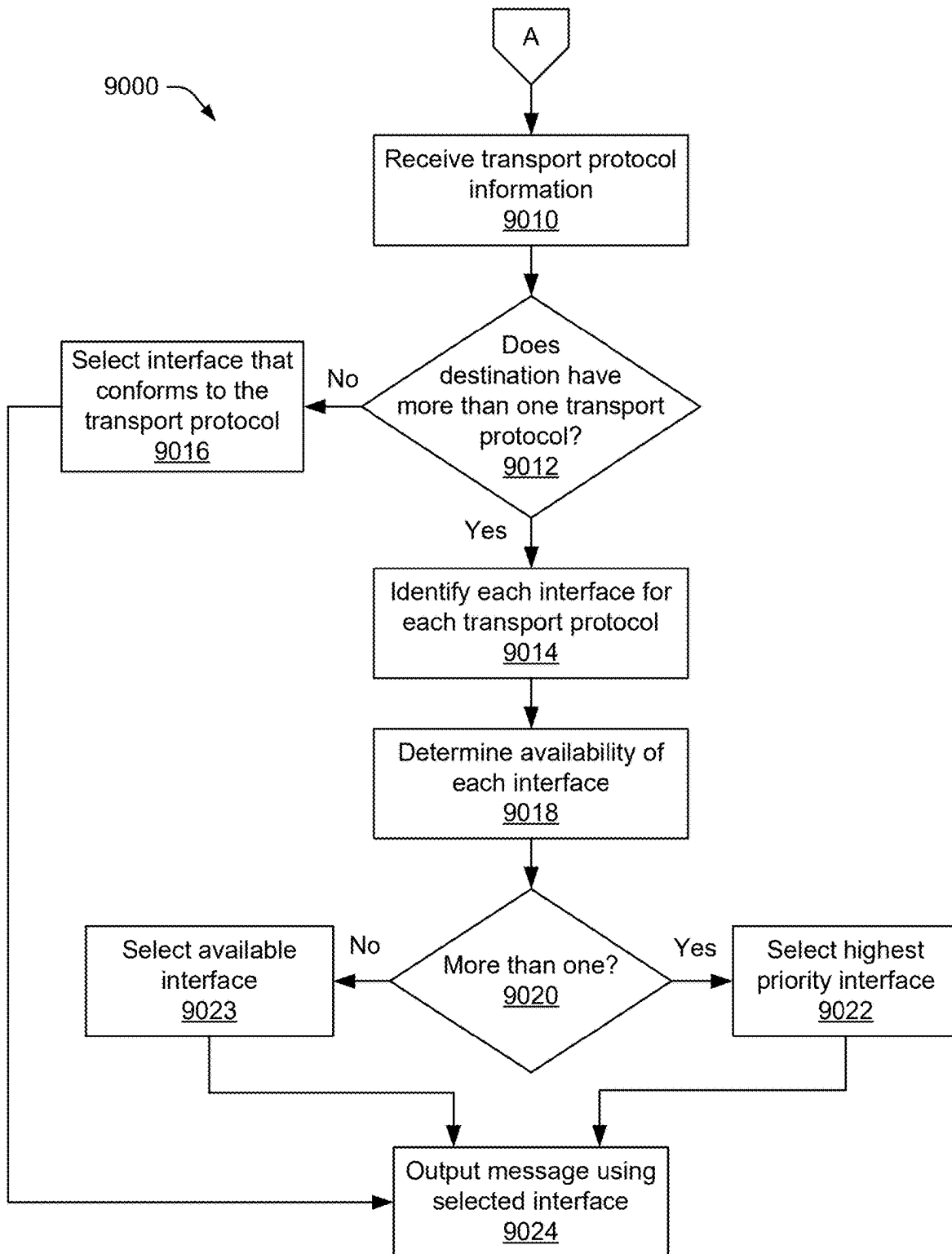


Fig. 9

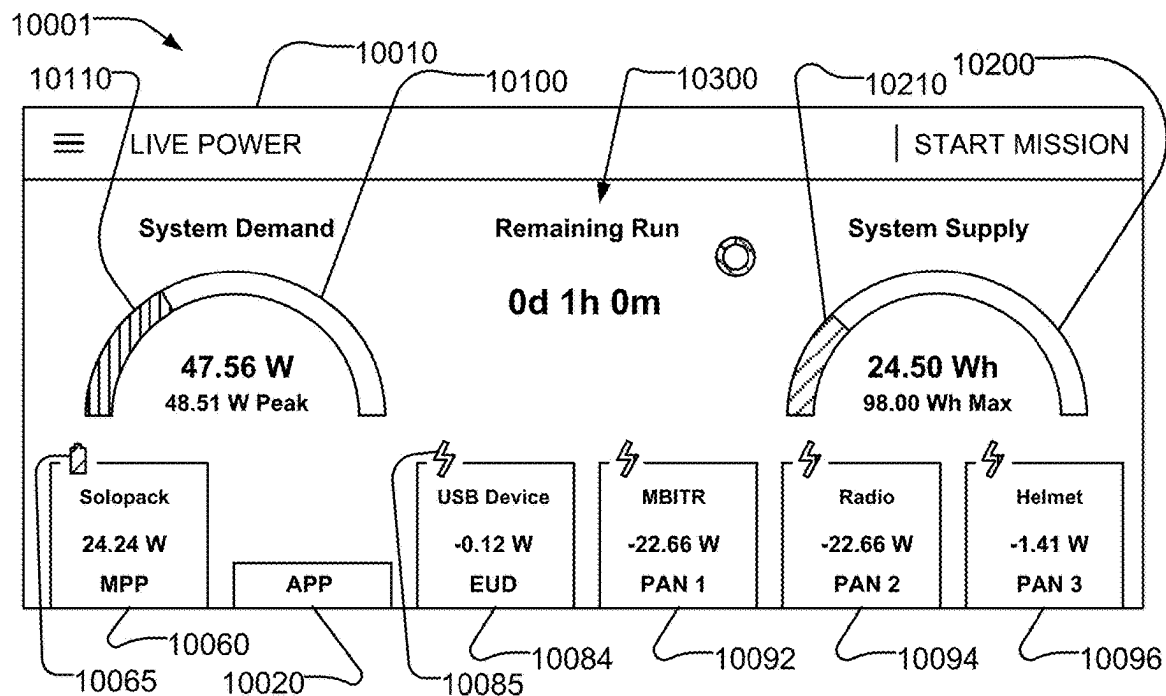


Fig. 10A

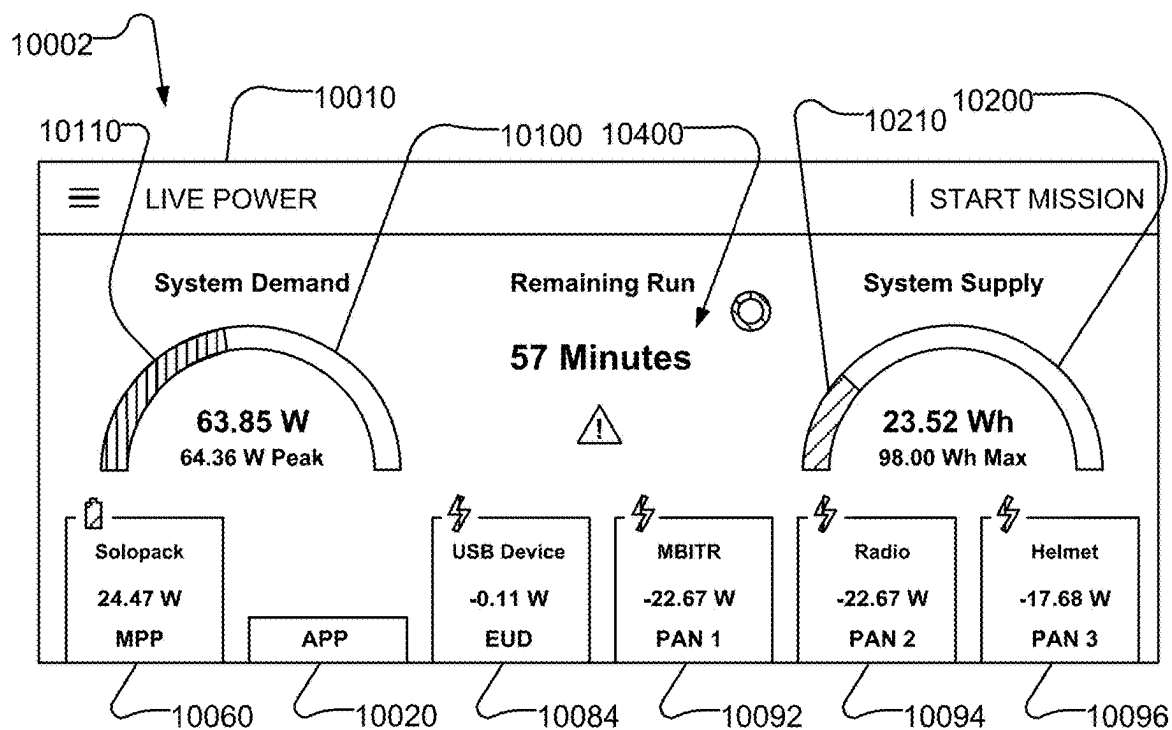


Fig. 10B

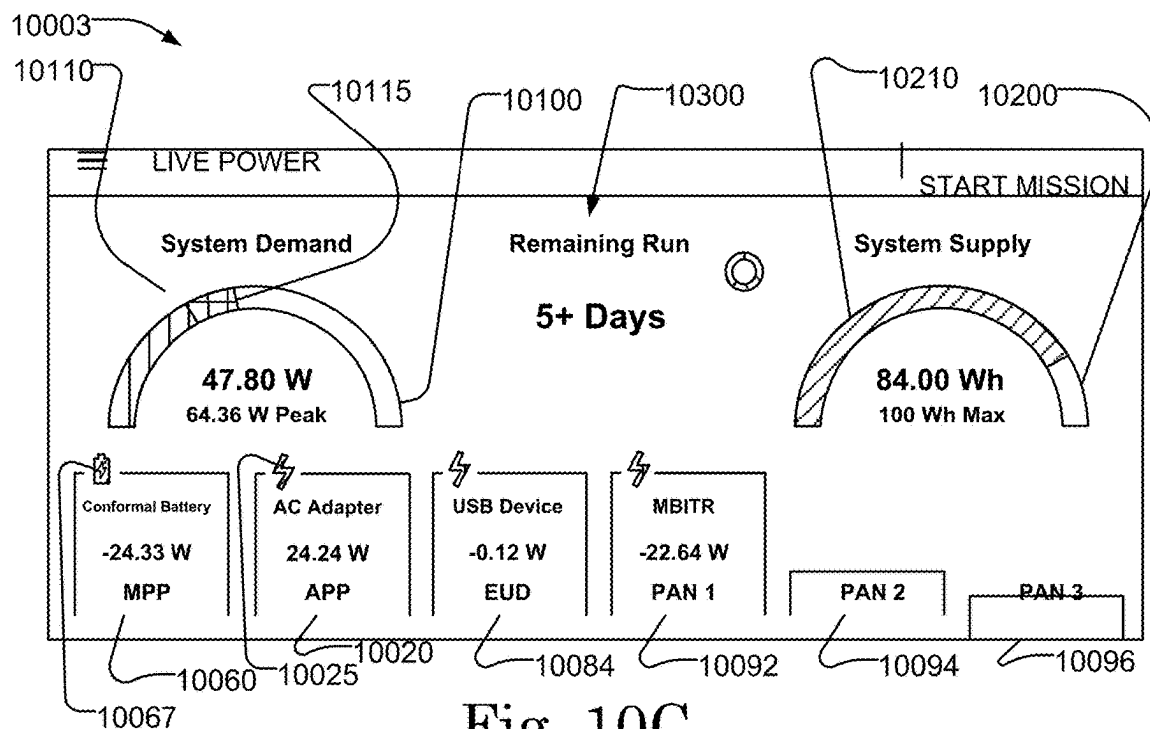


Fig. 10C

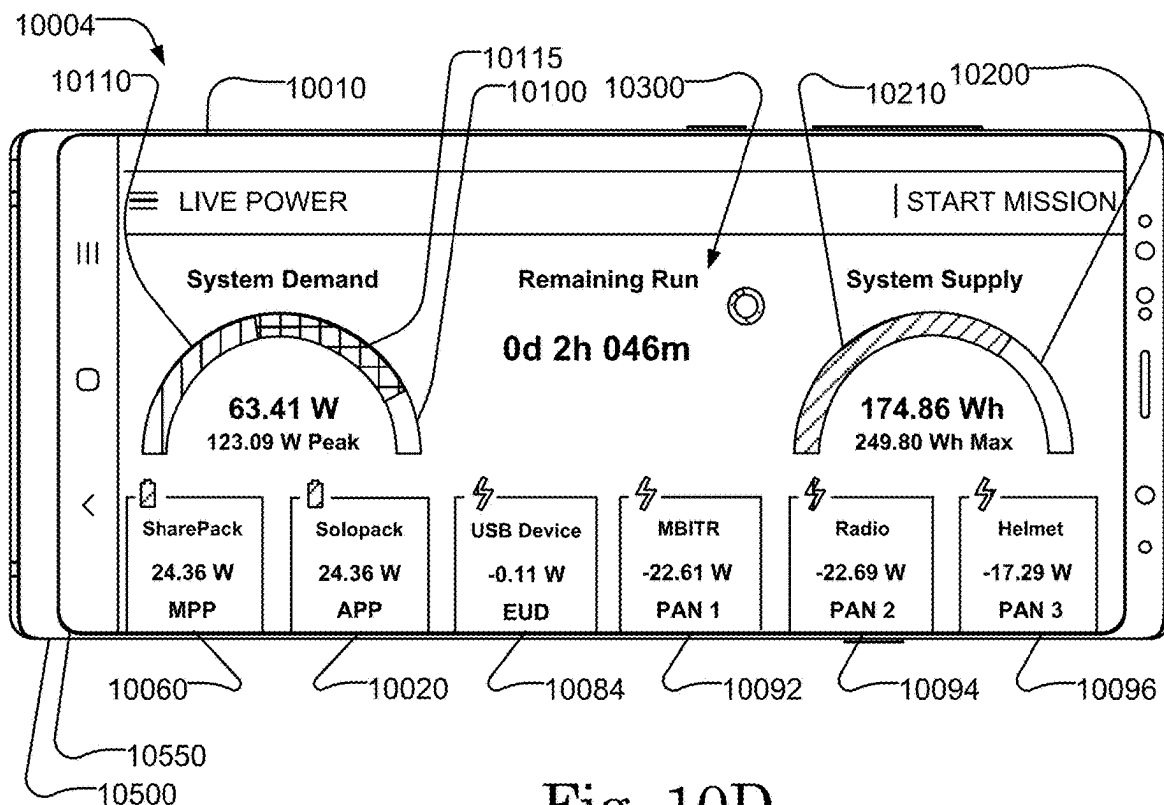


Fig. 10D

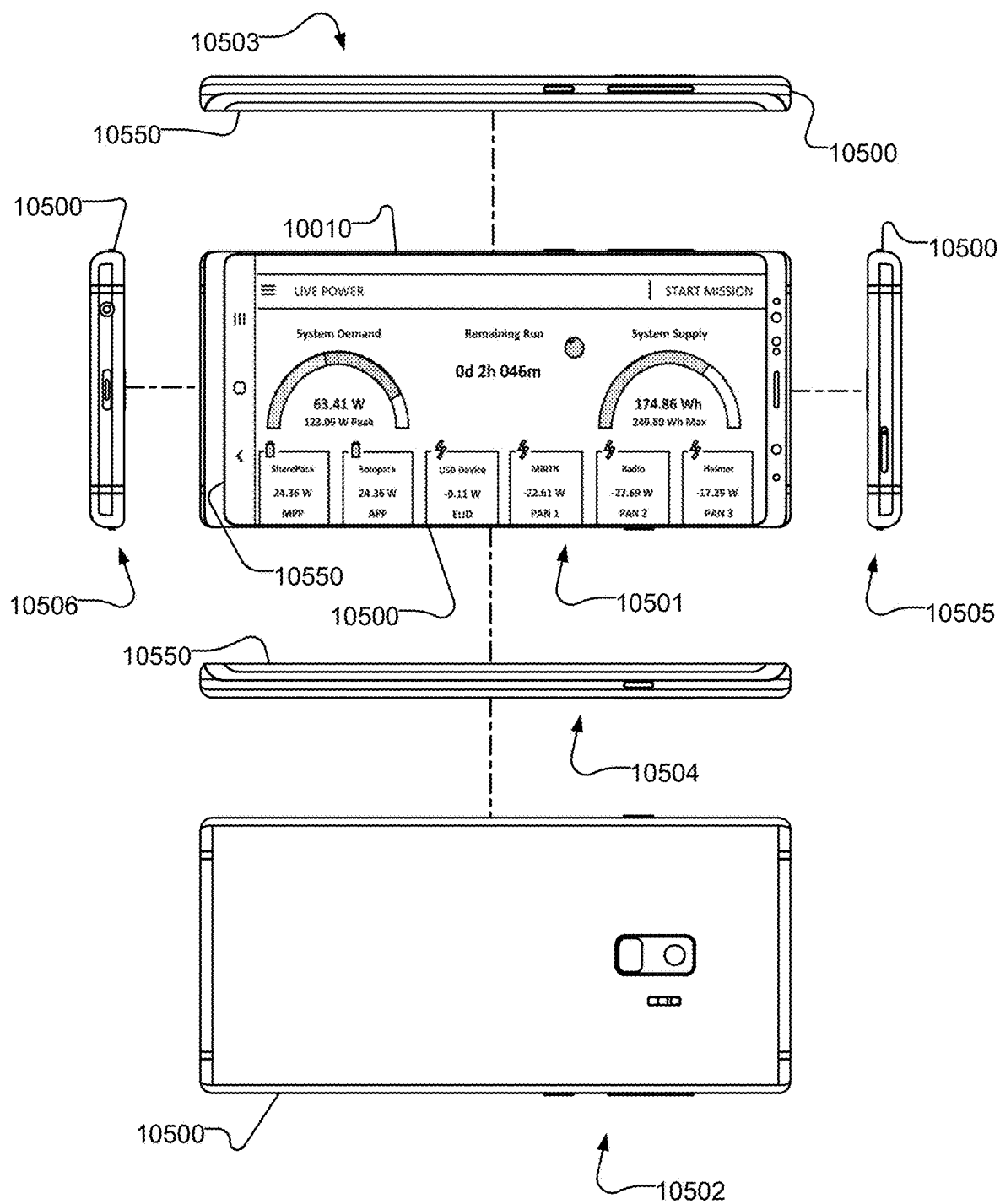


Fig. 10E

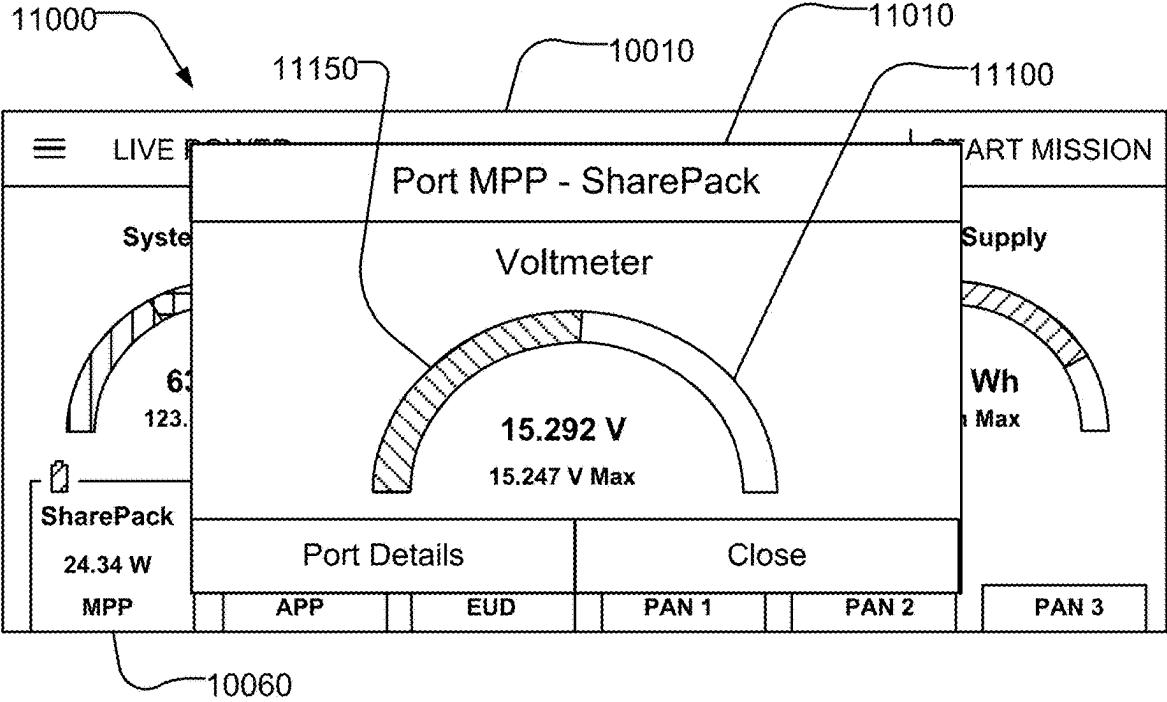


Fig. 11

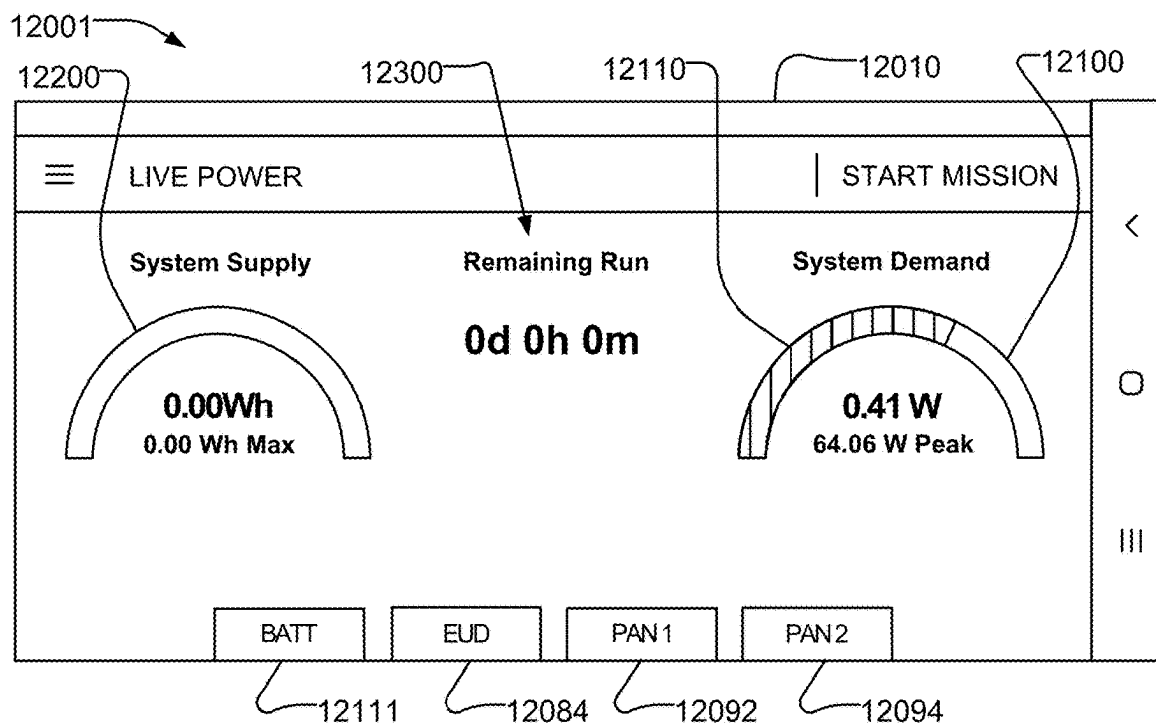


Fig. 12A

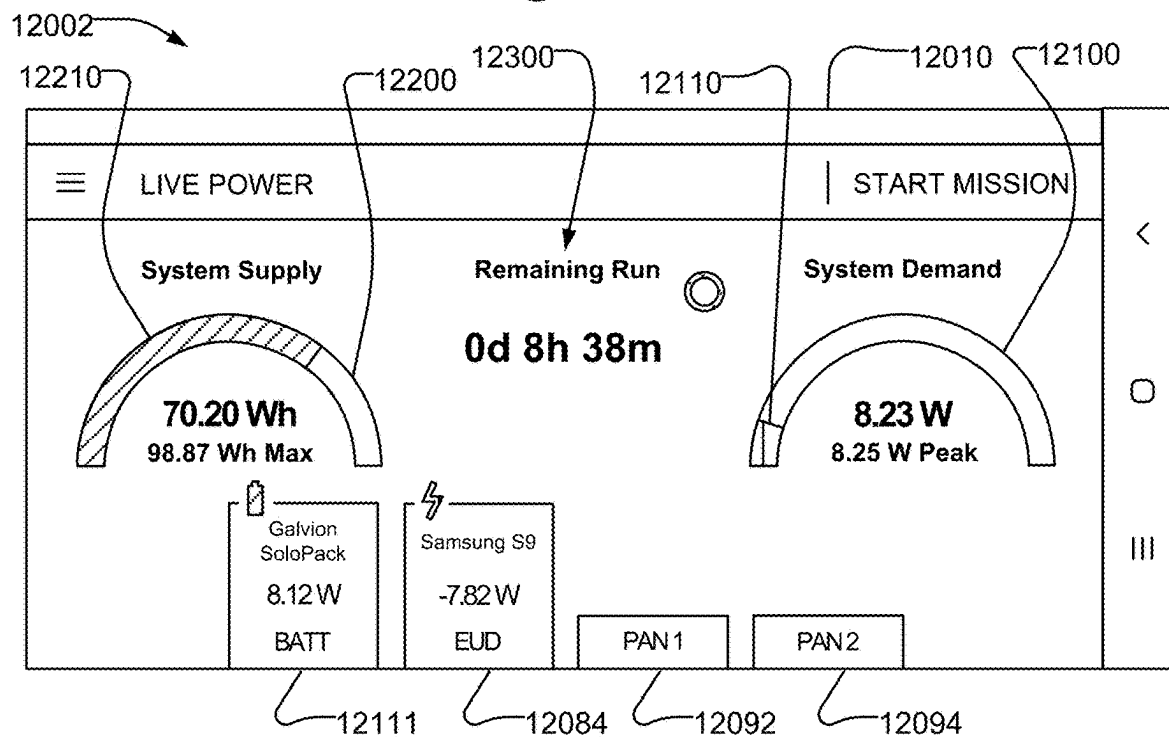


Fig 12B

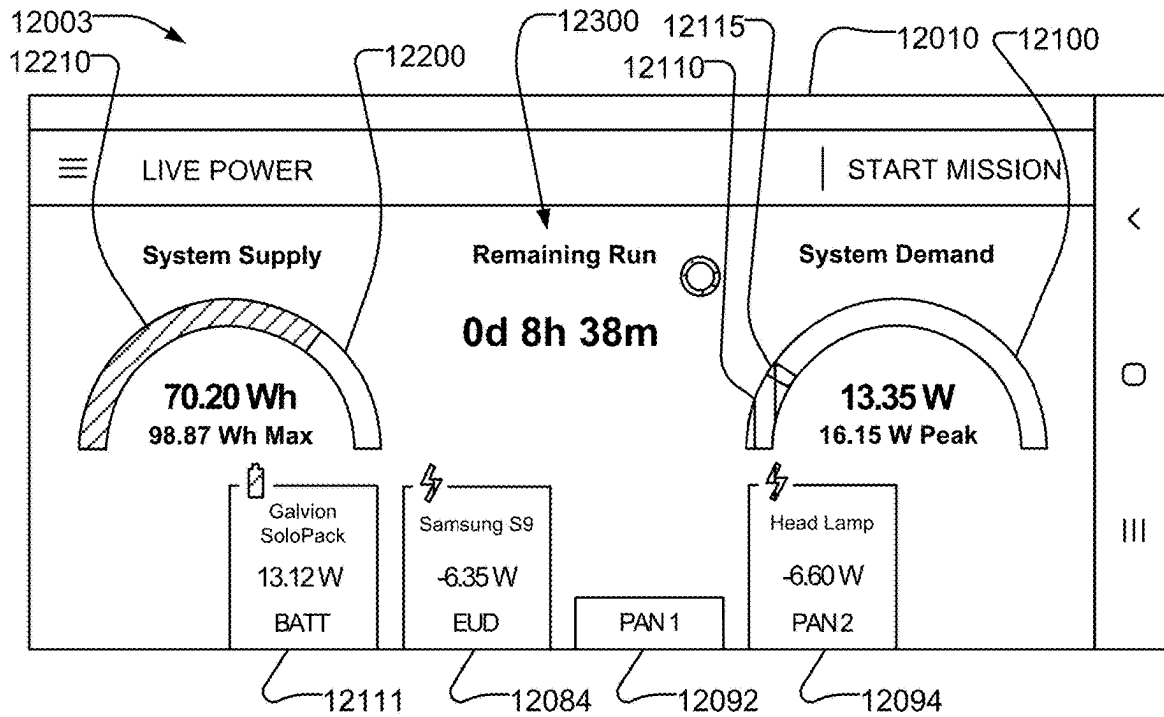


Fig. 12C

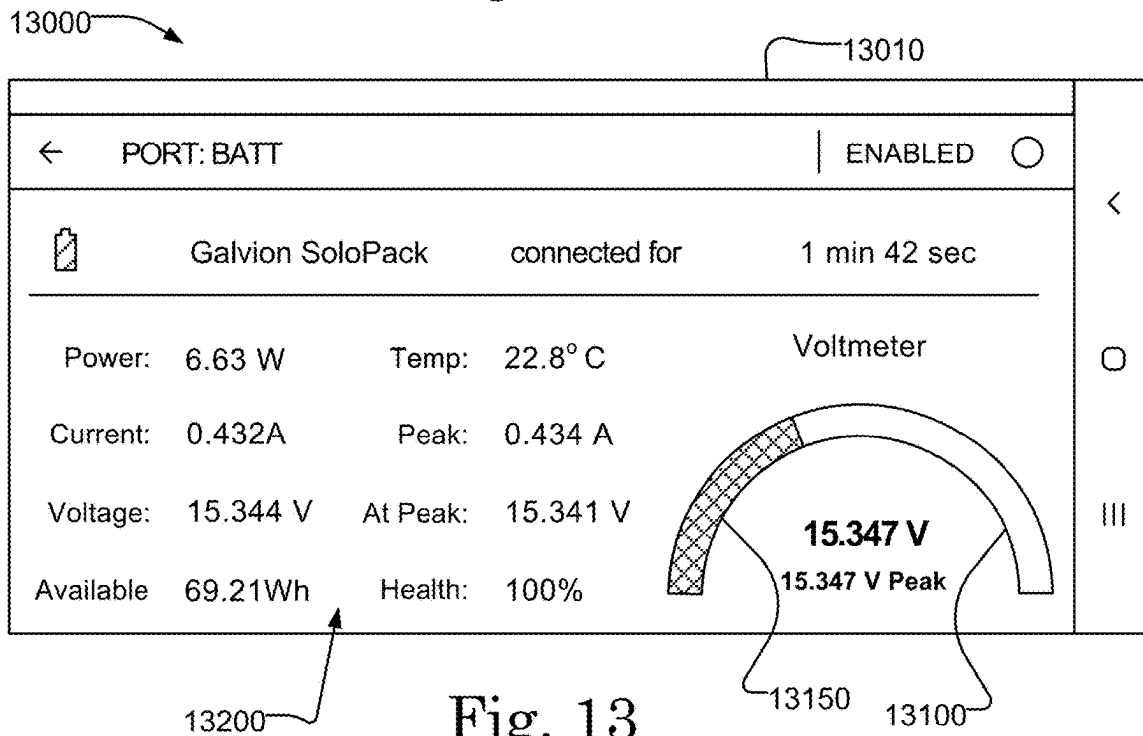


Fig. 13

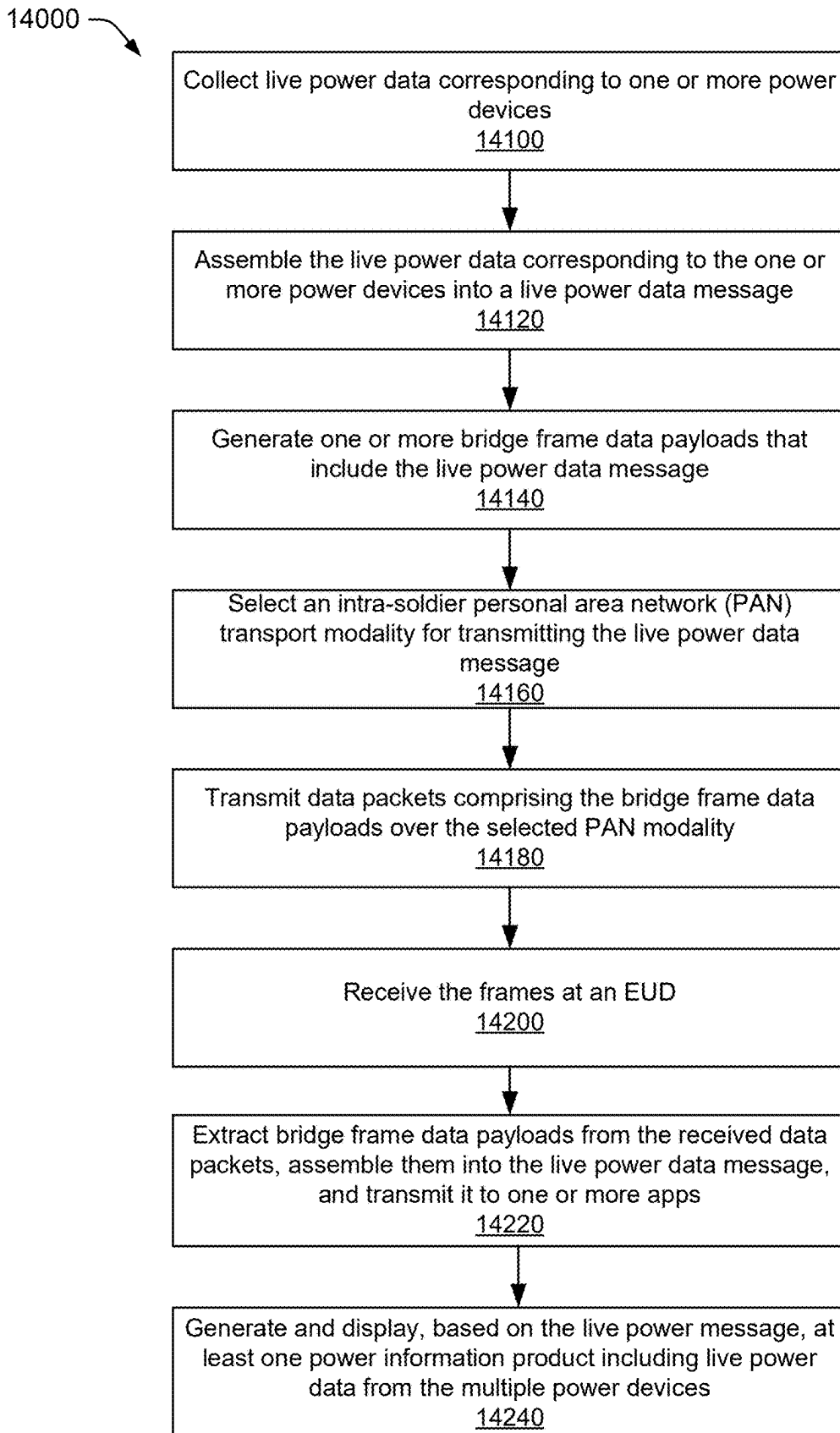


Fig. 14

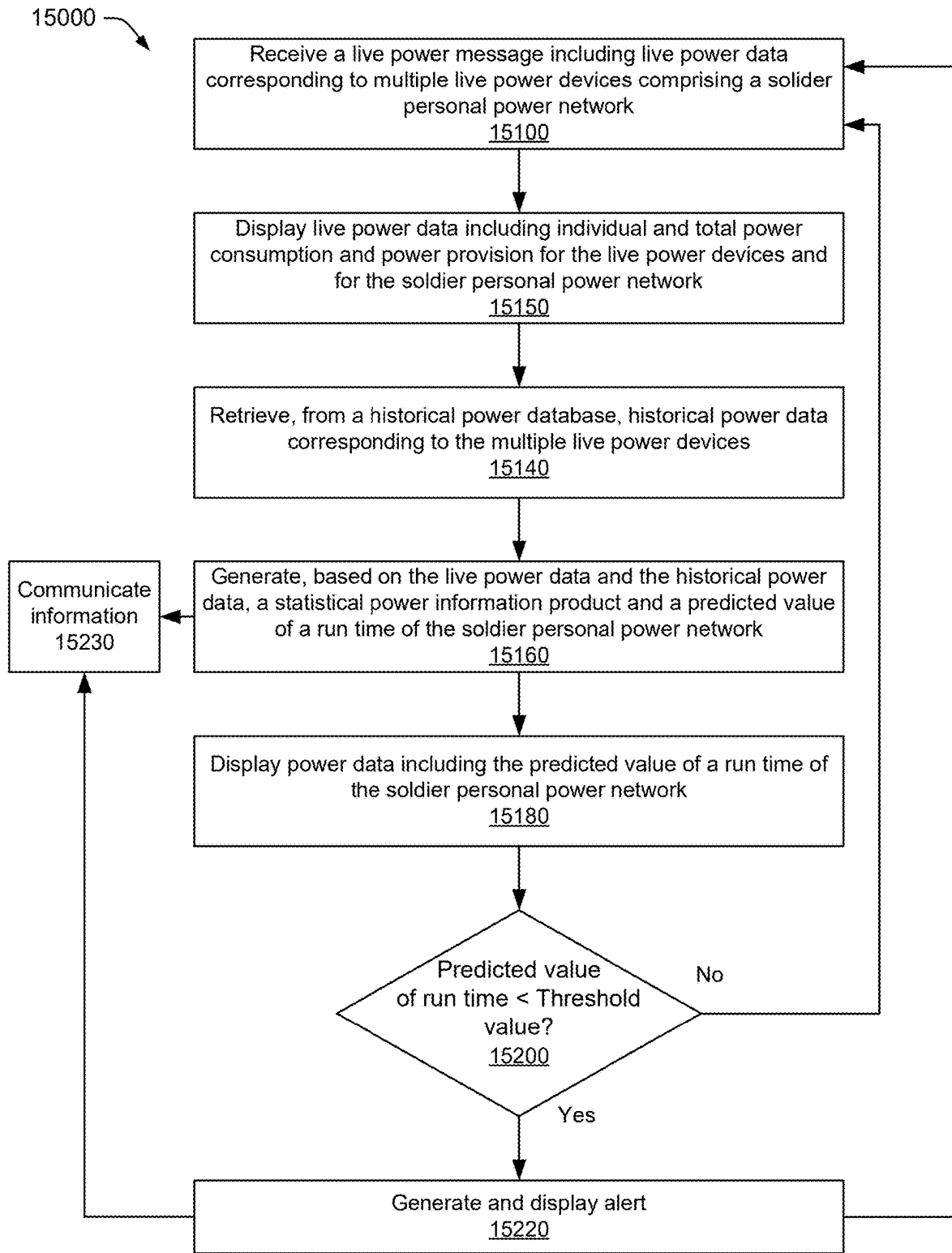
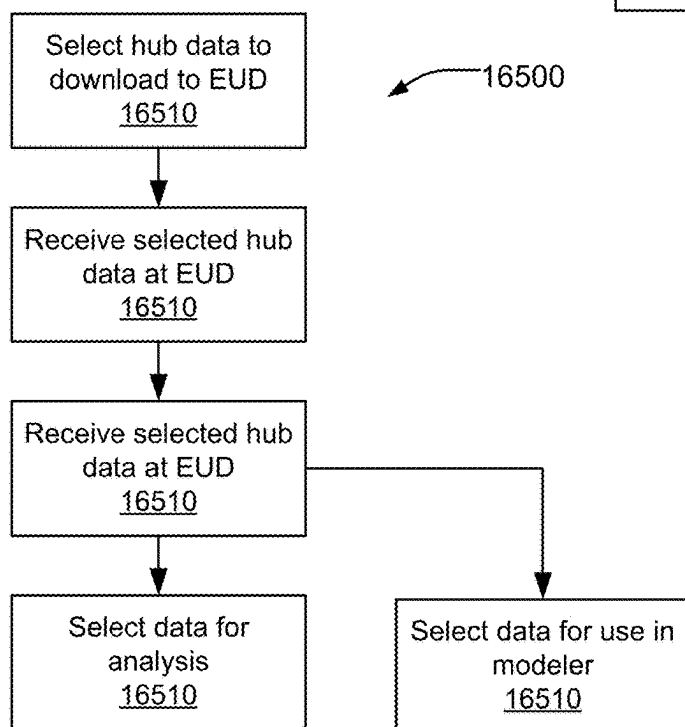
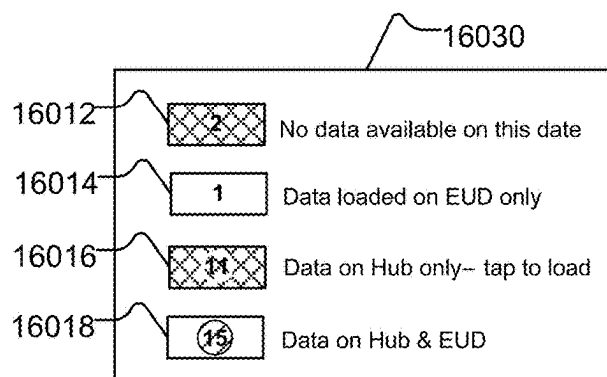
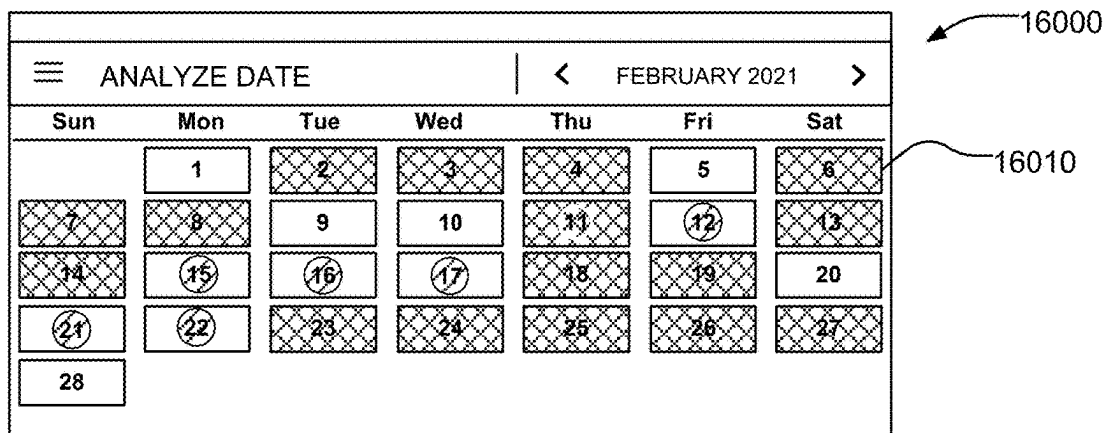


Fig. 15



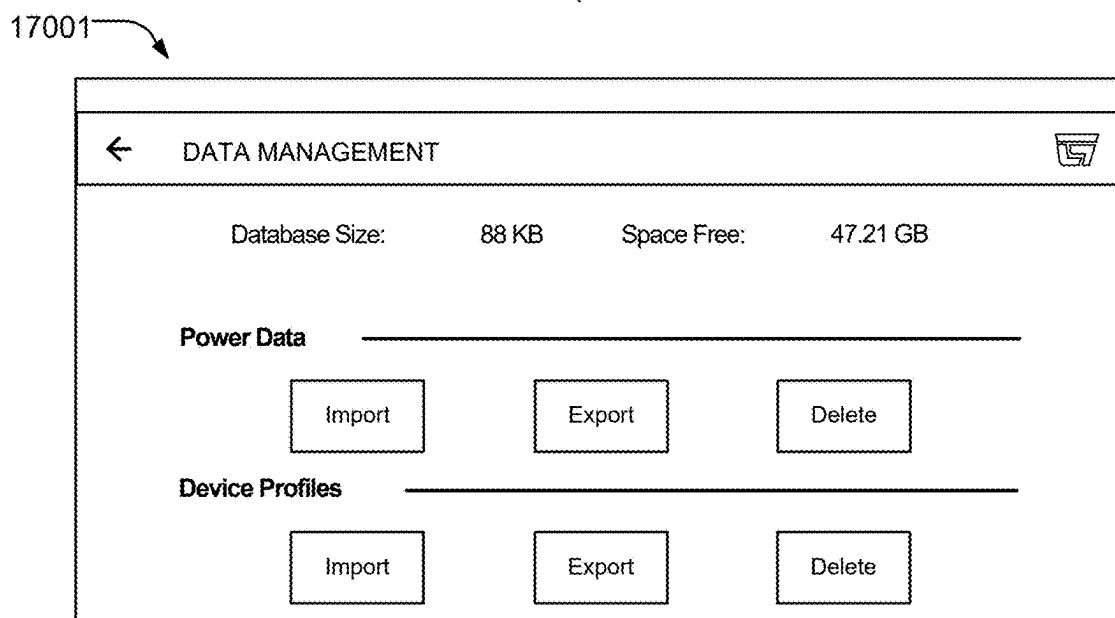


Fig. 17

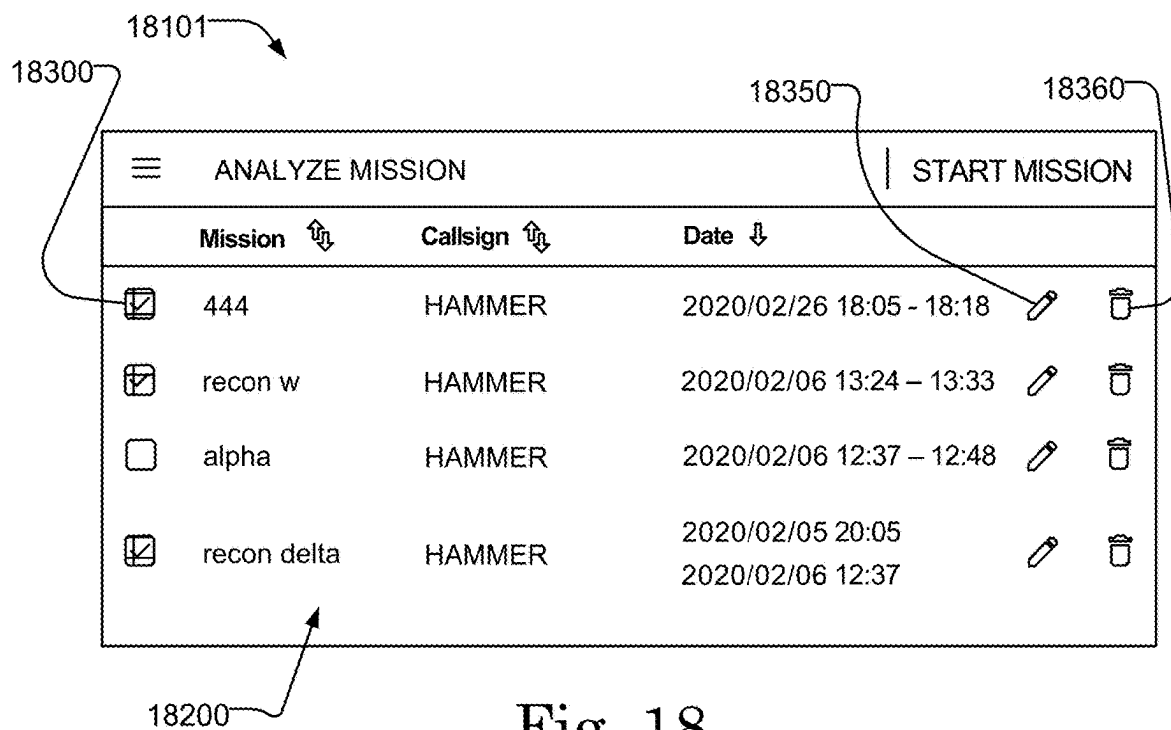


Fig. 18

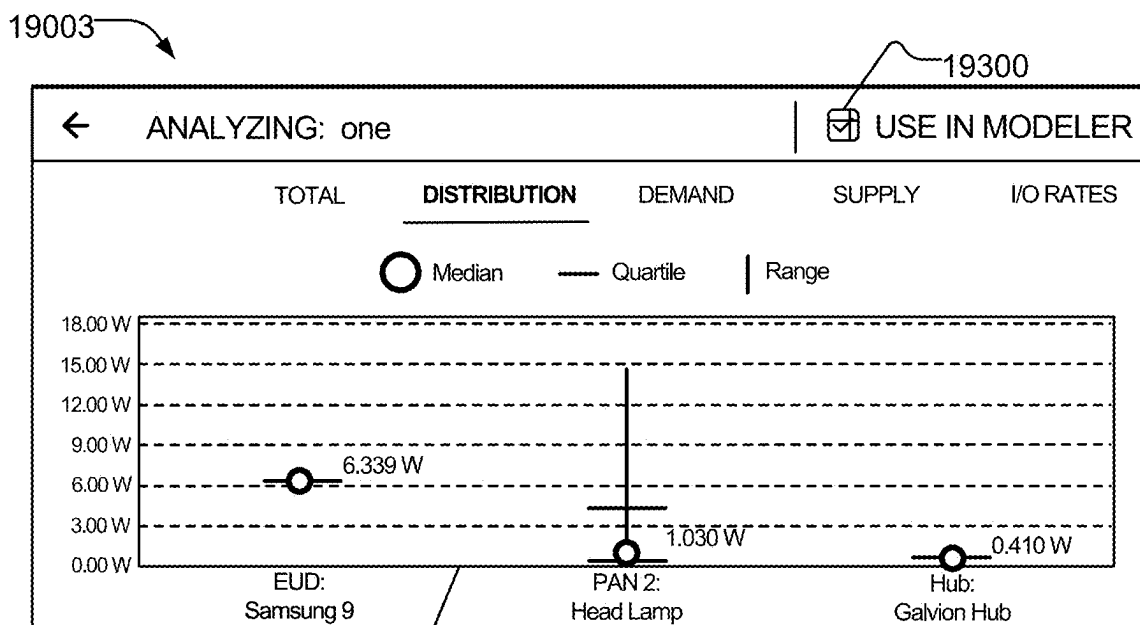


Fig. 19

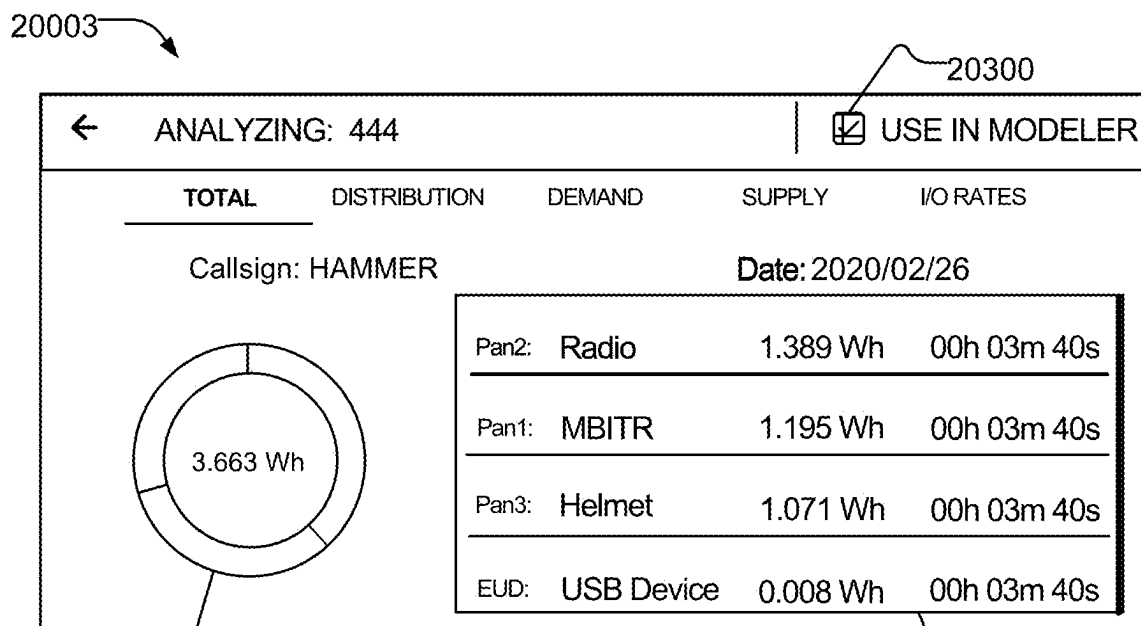


Fig. 20

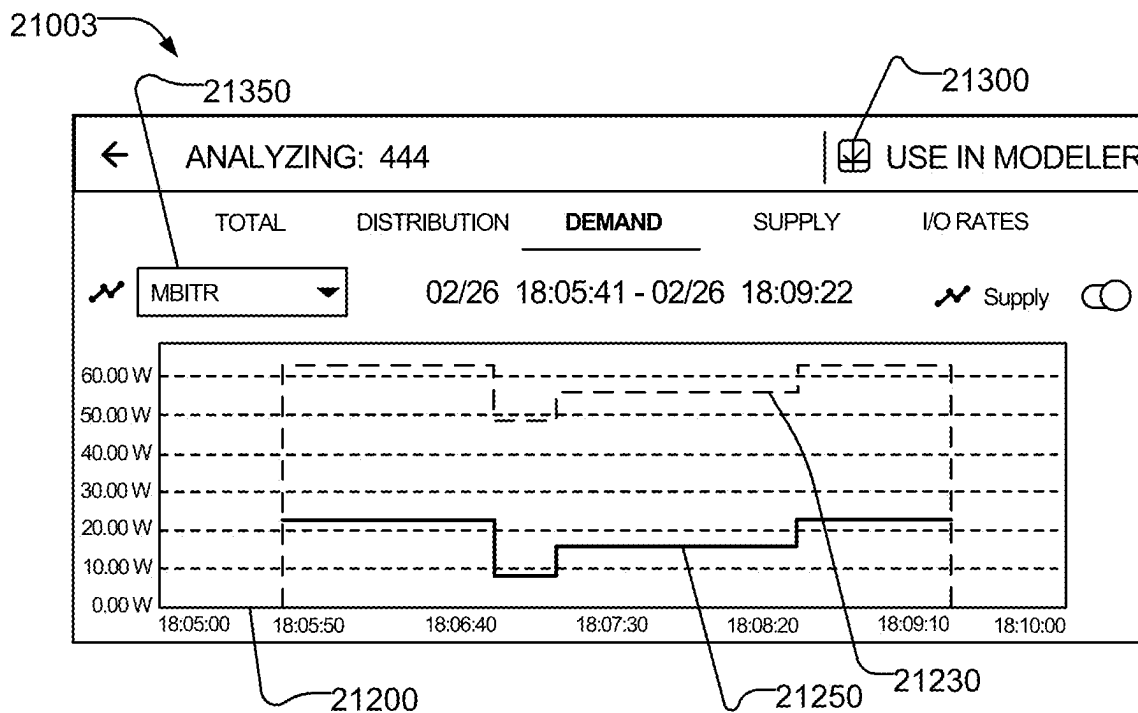


Fig. 21

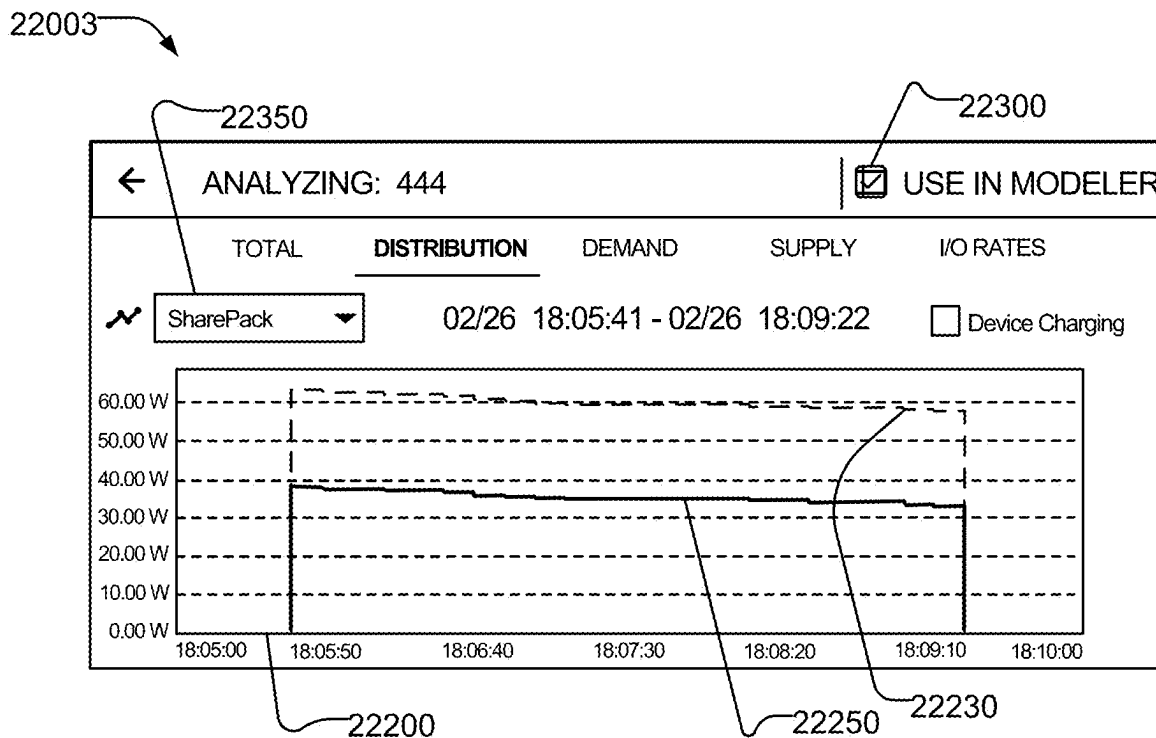


Fig. 22

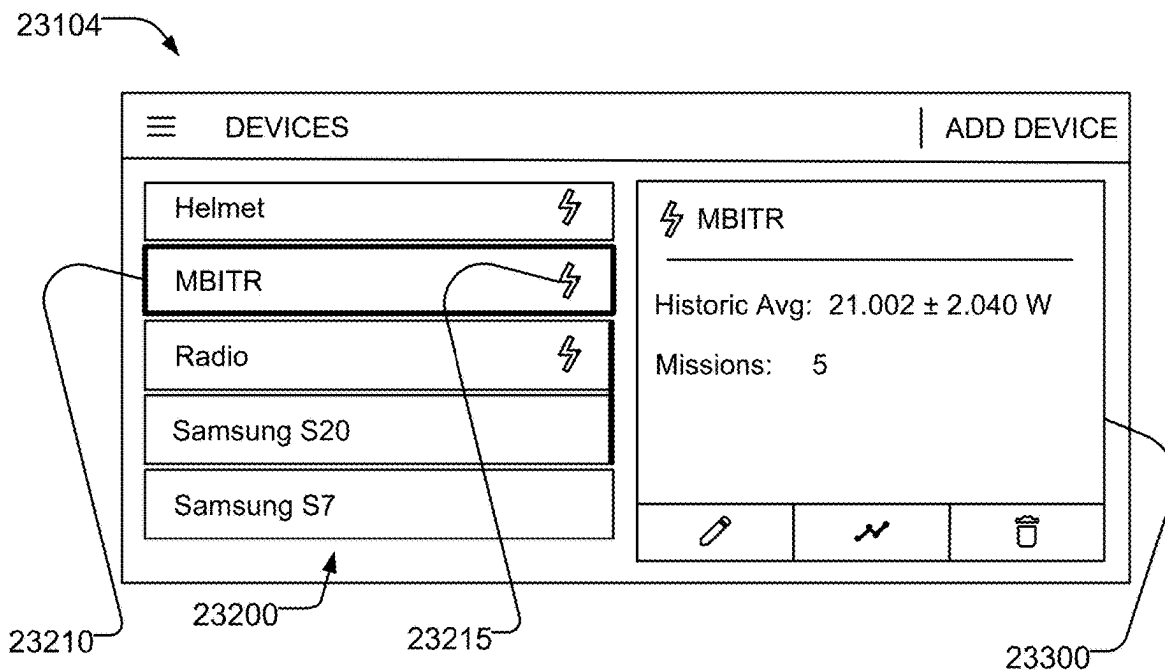


Fig. 23

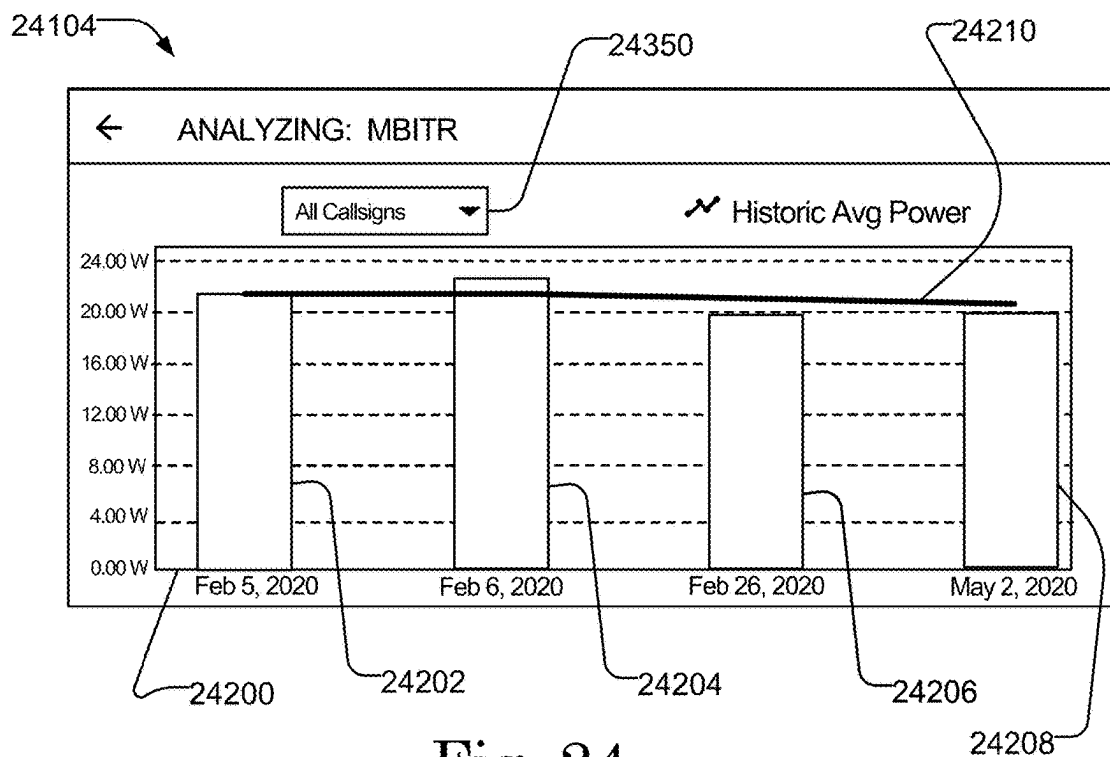


Fig. 24

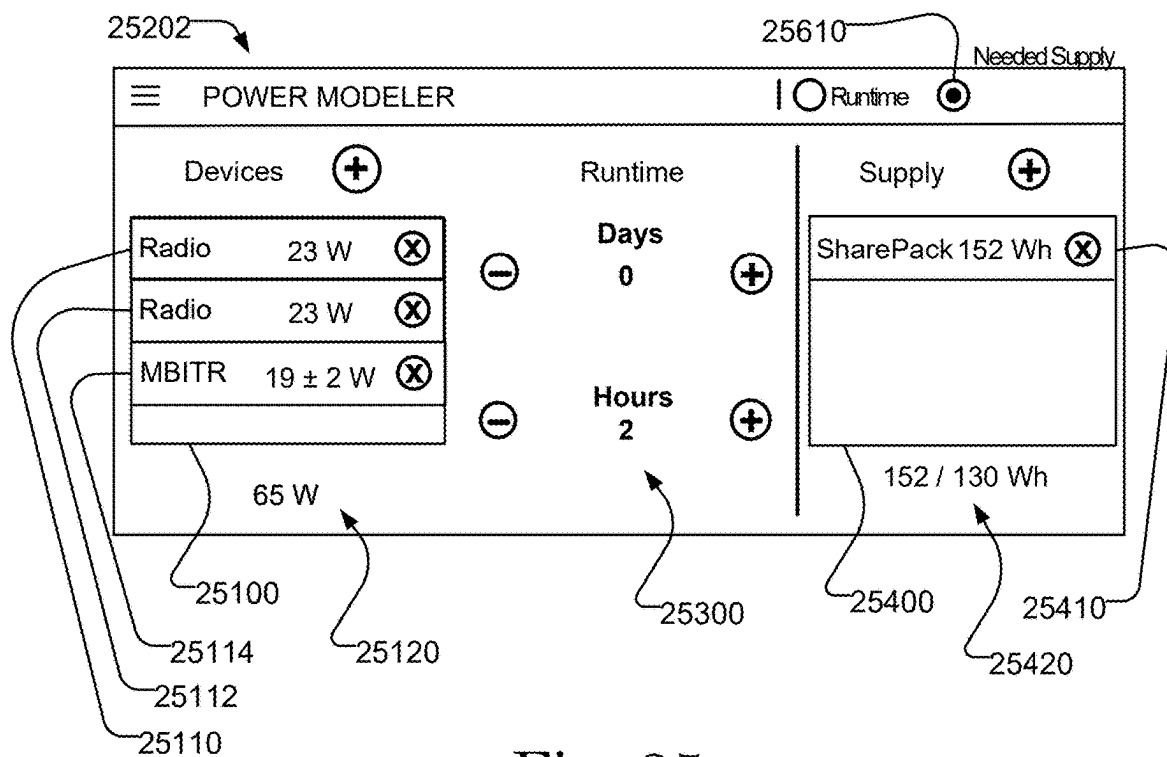


Fig. 25

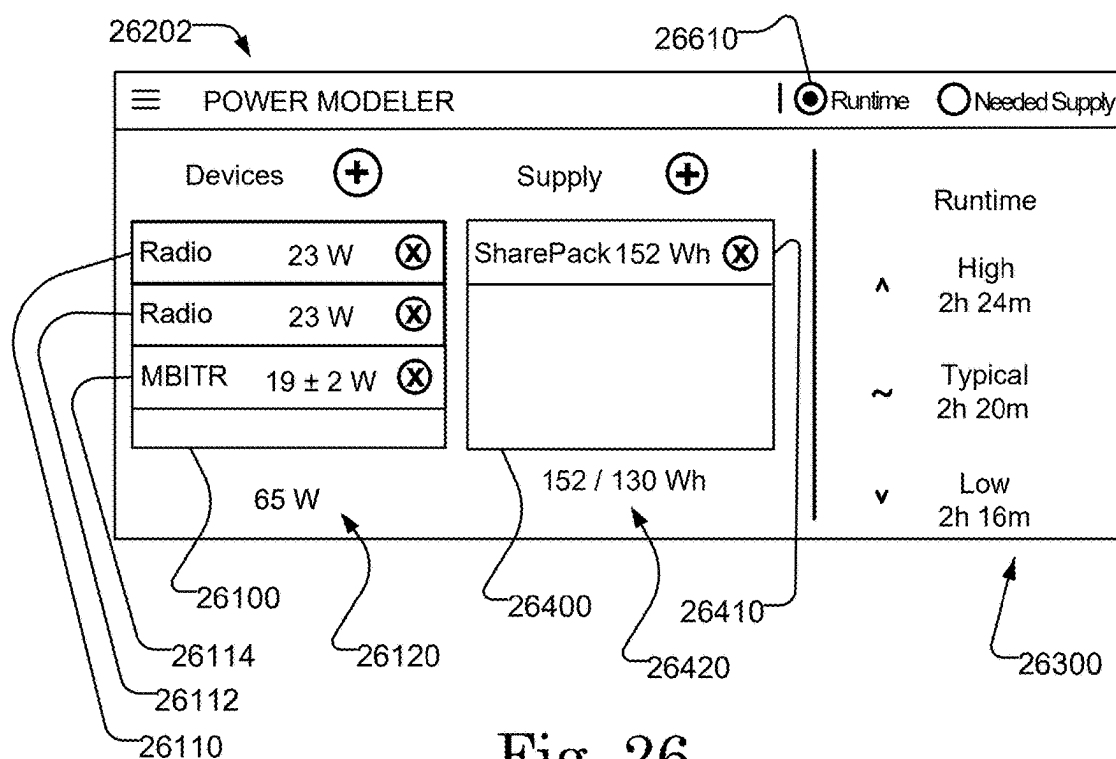


Fig. 26

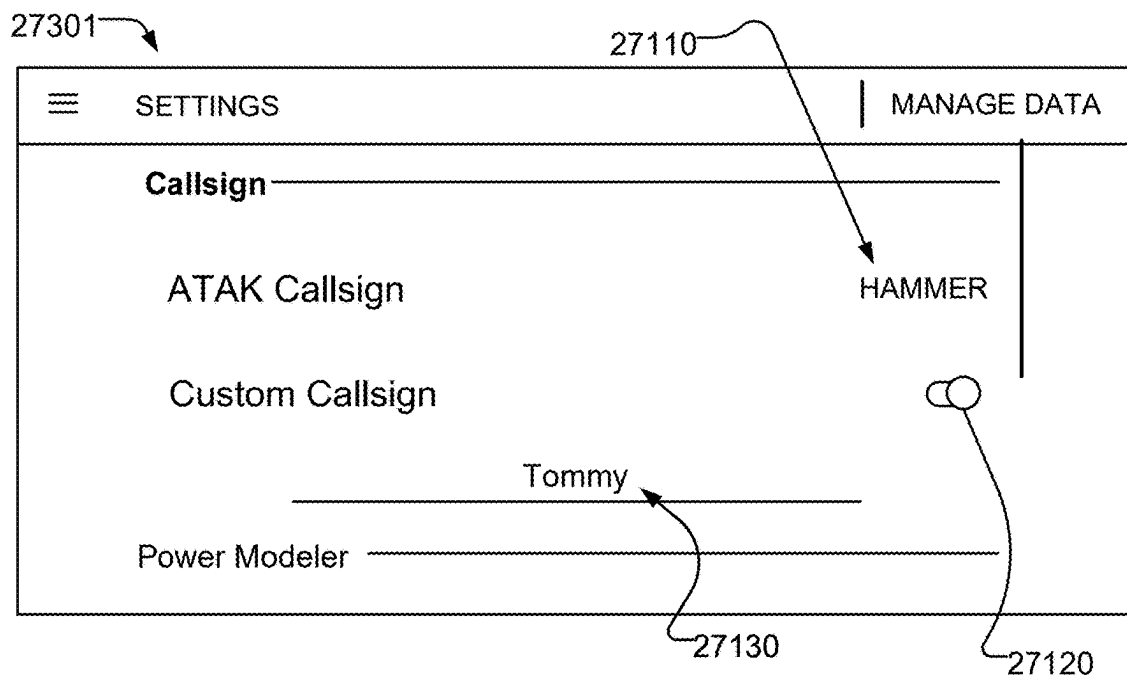


Fig. 27

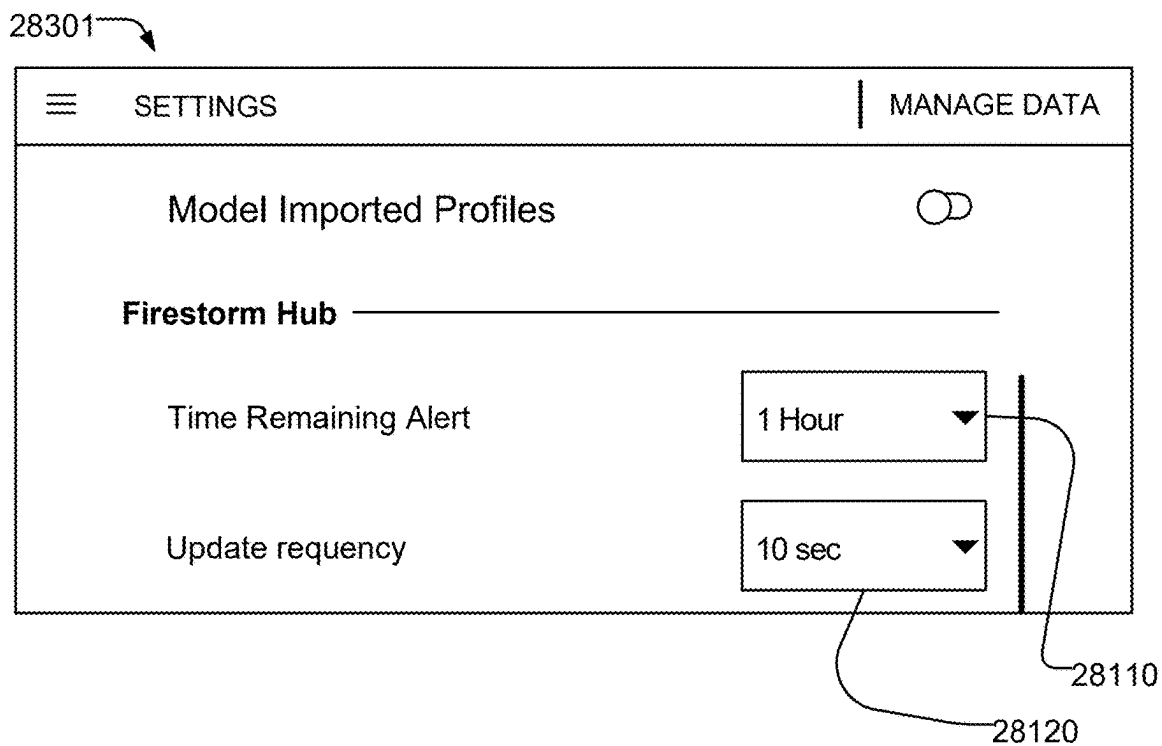


Fig. 28

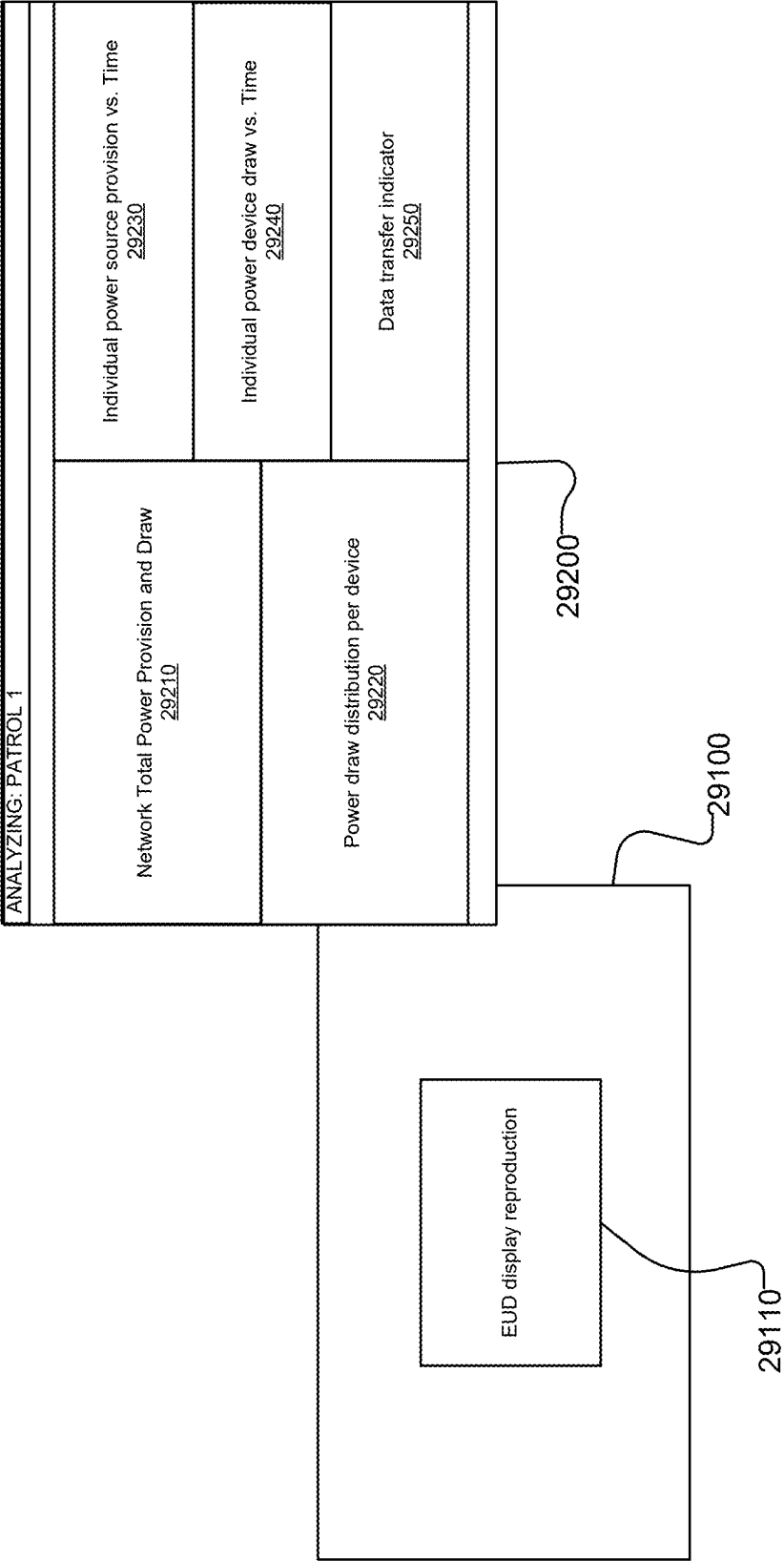


Fig. 29

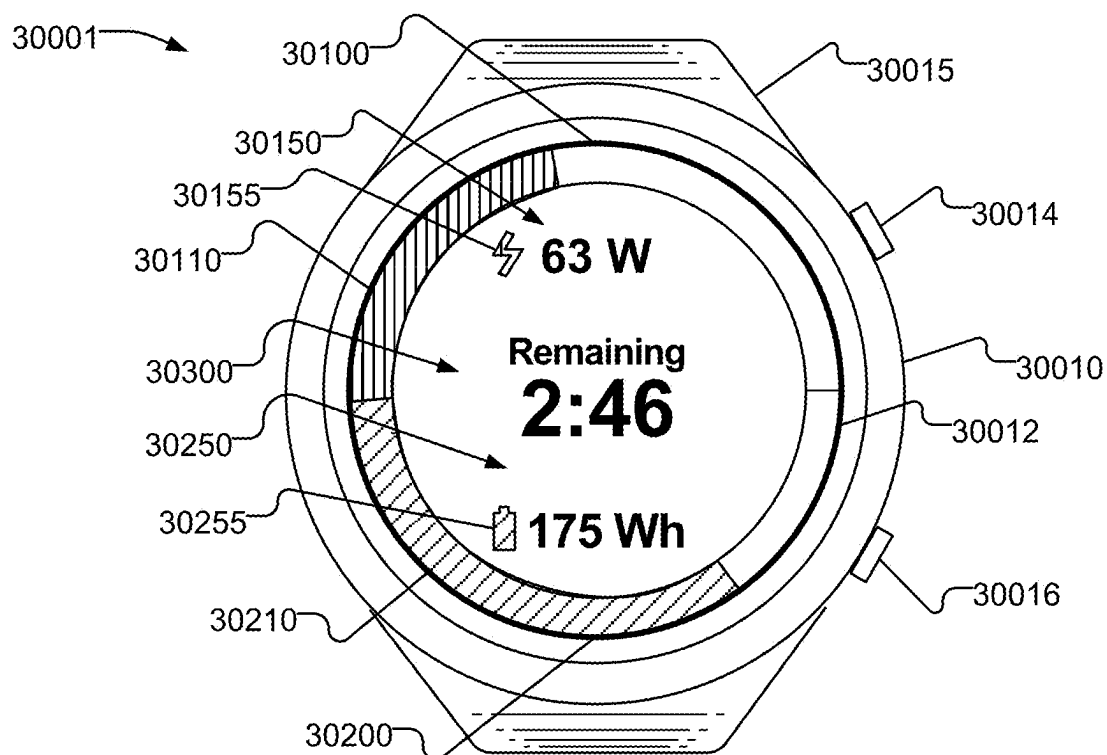


Fig. 30

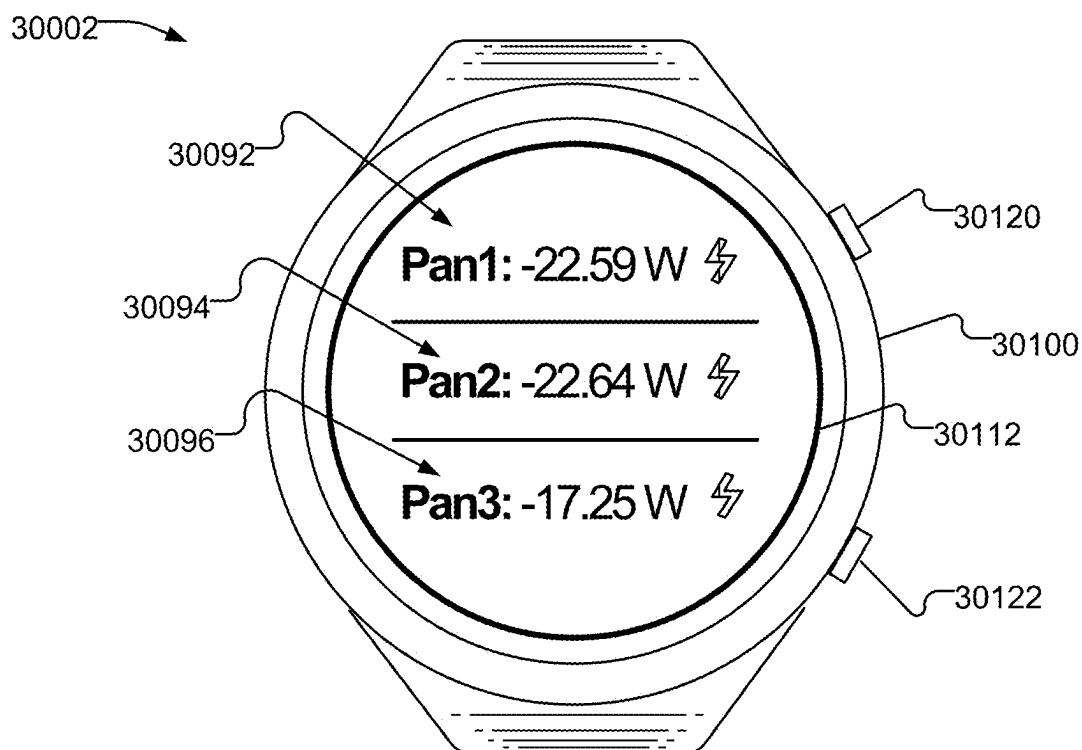


Fig. 31

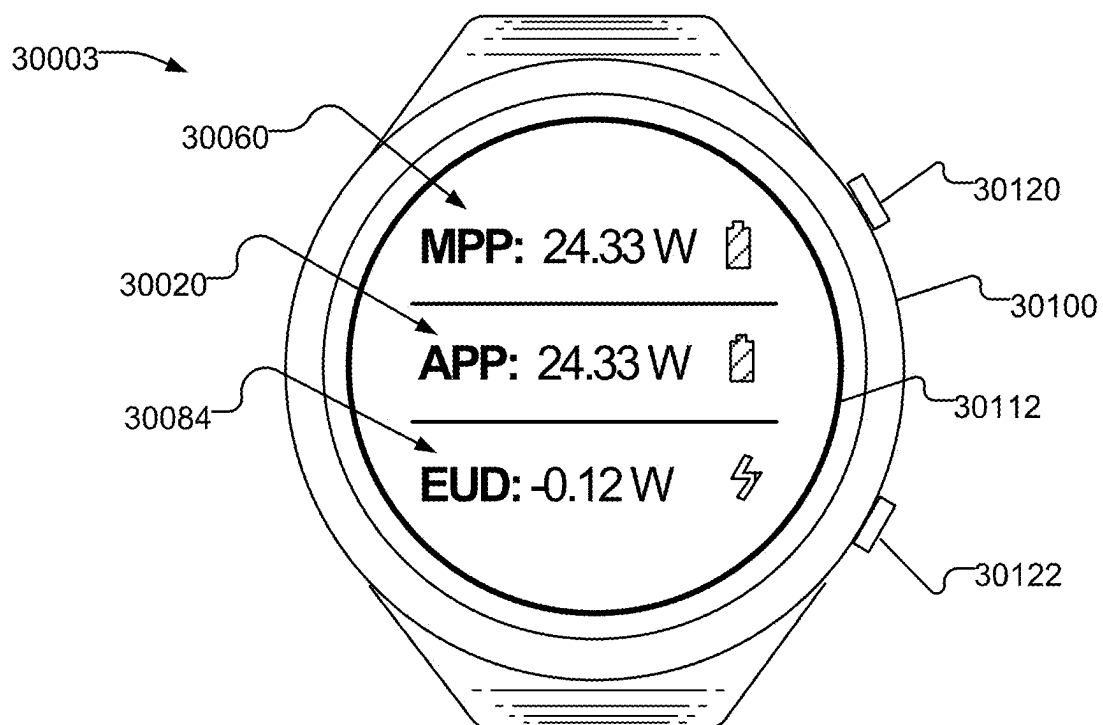


Fig. 32

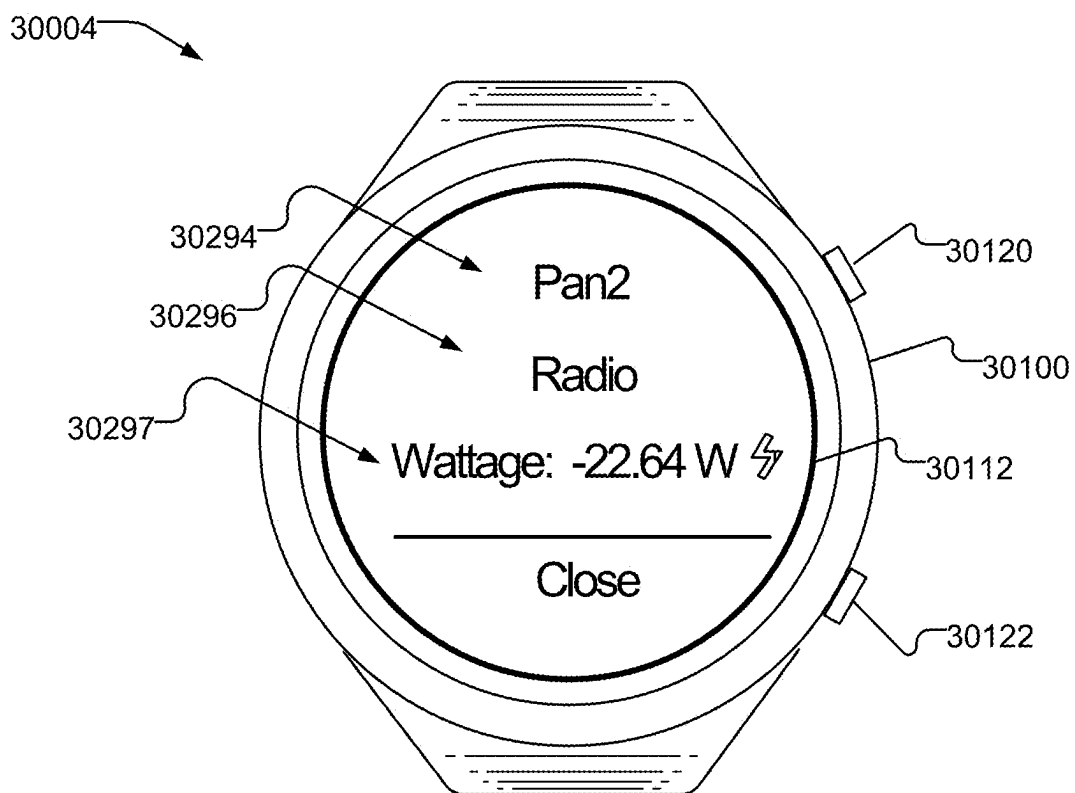


Fig. 33

POWER TRACKING USING A SOLDIER NETWORK MIDDLEWARE SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This present application is a continuation of and claims the benefit of U.S. patent application Ser. No. 18/317,629 filed on May 15, 2023, which claims the benefit of priority of U.S. Provisional Application No. 63/342,378 filed on May 16, 2022, and U.S. Provisional Application No. 63/397,044 filed on Aug. 11, 2022, the disclosures of which are incorporated herein by reference in their entirety.

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FIELD OF THE INVENTION

[0003] The inventive technology described herein includes a middleware messaging system useful for communicating messages between components of a soldier-portable (or intra-solider) personal area network (PAN).

BACKGROUND

[0004] Most, or all, message-oriented middleware is designed to operate on a TCP-IP network, the Bridge system is designed to operate on a multi-protocol PAN, and in particular on a multi-protocol solider PAN that includes at least on end user device operating an Android operating system and one or more Android applications. The technology described herein enables solutions to a number of problems that have not been previously solved including, but not limited to, a need to communicate and to share data with multiple applications in the Android OS environment and the need to communicate messages using multi-protocol routing one a solider multi-protocol PAN.

[0005] Furthermore, known solutions operate at the hardware and/or electronics layer and therefore require knowledge of details of devices that are sending and receiving data to implement and in some cases include device-specific solutions. The disclosed messaging system is designed to operate at the application layer. This enables a generic solution for multiple devices, electronics, and applications that does not require specific knowledge of what is occurring at a hardware and electronics level. The disclosed messaging system is generalized such that it can package and communicate information regardless of what particular device or devices generated the information.

SUMMARY

[0006] The inventive technology described herein includes a middleware messaging system useful for communicating messages between components of a soldier-portable (or intra-solider) personal area network (PAN).

[0007] In general, according to one aspect, the subject technology features a system that includes a middleware messaging system, called herein a Bridge system. The Bridge system includes bridge service modules, preferably in the form of APIs that are used to convert messages generated by an application running on a component of the PAN or received from a source outside of the PAN to a bridge message having a middleware message format. Bridge messages with the middleware message format are distributed to one or more devices connected to the PAN.

[0008] Whereas most, or all, message-oriented middleware is designed to operate on a TCP-IP network, the Bridge system is designed to operate on a multi-protocol PAN, and in particular on a multi-protocol solider PAN that includes at least on end user device operating an Android operating system and one or more Android applications. The technology described herein enables solutions to a number of problems that have not been previously solved including, but not limited to, a need to communicate and to share data with multiple applications in the Android OS environment and the need to communicate messages using multi-protocol routing one a solider multi-protocol PAN.

[0009] The bridge system operates across multiple communication and network protocols, including, for example, Universal Serial Bus (USB), Intra-Soldier Wireless (ISW) network, Bluetooth (BT), and Wi-Fi. If the PAN includes a multi-protocol network, the bridge messages can be distributed over the PAN using one or more of the PAN's network protocols. For example, in a PAN that includes USB connections, ISW network connections, and BT connections, bridge messages can be distributed over one or more of USB, ISW, and BT. In this manner, the Bridge system advantageously provides a multi-protocol router wherein messages can be communicated between applications and devices regardless of the network protocol or protocols used to interconnect the devices. In the case where more than one network, each with a different network protocol, is available for distribution of a bridge message, the Bridge system can select a particular network over which to communicate the bridge message, for example based on availability, preferred networks, network load, or other factors.

[0010] Whereas known solutions operate at the hardware and/or electronics layer and therefore require knowledge of details of devices that are sending and receiving data to implement and in some cases include device-specific solutions. The disclosed bridge system is designed to operate at the application layer. This enables a generic solution for multiple devices, electronics, and applications that does not require specific knowledge of what is occurring at a hardware and electronics level. The disclosed bridge system is generalized such that it can package and communicate information regardless of what particular device or devices generated the information.

[0011] The Bridge service includes different routing rules for each transport type. For USB routing, the Bridge system explicitly repeats a message for each USB connection, while for a wireless network the Bridge system sends a once because the wireless protocol sends to all devices. Further, to support multiple device needs, the bridge service modules can implement methods to send a message to a specific device without sending the message to other devices. This can be used, for example, to query a charge status from a battery.

[0012] The bridge service modules are also used to decode received bridge messages, e.g., to convert bridge messages received by a device on the PAN to a format that can be communicated to applications running on a device and to convert messages generated by an application running on the device to the bridge message format. The bridge service modules can be used to convert bridge messages to a format that can be communicated to and from devices outside of the PAN, for example to devices connected to a local area network (LAN) to which at least one component of the PAN is also connected.

[0013] In some examples, the technology described herein provides enhanced security. Because a bridge message payload is agnostic to data contained therein, the payload can be encrypted, thereby reducing the ability of a bad actor to extract useful information from intercepted bridge messages.

[0014] Example bridge messages generated, communicated, and decoded by devices implementing the novel technology disclosed herein include live power messages which carry live power usage and provisioning information corresponding to each of multiple power devices providing power to or receiving power from the PAN. Live power and provisioning information includes values representing near real time measurement of power draw and provision by one or more power devices that provide or receive power.

[0015] In a particular exemplary embodiment, the novel middleware is used to generate bridge messages, including live power messages, and communicate the bridge messages from a hub to an end user device (EUD) in USB environment. Data from a USB device is typically available to only one Android EUD application at a time. Bridge becomes the single point of communication between devices, which enables multiple applications to communicate to a device. Bridge relays the communications to multiple applications. For example, a power application may probe a radio for battery data, while a communications

[0016] application may change the radio channel. The result is multiple smaller applications may perform functions without needing to coordinate in a monolithic application.

[0017] The novel middleware technology described herein enables the EUD to communicate with a single device, i.e. the hub, to receive live power usage data, encoded in bridge messages, or from multiple devices simultaneously. Bridge messages can be similarly used to communicate other types of data, for example audio data, from multiple devices simultaneously. Multiple applications running on an EUD, or other device can each receive data from bridge messages. The EUD is able to display and use live power data from multiple devices that would not be available without the bridge message and middleware technology described herein. This enables a user to better understand power consumption of various pieces of equipment and to better manage a personal power network accordingly.

[0018] Bridge messages can also be used to encode any other type of useful information, for example USB sound data or other audio communication data, still image or video data, IP protocol data, and data packets from one or more other middleware, LAN, or WAN messaging protocols, for example MQTT middleware messages.

[0019] The above and other features of the subject technology including various novel details of construction and combinations of parts, and other advantages, will now be

more particularly described with reference to the accompanying drawings and pointed out in the claims. It will be understood that the particular method and device embodying the subject technology are shown by way of illustration and not as a limitation of the subject technology. The principles and features of this subject technology may be employed in various and numerous embodiments without departing from the scope of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] In the accompanying drawings, reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale; emphasis has instead been placed upon illustrating the principles of the subject technology. The features of the present subject technology will best be understood from a detailed description of the subject technology and example embodiments thereof selected for the purposes of illustration and shown in the accompanying drawings.

[0021] Various embodiments are depicted in the accompanying drawings for illustrative purposes, and should in no way be interpreted as limiting the scope of the inventions. In addition, various features of different disclosed embodiments can be combined to form additional embodiments, which are part of this disclosure. Any feature or structure can be removed or omitted. Throughout the drawings, reference numbers can be reused to indicate correspondence between reference elements.

[0022] FIG. 1 depicts a photographic rendering of an embodiment of a uniform messaging platform (UMP) according to the technology disclosed herein.

[0023] FIG. 2A depicts a perspective view of a first embodiment of exemplary power hub according to the technology disclosed herein.

[0024] FIG. 2B exemplary schematic diagram of a first embodiment of a system according to the technology disclosed herein.

[0025] FIG. 3 depicts an exemplary schematic diagram of a second embodiment of a system according to the technology disclosed herein.

[0026] FIG. 4 depicts an exemplary schematic diagram of a third embodiment of a system according to the technology disclosed herein.

[0027] FIG. 5 depicts an exemplary schematic diagram of a fourth embodiment of a system according to the technology disclosed herein.

[0028] FIG. 6 depicts an exemplary schematic diagram illustrating message flow according to the technology disclosed herein.

[0029] FIG. 7 is a process flow describing a first method of operation of one or more systems according to the technology described herein.

[0030] FIG. 8 is a process flow describing a second method of operation of one or more systems according to the technology described herein.

[0031] FIG. 9 is a process flow describing a third method of operation of one or more systems according to the technology described herein.

[0032] FIG. 10A depicts a first configuration of a first exemplary user interface (UI) layout for displaying power data according to the technology disclosed herein;

[0033] FIG. 10B depicts a second configuration of the first exemplary user interface (UI) layout of FIG. 10A.

[0034] FIG. 10C depicts a third configuration of the first exemplary user interface (UI) layout of FIG. 10A.

[0035] FIG. 10D depicts a fourth configuration of the first exemplary user interface (UI) layout of FIG. 10A shown on a display screen of an exemplary end user device (EUD) according to the subject technology.

[0036] FIG. 10E depicts multiple views of the EUD of FIG. 10D.

[0037] FIG. 11 depicts an exemplary user interface (UI) layout for displaying power data corresponding to a power port according to the technology disclosed herein.

[0038] FIG. 12A depicts a first configuration of a second exemplary user interface (UI) layout for displaying power data according to the technology disclosed herein;

[0039] FIG. 12B depicts a second configuration of the second exemplary user interface (UI) layout of FIG. 12A.

[0040] FIG. 12C depicts a third configuration of the second exemplary user interface (UI) layout of FIG. 12A.

[0041] FIG. 13 depicts an exemplary user interface (UI) layout for displaying power data corresponding to a battery device port according to the technology disclosed herein.

[0042] FIG. 14 is a process flow describing a fourth method of operation of one or more systems according to the technology disclosed herein.

[0043] FIG. 15 is a process flow describing a fifth method of operation of one or more systems according to the technology disclosed herein.

[0044] FIG. 16A depicts an exemplary user interface (UI) layout for managing power data according to the technology disclosed herein.

[0045] FIG. 16B depicts information correspond to the UI layout of FIG. 16A.

[0046] FIG. 16C is a process flow for managing power data according to the technology disclosed herein.

[0047] FIG. 17 depicts an exemplary user interface (UI) layout for managing power data according to the technology disclosed herein.

[0048] FIG. 18 depicts an exemplary user interface (UI) layout for displaying and analyzing power data according to the technology disclosed herein.

[0049] FIG. 19 depicts an exemplary user interface (UI) layout for displaying and analyzing power data according to the technology disclosed herein.

[0050] FIG. 20 depicts an exemplary user interface (UI) layout for displaying and analyzing power data according to the technology disclosed herein.

[0051] FIG. 21 depicts an exemplary user interface (UI) layout according to the technology disclosed herein for displaying and analyzing power data.

[0052] FIG. 22 depicts an exemplary user interface (UI) layout according to the technology disclosed herein for displaying and analyzing power data.

[0053] FIG. 23 depicts an exemplary user interface (UI) layout according to the technology disclosed herein for displaying historical power data corresponding to a selected power device.

[0054] FIG. 24 depicts an exemplary user interface (UI) according to the technology disclosed herein for displaying historical power data corresponding to a selected power device across a plurality of missions.

[0055] FIG. 25 depicts an exemplary user interface (UI) layout according to the technology disclosed herein for modeling and predicting a required power supply of a system according to the technology disclosed herein.

[0056] FIG. 26 depicts an exemplary user interface (UI) layout for modeling and predicting system runtime of a system according to the technology disclosed herein.

[0057] FIG. 27 depicts an exemplary user interface (UI) layout for selecting settings of a system according to the technology disclosed herein.

[0058] FIG. 28 depicts an exemplary user interface (UI) layout of a system according to the technology disclosed herein for selecting settings.

[0059] FIG. 29 depicts an exemplary user interface (UI) layout of a system according to the technology disclosed herein ported to a desktop PC monitor.

[0060] FIG. 30 depicts a first exemplary user interface (UI) layout for displaying power data on a smart watch according to the technology disclosed herein.

[0061] FIG. 31 depicts a first configuration of a second exemplary UI layout for displaying power data on a smart watch according to the technology disclosed herein.

[0062] FIG. 32 depicts a second configuration of the second exemplary UI layout for displaying power data on a smart watch of FIG. 31.

[0063] FIG. 33 depicts a third exemplary user interface (UI) layout for displaying power data on a smart watch according to the technology disclosed herein;

DESCRIPTION OF SOME EMBODIMENTS OF THE INVENTION

Detailed Description of the Invention

[0064] As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the singular forms of the articles “a”, “an” and “the” are intended to include the plural forms as well, unless expressly stated otherwise. It will be further understood that the terms: includes, comprises, including and/or comprising, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence of addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Further, it will be understood that when an element, including component or subsystem, is referred to and/or shown as being connected or coupled to another element, it can be directly connected or coupled to the other element or intervening elements may be present.

[0065] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit comprising hardware may also perform one or more of the techniques of this disclosure.

[0066] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various operations and functions described in this disclosure. In addition, any of the described

units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware or software components or integrated within common or separate hardware or software components.

[0067] The techniques described in this disclosure may also be embodied or encoded in a computer-readable medium, such as a computer-readable storage medium, containing instructions.

[0068] Instructions embedded or encoded in a computer-readable medium may cause a programmable processor, or other processor, to perform the method, e.g., when the instructions are executed. Computer-readable media may include non-transitory computer-readable storage media and transient communication media. Computer readable storage media, which is tangible and non-transitory, may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a CD-ROM, a floppy disk, a cassette, magnetic media, optical media, or other computer-readable storage media. It should be understood that the term “computer-readable storage media” refers to physical storage media, and not signals, carrier waves, or other transient media.

[0069] The terms “bridge service”, “bridge service module” and “uniform messaging service” have been used interchangeably throughout the disclosure. The term “the bridge”, “bridge system” and “uniform messaging platform” have been used interchangeably throughout the disclosure. The terms “bridge message”, “middleware message format”, “bridge formatted message(s)” and “uniform bridge message” have been used interchangeably throughout the disclosure. The terms “component”, “device” and “end-point” have been used interchangeably throughout the disclosure.

[0070] Referring now to FIG. 1, a perspective view of a uniform messaging platform (UMP) **1001** is shown. The UMP **1001** includes a uniform messaging service (UMS) which provides seamless messaging between devices operatively connected to a network system. The UMP includes a device interface **1998**, at least one processor **1071** and at least one memory **1072**. The at least one memory includes a profile database **1701**, a library **1703**, and a uniform messaging application (UMA) **1220** stored therein. In some embodiments, the at least one memory additionally includes a multi-protocol routing application (MPRA) **1222**. The device interface is communicatively coupled to at least one component (i.e., end-point) and at least one network.

[0071] The uniform messaging service (“UMS”) **1116** may be used on a network system, such as, for example, any of the network systems illustrated in FIG. 2B (i.e., **2200**), FIG. 3 (i.e., **3200**), and FIG. 4 (i.e., **4000**), to provide seamless communication between heterogeneous components connected to the network system. One or more of the components (i.e., end-points) connected to the network system may include the uniform messaging platform **1001** which provides the UMS **1116**. For example, in the network system illustrated in FIG. 2B, the uniform messaging plat-

form **1001** is only integrated in EUD **2110**. In the network system illustrated in FIG. 3, the uniform messaging platform **1001** is integrated in EUD **2110** and Hub **3001**. In the network system illustrated in FIG. 4, the uniform messaging platform **1001** is integrated in EUD **2110**, Hub **4001**, and Helmet system **4175**.

[0072] Any component connected to the same network that has the uniform messaging platform **1001** integrated therein is understood to include the UMS **1116**. Any component connected to the same network system that includes the UMP **1001** is a bridge-compatible device. Conversely, any component connected to the same network that does not include the UMP **1001** is a non-bridge-compatible device.

[0073] Referring now to FIG. 2A perspective view of a first hub (HUB 1) **2901** is shown. The first hub is configured for communicating power and data signals between devices operatively connected to the hub.

[0074] The first hub **2901** is a six-port hub that includes a hub enclosure **2940**, three device ports (i.e., **2992**, **2994**, **2996**), an end user device (EUD) port **2984**, a main power port **2960**, and an auxiliary power port **2910**. It is understood that alternative embodiments of a hub can include more or fewer device ports, power ports, and EUD ports without deviating from the technology disclosed herein. In a particular alternative embodiment, a four port hub includes a main power port **2960**, an EUD port **2984**, and two device ports **2992**, **2996**. In preferred embodiments, a hub includes at least one power port, one EUD port, and one or more device ports.

[0075] The hub enclosure **2970** protects internal components, including processors and other electronics, from, for example, water, weather, and mechanical damage.

[0076] Each device port includes a power device interface **2990** for mechanically and electrically connecting with a power device. Power devices include power loads, power sources, and devices that can function either a power load or power source depending on circumstances, for example rechargeable batteries.

[0077] The device ports (i.e., **2992**, **2994**, **2996**) are typically used for connecting power loads although in some embodiments they can be used for connecting power loads and/or sources. Each device port includes a device port cable **2950** that mechanically and electrically connects the power device interface **2990** to the hub enclosure **2970** and electrically connects the power device interface to the electronics contained within the hub enclosure. Each device port cable is connected to the hub enclosure at a hub/cable interface **2940**. Each device port is configured to provide power and data connections between the hub and a connected power device.

[0078] Each power port (i.e., **2910**, **2960**) includes a power supply interface (i.e., **2925**, **2922**) for mechanically and electrically connecting with a power supply, e.g., a power source or a rechargeable battery that is providing power. Each power port includes a power port cable (**2930**, **2932**) that mechanically and electrically connects the power supply interface to the hub enclosure **2970** and electrically connects the power supply interface to the electronics contained within the hub enclosure. Each power port cable is connected to the hub enclosure at a hub/cable interface **2940**. Each power port is configured to provide power and data connections between the hub and a connected power source.

[0079] The EUD port **2984** includes an EUD interface **2982** for mechanically and electrically connecting with an

EUD. Each EUD port includes an EUD port cable **2980** that mechanical and electrically connects the EUD interface **2982** to the hub enclosure **2970** and electrically connects the EUD interface to the electronics contained within the hub enclosure. Each EUD port cable is connected to the hub enclosure at a hub/cable interface **2940**. Each EUD port is configured to provide power and data connections between the hub and a connected EUD.

[0080] In some exemplary, non-limiting, embodiments, the first hub **2901** is configured to operate using a USB protocol and each of the device, EUD, and power ports are configured as USB-compatible ports. Alternatively, one or more of the main power port **2960** and auxiliary power port **2910** may be configured to interface with a non-USB power source.

[0081] In some embodiments, the hub **2901** is configured to be used to form an intra-soldier personal area network (PAN), for example a USB network that includes one or more of connected power devices, power supplies, and EUD. The PAN is not limited to a USB network, for example the PAN can include a network using one or more networking protocols, including USB, Bluetooth (BT), Wi-Fi, and Intra Soldier Wireless (ISW), for example an ISW network as provided by Alereon.

[0082] The first hub **2901** is configured to perform full voltage and current monitoring of external power devices **2901**, including power loads and power sources, electrically connected with one or more of the device ports and power ports. The first hub is further configured to log power characteristics of the connected power devices, for example one or more of power provided to the hub, power drawn from the first hub, and length of time that power is drawn or provided.

[0083] The first hub **2901** is configured to draw power from a main power supply, for example a 5V power supply, or to scavenge from an auxiliary power supply. The first hub is configured to provide power to power devices and to the EUD. The first hub can be configured to provide power to each connected power device and EUD at a configurable voltage and/or current amplitude that can be set based on power requirements of a connected power device or EUD. In examples, the first hub provides power having a voltage of 8V to 24V, or 8V to 17V.

[0084] Referring now to FIG. 2B, a schematic representation of an exemplary embodiment of a first intra-soldier personal area network (PAN 1) **2200** is shown. The first intra-soldier PAN includes a second hub (HUB 2) **2001**. The second hub (HUB 2) is substantially similar to the first hub (HUB 1) **2901** shown in FIG. 2A. The second hub is electrically connected to multiple power devices (e.g., **2101**, **2103**), power supplies (e.g., **2020**, **2060**), a helmet system **2105** and EUD **2110**. Each device port (e.g., **2992**, **2994**, **2996**), auxiliary power port (AUX PP) **2910**, main power port (Main PP) **2960**, and EUD port **2984** may include a cable (e.g., **2930**, **2932**, **2950**, **2980**) that is fixedly attached to the hub enclosure **2070**, as illustrated in FIG. 2A. In an alternative embodiment, one or more of the ports are mounted on an exterior surface of the hub enclosure and may be connected directly to a power device, power supply, or EUD or may be connected thereto via a detachable cable.

[0085] An auxiliary power supply (AUX power supply) **2020** is connected to the auxiliary power port (AUX PP) **2910** and a main power supply **2060** is connected to the main power port (Main PP) **2960**. The EUD **2110** is connected to

the EUD port **2984**. An exemplary EUD **2110** includes a smartphone running an Android operating system (OS), for example a Samsung Galaxy S20. The EUD can function as master of a system that includes the second hub and power devices operably connected to the first PAN. Moreover, the EUD includes the uniform messaging platform (UMP) **1001**.

[0086] A power device, and more specifically in some embodiments a power load, is connected to each device port. Exemplary power devices include a first IP radio, for example a Multiband Inter/Intra Team Radio (MBITR) (**2101**) connected to the first device port **2992**, a smart scope **2103** (connected to the second device port **2994**), and a helmet **2105** comprising one or more power devices, for example a communication system, strobe, and/or one or more sensors, connected to the third device port **2996**. In alternative configurations, one or more different power devices can be connected to one or more of the device ports including, for example, a gesture control system, for example a Pison system produced by Pison, Boston, MA, targeting sights, or a GPS system.

[0087] The second hub **2001** includes a processor **2071** and associated memory **2072**. The second hub includes a power and data router **2075** that is electrically connected with each port and that is controllable by the processor **2071** to exchange power and communication signals with and between the ports. In some embodiments, the power and data router **2075** includes a USB hub. Double-headed arrows represent communication and power exchange pathways.

[0088] A scavenger converter **2074** is disposed between the auxiliary power port **2910** and the power and data router **2075**. The scavenger converter is a DC to DC power converter that can be controlled by the processor **2071** to convert a voltage of a power signal received from an auxiliary power supply **2020** to a voltage that is compatible with the power and data router. Embodiments can include an AC to DC power converter for scavenging power from an AC auxiliary power supply.

[0089] In an exemplary embodiment, the auxiliary power port **2910** is configured to receive power up to 50V and up to 5 A and can handle power up to 60 W. Internal thermal protection is associated with the auxiliary power port to prevent overloading. The processor **2071** implements a math based detection scheme to identify a type of a power source connected to the auxiliary power source, for example to determine whether the source is a solar blanket, and AC adapter, or a generator. In an exemplary embodiment, a stable power source connected to the auxiliary power port is ramped up until voltage collapse, following which the last stable is used as the current point to extract a highest sustainable power level from the power source.

[0090] The second hub **2001** includes a universal serial bus interface (USB IF) **2057** for routing messages over a USB transmission medium (e.g., cables and ports), a ultra-wideband interface (UWB IF) **2059** for routing messages over a ultra-wideband transmission medium (e.g., radio waves) and a Bluetooth interface (BLE IF) **2067** for routing messages over a Bluetooth transmission medium (e.g., radio waves).

[0091] Referring now to FIG. 3 a schematic diagram of a second intra-soldier personal area network (PAN 2) **3200** is shown. The second intra-soldier PAN **3200** includes an EUD **2110**, a third hub (HUB 3) **3001**, and multiple power devices (Power Device 1 through n) (i.e., **3101**, **3103**, **3105**, **310n**). In embodiments, and referring to FIGS. 2A and 2B, the third

hub **3001** is substantially similar to the first hub **2901** and to the second hub **2001** and shares characteristics thereof.

[0092] The hub and EUD together implement a novel middleware bridge (i.e., both components include a UMP **1001**), which is operable to standardize data communicated between the two devices and to standardize transport of the data, both of which are useful for quickly integrating the middleware across multiple devices.

[0093] The third hub **3001** includes a hub processor **2071** that is communicatively coupled to the memory **2072**. The memory includes a hub power log **3307**, which includes a file system for storing power-related information.

[0094] The third hub **3001** includes multiple device ports (Device ports 1 through n) (i.e., **3092**, **3094**, **3096**, **309n**), each of which is communicatively connected to the hub processor **2071**. The set of device ports can include one or more ports for connecting a power load, a power source, or a rechargeable battery, as similarly shown in FIG. 2A and FIG. 2B. In an exemplary embodiment, the device ports of the third hub include three or more ports for connecting power loads (e.g., **3101**, **3103**, **3105** and **310n**), at least one port (e.g., **2960**) for connecting a main power supply (e.g., **2060**), and at least one port (e.g., **2910**) for connecting an auxiliary power supply (e.g., **2020**), wherein the at least one port for connecting an auxiliary power supply is electrically connected to a scavenging power converter. The third hub includes an EUD port **2984** that is communicatively connected to the hub processor **2071**. Each device port, power port, and EUD port is configured for exchange of power and data signals between the hub and a connected power source, power device, or EUD.

[0095] Multiple power devices (power devices 1 through n) are each connected to a device port. Each power device is electrically and communicatively coupled to a device port and can either provide power to the third hub or receive power from the third hub. Each power device can also communicate via the device port to which it is connected, power data to the hub processor and exchange communication signals with the hub processor.

[0096] The EUD **2110** is electrically and communicatively coupled to the EUD port **2984** and can exchange power and communications with the third hub over the connection.

[0097] Together, the third hub **3001**, EUD **2110**, and the power devices (i.e., **3101**, **3103**, **3105**, and **310n**) comprise components (i.e., end-points) of the PAN **3200**. The EUD can function as master of a system that includes the third hub and power devices operably connected to the second PAN **3200**. The components comprising the PAN can communicate and exchange power signals using one or more networking protocols. In a particular embodiment, the PAN uses a USB protocol for power and communication exchange. The PAN can use one or more additional or alternative communication protocols (i.e., transport protocols), for example BT or ISW (intra-soldier) network (e.g., UWB). In some embodiments, the hub can receive and/or provide non-USB signals including power signals to and from devices that do not communicate power information to the hub.

[0098] The EUD **2110** includes an EUD processor **2114** and associated memory **3118**. The EUD processor is communicatively connected to the memory and at least one port or network interface (not shown) for communication over the PAN. The EUD **2110** includes a UMP **1001** and operates a power application (Power App) **3304**.

[0099] The power application **3304** includes a display module (DISPLAY) **2187** configured to generate UI information to be displayed on the EUD display screen. The power application includes an analysis module (Analysis) **2188** configured to generate power statistics data, and a modeler module (Modeler) **2189** configured to use one or more models or algorithms to make predictions regarding power data.

[0100] The memory **3118** of the EUD **2110** includes a historical power database (Historical Power DB) **3703** and an EUD power log **3705**. The EUD power log **3705** is configured to store power data received by the EUD from the third hub **3001**. The historical power database **3703** is configured to store historical power data corresponding to a plurality of power devices, including power devices that may be operative connected with the third hub. The historical power data can include power data that was collected from power devices, or power data corresponding to power devices, on previous missions, i.e. power data that was collected in the field under real use conditions, or power data that was collected from power devices under test conditions. The historical power data can include metadata such as, for example, temperature, location, and time of day corresponding to collection of the power data. In some embodiments, the historical power data includes data collected and stored on a per-device basis. The historical power data can include power data pooled by device type. The historical power database is typically provisioned with historical power data prior to deployment of the EUD **2110** on a mission that includes the generation and collection of live power data. The historical power database can be augmented with newly collected live power data either during a mission or after a mission is completed.

[0101] In an embodiment, each power device (i.e., **3101**, **3103**, **3105** and **310n**) communicates raw power data messages **3300** to the hub processor **2071**. Exemplary raw power data include power that the power device (e.g., a load or rechargeable battery) is consuming, power that the power device (e.g., a source or power supply) is providing. For example, each power device provides a stream of data including instantaneous power consumption or power supplied by the power device to the hub processor. Exemplary raw power messages can further include data reported by a smart battery, e.g., state of charge (SoC) and state of health (SoH). The raw power messages can further include ancillary data or metadata such as a device identified and data that may be relevant for power device performance, for example temperature. Raw power data messages include an identifier of a device sending the message, for example a USB ID, hardware identifier, serial number, provisioned device ID, or the like.

[0102] In an embodiment, the hub processor **2071** probes each device port (i.e., **3092**, **3094**, **3096** and **309n**) and, in response, receives raw power data **3300** from each power device that is operably connected to the corresponding device port. The hub processor can be configured to retrieve raw power data messages at a default time interval or at an interval set by a user. For example, the hub processor can be configured to probe each device port at intervals of 1 second, 5 seconds, 30 seconds, 1 minute, 5 minutes, or at another desired interval. In addition or alternatively, the hub processor can be configured to determine power characteristics of power devices by monitoring voltage, current, and/or power signals provides by one or more power sensors (not

shown) in communication with the hub processor. In this manner, the hub processor can determine power characteristics of power devices independently of communications received from the power devices, for example of non-USB power devices.

[0103] The hub processor 2071 writes power data corresponding to each power device to the hub power log 3307. The hub processor may tag the power data with information including an identity of the power device, a mission ID, a callsign of a user of the third hub, and time information. The hub processor may also write additional information to the hub power log, including for example a temperature associated with power data, a date, time of day, battery state of health, battery state of charge, or any other information including information communicated to the hub processor by one or more of the power devices.

[0104] The hub processor 2071 and EUD 2110 include the uniform messaging platform 1001 to communicate live power messages. The hub processor of the third hub operates the UMP to generate live power data messages. The live power data messages 3500 include uniform bridge messages with power data corresponding to one or more of the power devices and a bridge header. The power data included in the live power data messages are generated by the hub processor based on raw power data received from the power devices.

[0105] An exemplary live power data message 3500 includes live power data corresponding to multiple power devices. In an embodiment, the hub processor 2071 uses the UMP 1001 to generate live power messages based upon raw power data messages received from power devices by combining data from multiple raw power messages into one or more live power messages, each including data from multiple power devices. In addition, or alternatively, the hub processor generates live power data messages based on raw power data that the hub processor has saved to the hub power log 3307.

[0106] In embodiments, the hub processor 2071 generates and communicates live power data messages 3500 that include power data corresponding to each power device connected to a device port. In this manner, the hub processor combines multiple streams of power data, each stream corresponding to a different power device, into a single stream of live power message data that includes power data from each power device connected to the third hub.

[0107] In some embodiments, the third hub 3001 uses USB for exchange of power signals between power devices and uses USB or another communication protocol for communicating live power data messages. In a preferred embodiment, the third hub 3001 preferentially uses USB for communication live power data messages but can use ISW if USB is unavailable and can use BT if both USB and ISW are unavailable. The hub processor can generate and communicate live power data messages at configurable intervals, for example at an interval corresponding to an interval at which the hub processor probes the device ports.

[0108] The EUD processor 2114 receives live power data messages from the third hub 3001 and operates the EUD bridge application (i.e., UMP 1001) to decode the received live power data messages, for example by removing bridge headers from live power data payloads and assembling the data payloads into messages that can be broadcast to applications running on the EUD. The EUD processor 2114 extracts power data corresponding to each of the multiple power devices from the live power data messages 3500. The

EUD processor can communicate the power data to the power app 3304 and can store power data in the EUD power log 3705.

[0109] The power app display module 2187 is configured to generate display data that includes live power data corresponding to power devices operatively connected to the third hub 3001. The analysis module 2188 of the power app 3304 is configured to generate power statistics, for example average and peak power consumption and provision.

[0110] In an exemplary embodiment the power app 3304 displays an average power drawn by power loads connected to the third hub, a peak total power drawn by the power loads, and a bar chart indicating live power draw of each of the power loads. In additional embodiments, the power app is configured to display average and peak power supplied by one or more power sources and power provided by each power source.

[0111] The power app 3304 can communicate with the EUD memory 3118 to access the historical power database 3703. In a particular embodiment, the power app retrieves, from the historical power database, historical power data corresponding to each of the power devices connected to the third hub and/or historical power data corresponding to power devices from which the power app has received power data. The power app may perform automatic filtering of power data, for example, to filter power data with values that are outside of expected bounds. In some embodiments, the historical power data is filtered by relevance. For example, if a mission occurs during winter, the power app can select historical power data with associated temperature data that is similar to temperature during the mission, which may be selectable based on co-stored temperature data. In other embodiments, the historical power database 3703 can be selectively provisioned with power data corresponding to conditions of a particular mission and power data can be filtered, for example to remove out-of-bounds values, prior to being saved to the historical power database.

[0112] The modeler module 2189 of the power app 3304 operates one or more predictive algorithms using live power data from one or more power devices and historical power data corresponding to the one or more power devices to generate predictions. In an exemplary embodiment, the modeler generates a prediction of remaining run (i.e., remaining powered time) based on live power data including power remaining from a power source (e.g., state of charge or a battery) and power drawn by power loads; and historical power data including historical power consumption and power provision data corresponding to connected power devices.

[0113] In embodiments, the power app 3304 can display a predicted remaining run time and update the displayed predicted time as new live power data is received from the third hub. In embodiments, the power app can generate one or more alerts based on one or more of live power data, historical power data, and predictions. For example, the power app can display an alert when a predicted remaining run time falls below a threshold value.

[0114] Referring now to FIG. 4, a schematic representation of a first network system (System 1) (4000) is shown. The first network system includes an intra-soldier personal area network (PAN) 4200 and an inter-soldier Wi-Fi network 4595. The PAN includes a fourth hub (HUB 4) 4001, and EUD 2110, and multiple power devices (Power Devices 1 through N) (i.e., 4101, 4103, 4105).

[0115] The components of the PAN are communicatively coupled to each other over at least one intra-soldier communication network, for example one or more of a USB network, an ISW network, and a BT network. Components of the PAN can also be communicatively coupled with external, to the PAN, devices, services, etc. over a LAN, for example over an inter-soldier Wi-Fi network.

[0116] The inter-soldier Wi-Fi network is configured for communication between the PAN, components of which are worn or carried by a first soldier, and components comprising one or more other PANs, each associated with another soldier, and other Wi-Fi enabled devices. The inter-soldier Wi-Fi network includes Wi-Fi device 1 and Wi-Fi device 2, which can include a hub comprising another PAN or any other Wi-Fi-enabled device. Arrows indicate communication of messages, including for example live power data messages and more generally any communication message, between components of the system.

[0117] In embodiments, the fourth hub 4001 is substantially similar to the first, second, and third hubs illustrated in FIGS. 1 through 3. The fourth hub (HUB 4) includes a UMP 1001, a hub processor (host processor) 2071, an associated memory 2072, a USB hub 4205, a Wi-Fi radio 4206, and an ISW radio 4203. The USB hub 4205 is configured to relay communication and power signals to, from, and between the power devices and the EUD. In embodiments, the fourth hub includes, in addition or alternatively to the USB hub, Wi-Fi radio, and ISW radio, one or more devices or modules for communication using another protocol, for example BT. The fourth hub includes multiple device ports and at least one EUD port (not shown).

[0118] The power devices (4101, 4103 and 4105) can include one or more power loads, power sources, or rechargeable batteries. Example power loads include, but are not limited to, a MBITR, a helmet, and another power device. Example power sources include, but are not limited to, a main power supply and an auxiliary power supply.

[0119] The EUD 2110 includes an EUD processor 2114 and an associated memory 2118, a Wi-Fi radio 4119 and one or more ports or network interfaces (not shown) for communication over the PAN 4200. The EUD processor operates a bridge service (UMS 1116) and multiple applications (App A, App B). Referring to FIG. 3, an exemplary application includes a power app 3304. Additional exemplary applications can include an ATAK application, a communication application, for example an application that converts audio to text, and one or more applications for interfacing and interacting with one or more power devices.

[0120] Referring now to FIG. 4, an example uniform bridge message flow is shown. The example bridge message flow uses the UMP 1001. The bridge message flow begins at the circle labeled "Start" and is indicated by arrows connecting various components, applications, and modules of the PAN and inter-soldier Wi-Fi network. The UMP 1001 includes loosely coupled integration between devices, including power device and the EUD, and applications and functions as a multi-protocol message router to route messages between devices and between applications and services.

[0121] A first application (App A) operating on the EUD generates an application message (Message). The application message can be a message having a particular format, for example MQTT message format, or a message for any

transport protocol that the application is capable of using, for example Android OS broadcast, USB, ISW, BT, or Wi-Fi.

[0122] The UMP 1001 operating on the EUD processor generates a bridge message that includes the application message as a bridge message payload. The bridge service module (i.e., UMS 1116) inserts the applications message as-is into a payload section of a bridge message and adds, to the bridge message, a header section that includes information that indicates how the middleware should route the bridge message. The bridge service module may split the application message into bridge message payloads of multiple frames to make a larger application message compatible with size restrictions associated with a particular transport protocol used for communication of the bridge message or messages. In this case, the bridge message payloads are reassembled into the original application message on the receiving side.

[0123] The EUD bridge service (i.e., UMS 1116) communicates, via the USB hub, the middleware formatted bridge message to a host bridge service module operating on the host processor. The EUD 2110 can select a particular transport protocol for sending uniform bridge messages. In an example, the EUD bridge service generates a plurality of bridge frames containing the application message and the bridge frames are broadcast from the EUD in USB data packets. In an example, EUD can send the bridge frames across ISW, BT, or Wi-Fi protocols. In a particular example, the EUD hosts a USB device and a BT-connected device concurrently. The middleware ensures that all connected devices receive messages over the highest priority connection between the EUD and each of the devices. In a preferred embodiment, the EUD preferentially sends the middleware formatted bridge messages over USB. If USB is not available the EUD sends the messages over a soldier wireless network such as an ISW network, and if neither USB nor soldier wireless are available, EUD sends them over BT. In a similar manner, a bridge service operating on another device, for example on the hub or on a power device, can send bridge frames across a protocol selected from one or multiple available protocols.

[0124] The host bridge service module receives middleware formatted bridge messages from the EUD bridge service and broadcasts the middleware formatted bridge messages to other devices comprising the PAN, for example to each of power devices 1 through N and to the EUD. The host processor can send the middleware formatted bridge messages over an available intra-soldier communication network, for example of USB, ISW, or BT (in example descending order of preference).

[0125] Referring now to FIG. 2, in some embodiments a hub (e.g., HUB 2) a hub processor may not act as a USB host or alternatively, a hub may not include a processor. In these embodiments, a host bridge service module (i.e., UMS 1116) may be installed on one or more devices connected to the hub, for example on an Android OS EUD connected to the hub. The software selects one of the devices that includes a bridge service module to act as host.

[0126] Referring once again to FIG. 4, if a power device or EUD 2110 that receives a middleware-formatted bridge message includes a bridge service (i.e., UMS 1116), the device or EUD bridge service decodes the bridge message, i.e., removes the bridge header and extracts the data payload that includes an application message, or a portion thereof. In other words, the EUD bridge service receives, from the hub,

one or more bridge messages frames according to the technology disclosed herein. The EUD bridge application extracts an application message from one or more bridge messages frames and reassembles the application message if it was received as multiple portions in multiple frames. The application message is reassembled from the frames, even if frames are not received in order or interspersed with frames from other messages. The application message can then be passed to one or more applications on the EUD. For example, the EUD receives a bridge message frame (i.e., uniform bridge message) from the host processor **2071** and operates the EUD bridge service to extract, from the frame, an application message that can be communicated to one or more applications operating on the EUD. In the example illustrated in FIG. 4, Application B, operating on the EUD, receives the message from the EUD bridge service and processes information contained therein. In this manner, applications interact with the EUD bridge service to receive data that was encoded in bridge messages.

[**0127**] In some embodiments, an EUD **2110** includes a hub API **4117**, which it can use to communicate to the hub to set various system properties, including but not limited to, system recording period, system date, system time, system license, system calibration parameters. The EUD **2110** can also use the hub API to communicate to the hub to request and receive various data, including but not limited to, system recording period, system date, system time, system license, names of the full set of hub stored datalogs by date and time, or the full content of a specific datalog. The EUD can also use the hub API to communicate to the hub to request the hub to delete a specific datalog or to request the hub to delete all datalogs.

[**0128**] If the host processor **2071** receives one or more bridge messages that include, in a bridge header field, an indication that the messages should be transmitted over Wi-Fi, the host bridge service generates Wi-Fi packets that include the bridge message payload content and transmits the Wi-Fi packets to Wi-Fi devices (Wi-Fi device 1, Wi-Fi device 2) over the inter-soldier Wi-Fi network. In exemplary embodiments, only the host processor is configured to send messages out of the PAN, e.g., over Wi-Fi or over another LAN or WAN. In examples, the host processor can send messages an external, relative to the PAN, device, for example a Wi-Fi device, which is part of an intra-squad, intra-platoon, or intra company network. Advantageously, bridge messages that include power data can be communicated from one or more PANs to one or more external devices. The power data from one or more PANs can be analyzed on an external device, either on a per-PAN basis or as data aggregated from the multiple PANs, and in some examples combined with data from a source external to a PAN, for example from a weather forecast data source. An example analysis of this type includes detecting an impending low power state within a group of soldiers, for example within a squad. In addition, a hub can receive message data from an external device, for example historical power data to seed a local data store comprising historical power data with aggregated historical power data.

[**0129**] In some embodiments, another device connected to the PAN **4200**, for example the EUD, includes a Wi-Fi radio **4119** and can send Wi-Fi messages out of the PAN, for example to Wi-Fi device 2 (e.g., a device connected to **4595**). Referring now to FIG. 2, this is useful in the case where a hub (e.g., HUB 2) **2001** does not include a Wi-Fi

radio. In an example wherein multiple Wi-Fi enabled devices are connected to a PAN, a single Wi-Fi enabled device is selected for communication, via Wi-Fi, with devices outside of the PAN. In an example, multiple bridge service module instances communicate to select the one Wi-Fi enabled device that acts as host between the PAN and a LAN. This is advantageous to prevent duplicate messaging if multiple Wi-Fi enabled devices on the PAN were to each communicate messages to the LAN.

[**0130**] Referring now to FIG. 5 is a schematic representation of a second network system (system 2) (**5000**) is shown. The second network system includes an intra-soldier PAN **5200**, an inter-soldier Wi-Fi network **5601**, and a MQTT broker **5605**. MQTT is a server-based middleware used to integrate the PAN with TCP-IP networks, for example across a battlefield. The MQTT broker provides integration with a wide area network WAN for exchange of MQTT messages. The PAN **5200** includes a host processor **2071**, a UMP **1001** and a USB hub **4205** controlled by the host processor. The EUD **2110** is in communication with the host processor **2071** via the USB hub **4205**.

[**0131**] An application (App C) operating on the EUD **2110** generates a MQTT message **5118**. The EUD bridge service (UMS **1116**) generates one or more bridge messages **5115** with the MQTT message as bridge message payload and with added bridge headers. Bridge-specific information, for example a message target and indication of payload type, are included in the headers of the bridge-formatted messages. Bridge message payloads are generic, for example holding one or more MQTT messages as shown. The EUD bridge service communicates the bridge formatted message(s) to the host processor **2071** via the USB hub. In an alternative implementation, the EUD can communicate the bridge-formatted message(s) to the host processor over a different PAN communication network, for example over ISW or BT.

[**0132**] The Host processor receives the bridge-formatted messages from the EUD. The host bridge service removes the bridge header(s) and sends the MQTT message over inter-soldier Wi-Fi to the MQTT broker.

[**0133**] The host processor can receive an MQTT message **5119** from the MQTT broker **5605**. When this happens, the host bridge service (i.e., UMS **1116**) generates one or more bridge-formatted messages **5113** that include the received MQTT message as bridge message payload(s) and with added bridge headers. The host processor transmits the bridge-formatted message(s) to the EUD **2110**. The EUD bridge service removes the bridge header(s) from the bridge formatted message(s) and provides the received MQTT message payload **5117** to the application (App C).

[**0134**] A method implemented by a system similar to the second system (system 2) **5000** can be used to encapsulate message payloads with native formats other than MQTT in bridge-formatted messages, communicate the bridge-formatted messages through the PAN, remove bridge headers from the message payloads, and transmit the message payloads in native format over one or more communication networks or provide message payloads in native format to one or more applications (e.g. App C).

[**0135**] Referring now to FIG. 6, a diagram including exemplary, non-limiting, details of operations carried out by components implementing exemplary inventive middleware according to embodiments of the technology disclosed herein is shown. FIG. 6 illustrates operations carried out on

a device such as an EUD **2110** or power device or on a hub or host processor, with reference to FIGS. **1** through **4**.

[0136] The UMP **1001** is designed to isolate messaging APIs (e.g. a Bridge API) from hardware and to isolate the messaging API from message transport protocols. Novel aspects of the disclosed UMP **1001** include, but are not limited to, rules for handling data packets, routing of data packets, and device queues. A particular embodiment of the technology is designed specifically for Android OS and for meeting the needs of a multi-protocol soldier PAN.

[0137] A consumer application includes a bridge library **1703** with a bridge API (e.g., **6704** and **6705**) and a broadcast listener module. The consumer application represents an application operating on a host or hub processor, an application operating on a power device, or an application running on an EUD, for example and referring to FIG. **4** and FIG. **5**, applications (App A, App B, and App C).

[0138] The bridge service is operable to generate bridge frames that include a message generated by the consumer application and a bridge header. The bridge service is further operable to process the bridge frames for transport routing, and transmit the bridge frames over a selected transport protocol, for example one or more of USB, ISW Wi-Fi, or BT.

[0139] The bridge service is also operable to receive packets or other data structures that include bridge frames over a transport protocol (e.g. USB, ISW, Wi-Fi, or BT), extract the bridge frames from the received packets or other data structures, assemble data payloads from the frames into a message, and broadcast the message to one or more applications, for example using Android OS broadcast.

[0140] Bridge headers include information that may be used by the bridge service for processing bridge messages. Exemplary information in a bridge header includes sending device ID, and optionally a target device ID. The middleware operates using a Publish/Subscribe model wherein messages are broadcast to all connected devices but can support Request/Response messaging based on information included in the header, e.g., target device ID. In an embodiment, when a device sends a targeted message, the bridge sets up a temporary listener service to listen for a reply from the intended target device and to receive the full response before sending the response to the requester.

[0141] The bridge service includes a number of rules including, for example: rules for how to handle packets based on information in headers; rules for decoding and assembling data from packets; rules for processing targeted packets; and rules for routing packets based on type of device that is processing the packets. The rules can include different routing rules for each transport type. In some embodiments, the rules include rules for filtering data, for example based on data content of packets, and rules for routing data to particular devices, for example based on data content of packets.

[0142] If the bridge service is operating on a host processor or if the bridge service is operating on a device (e.g., an EUD or power device) and there are no other devices connected to a PAN comprising the device, bridge frames may be added to an incoming frame queue on the device. If the bridge services is operating on a host processor, bridge frames extracted from incoming messages (e.g., from USB, Alereon, Wi-Fi, or BT messages) may be added to a device queue to be broadcast from the host processor over one or more of USB, Alereon, Wi-Fi, and BT.

[0143] Turning to FIG. **7**, the steps described in flow diagram **7000** describe exemplary functions performed by the UMS **1116**. The UMS **1116** initially discovers any component(s) (this includes services offered by the component(s)) connected to the network system on which the UMS is operating, at step **7100**. The UMS may use a discovery protocol, such as, for example, service discovery protocol (SDP), simple service discovery protocol (SSDP), universal plug n play (UPnP), service location protocol (SLP), universal discovery description and integration (UDDI), secure persuasive discovery protocol (SPDP), etc.

[0144] The discovery protocol allows the UMS **1116** to obtain configuration and profile data for the discovered component (Step **7102**). For example, using a discovery protocol, the UMS may send an inquiry to a discovered component on the network requesting whether the component is enabled for UMS. The configuration and profile data for the discovered component(s) may include device identification information, one or more transport protocols used by the component, data routing rules, location/port number, data syntax/format, whether the component is bridge compliant or non-bridge-compliant, data routing rules, etc.

[0145] At steps **7104-7106**, the UMS **1116** creates a device profile for the component including the configuration and profile data and stores the device profile in the profile database. In addition to the profile database, the uniform messaging platform includes a library **2703** which has a collection of APIs used by the UMS for converting messages received on the network.

[0146] In a heterogeneous network system, (e.g., the network systems illustrated in FIGS. **2-4**), the devices connected to the network system may operate in different computing environments, and thereby may have different operating systems, programs, processors, transport protocols, transmission mediums, messaging protocols, etc. Thus, when a message is created by a device on the network system, the message is created in a format native to the device (i.e., using the specific protocols and resources that govern the device). Accordingly, a native source-format message is a message formatted consistent with the computing environment of the source device (i.e., end-point), and a native destination-format message is a message formatted consistent with the computing environment of the destination device (i.e., end-point).

[0147] The UMS **1116** disclosed herein allows UMS-enabled devices (i.e., devices integrated with the uniform messaging platform) to communicate with one another using a uniform bridge message. The format of the uniform bridge message is different from the native format of any source or destination. The advantages in using the uniform bridge message include providing a straight-forward and efficient means for messaging, reducing single point bottleneck and minimizes latency. Since, conversion to a uniform bridge message is straight forward, it reduces the processing time necessary to perform messaging between two UMS-enabled devices. Each UMS-enabled device is able to encode and decode uniform bridge messages.

[0148] When a received message is intended for a UMS-enabled device (bridge-compliant device), the UMS **1116** converts the message into a uniform bridge message by inserting the message as-is into a payload section of the uniform bridge message and adding a header section that includes routing information. The UMS **1116** is able to use the received message as-is unless the size of the message

exceeds the size restrictions associated with a particular transport protocol for communication of the uniform bridge message(s). In this instance, the UMS **1116** may split the large message into multiple uniform bridge message payloads having the same header information for transmission to a destination UMS-enabled device. A destination UMS-enabled device is able to decode the uniform bridge message (s) and perform further processing.

[0149] A uniform bridge message cannot be encoded or decoded by a device that is not UMS-enabled. Thus, a uniform bridge message is not generated for a device that is not UMS-enabled. To that end, when a received message is intended for a device that is not UMS-enabled (i.e., non-bridge-compliant), the received message will be in a format native to the source device. Hence, the UMS **1116** will not be able to use the message as-is but will instead have to convert the message into the format native to destination, which inherently requires more processing than generating a uniform bridge message. The UMS **1116** uses the information in the profile database and the library to convert the message into the format native to the destination.

[0150] Turning to FIG. **8**, the processes performed by the uniform messaging application (UMA) **1220** in the UMS **1116** are described in flow diagram **8000**. The UMA **1220** waits to receive a message from an end-point connected to the network system. At step **8002**, the UMA **1220** receives a message from a source (i.e., a source end-point) for transmission to a destination (i.e., a destination end-point). At step **8004**, the UMA **1220** determines the destination of the message from information included in the header of the message. At step **8006**, the UMA retrieves and examines the profile information for the destination from the profile database. Using the information retrieved from the profile database, the UMA **1220** is able to determine whether the destination is a bridge-compliant device (i.e., step **8008**) and is able to identify one or more transport protocols used by the destination device (i.e., step **8009**).

[0151] At step **8010**, the UMA **1220** converts the message into a universal bridge format, if the UMA **1220** determines that the destination is bridge compliant. If the UMA **1220** determines that the destination is not bridge compliant, at step **8012**, the UMA **1220** converts the message into a native destination format. At step **8014**, the converted message and the one or more transport protocols are provided to the multi-protocol routing application (MPRA) **1221** in the UMS **1116**.

[0152] Turning to FIG. **9**, the processes performed by the multi-protocol routing application (MPRA) **1221** in the UMS **1116** are described in flow diagram **9000**. At step **9010**, the MPRA **1221** receives one or more transport protocols (e.g., ultra-wideband (UWB), Bluetooth, USB, Wi-Fi, serial data) identified by the UMA **1220**. If the MPRA **1221** only receives one transport protocol (step **9012**), then at step **9016**, the MPRA **1221** selects the interface that conforms to the transport protocol. If the MPRA receives more than one transport protocol for the destination (i.e., step **9012**), then at step **9014**, the MPRA identifies an interface (e.g., UWB interface, SMBus, TCP/IP, USB, Bluetooth) for each transport protocol. At step **9018**, the MPRA determines the availability of each identified interface. If only one interface is available (step **9020**), then at step **9023** the MPRA selects the available interface for transmitting the converted message to the destination. If more than one of the interfaces is available (step **9020**), then at step **9022** the MPRA selects the

highest priority interface for transmitting the converted message to the destination. The highest priority interface may be selected based on preferred interface type, bandwidth, or other factors. At step **9024**, the MPRA outputs (i.e., routes) the converted message to the destination using the selected interface. In some exemplary embodiments, the MPRA **1221** uses different routing rules for each transport type to transmit messages to one or more destinations.

[0153] In some exemplary embodiments, the UMS **1116** may receive multiple native format messages simultaneously from multiple end-points connected to the network. In this instance, the UMS **1116** may convert the multiple native format messages collectively into one or more uniform bridge messages by combining the native format messages into one or more uniform bridge messages, with each uniform bridge message including data from each received native format message. Thereafter, the one or more uniform bridge messages are transmitted to a bridge compatible device.

[0154] Referring now to FIGS. **10A** through **13**, exemplary improved live power user interface (UI) layouts **10010**, **12010** according to the technology disclosed herein are shown. Embodiments (**10001**, **10002**, **10003**, and **10004**) of first exemplary live power UI layout **10010** corresponding to a six port hub is shown in FIG. **10A** through **11**. These UI embodiments may correspond, for example, to a six port hub such as Hub **2001** (see FIG. **2B**). Embodiments (**12001**, **12002**, and **12003**) of a UI layout corresponding to a second live power UI layout **12010** for a four port hub is shown in FIG. **12A** through **12C**. A live voltage UI layout **11000** for a power port of a six port hub is shown in FIG. **11**. A port detail UI layout **13000** for a battery port of a four port hub is shown in FIG. **13**.

[0155] The live power UI layouts **10010**, **12010** are advantageous over known systems in that they provide improved, intuitively understandable, display of power characteristics of individual power sources and loads, system power and system draw, predictions, and alerts. The UI layouts are generated, referring to FIG. **2A** and **2B**, by a power application (e.g. Power App **3304**) operating on an EUD, e.g. EUD **2110**, and are displayed on a display screen, e.g. **2112**, of the EUD.

[0156] The UI layouts **10010**, **12010** enable a user to monitor live power data on the EUD display, including power drawn from the hub and power supplied to the hub per port. This is useful in a number of ways. For example, a user can determine the actual power being consumed by each power device and, in response, can change one or more power device settings, for example radio power, to reduce power consumption. It is noted that information, including any of the disclosed UI layouts, e.g. **10010**, **12010**, are described herein in relation to a display of an EUD, which in the context of a military power and data system may include, for example, a Samsung S20 Tactical Edition smartphone. It is anticipated by the inventors, and would apparent to one skilled in the art, that the disclosed UI layouts and suitably modified or scaled version of the layouts, as well information contained therein, may be displayed on multiple types of devices that each comprising a means for displaying information to a user. Non-limiting examples of display devices on which the information may be displayed include a heads-up-display (HUD), a smart watch, and a tablet computing device. Exemplary smart watch layouts are disclosed herein in relation to FIGS. **30-33**.

[0157] Each of the illustrated live power UI layouts includes **10010**, **12010** an individual power display for each port of a hub that is in communication with the EUD and for each power device that is connected to the hub. The power information shown on the live power UI layouts may be generated by the power application **3304** as previously described in relation to FIG. 3. The live power can be updated at a default interval or at an interval configured by a user, for example live power data can be refreshed at an interval between 1 second and 5 minutes, for example every 5 seconds.

[0158] In an exemplary embodiment, a user can select a faster sampling interval for a particular port, for example an increased rate of sampling and display of live voltage data for a power port to which a solar blanket is connected. However the update interval, i.e. how often power data is recorded to a data store, for the particular port will not change, i.e. power data corresponding to the particular port will be stored at the same interval as other ports.

[0159] Referring to FIG. 10A through 10D, exemplary configurations **10001**, **10002**, **10003**, and **10004** of the first live power UI layout **10010** are shown. The first live power UI layout includes, across the bottom of the layout, six power display tabs, each representing a single port of the six port hub. In this example, individual power display tabs for each port of the six port hub include: main power port (MPP) **10060**; auxiliary power port (APP) **10020**, EUD port (EUD) **10084**, device port 1 (PAN 1) **10092**, device port 2 (PAN 2) **10094**, and device port 3 (PAN 3) **10096**. Referring now to FIGS. 2A, 2B, and 10A through 10D, power display tabs of embodiments **10001**, **10002**, **10003**, and **10004** of the first UI layout **10010** can correspond to device ports of power hubs **2000** and **2001** and can include display of information corresponding to one or more power devices that are connected to a corresponding hub device port. For example, MPP tab **10060** can correspond to MPP port **2960**, APP tab **10020** can correspond to APP port **2920**, EUD tab **10080** can correspond to EUD port **2984**, PAN 1 tab **10092** can correspond to PAN1 port, or device port 1, **2992**, PAN 2 tab **10094** can correspond to PAN2 port, or device port 2, **2994**, and PAN3 tab **10096** can correspond to PAN3 port, or device port 3, **2996**.

[0160] More or fewer power displays tabs can be shown on live power UI embodiments, depending on the number of ports that are included on a particular hub. For example, referring to FIG. 12A through 21C, exemplary configurations **12001**, **12002**, and **12003** of the second live power UI layout are shown. In exemplary configurations of the second live power UI layout, individual power display tabs for each port include: battery power port (BATT) **12111**, EUD port (EUD) **12084**, device port 1 (PAN1) **12092**, and device port 2 (PAN2) **12094**.

[0161] If no power device is connected to a particular port, a power display tab representing to port is shown grayed out and in a lowered position, for example tab (APP) **10020** as shown in FIG. 10A and tabs (BATT) **12111**, (EUD) **12084**, (PAN1) **12092**, and (PAN2) **12094** as shown in FIG. 12A.

[0162] If a power device is connected to a port, the corresponding individual power display tab is illustrated in a raised or “popped up” configuration and includes display of live power consumption or provision associated with the power device and an identifier of the power device, for example, referring now to FIG. 2A, 2B, 10A and 10B: MPP tab **10060** indicates that a SoloPack battery is connected to

the main power port **2960** and is providing 24.24 W of power; EUD tab **10084** indicates that a USB is device connected to the EUD port **2984** and is drawing 0.12 W of power; PAN 1 tab **10092** indicates that a MBTIR is connected to the first device port **2992** and is drawing 22.66 W; PAN 2 table **100094** indicates that a second radio is connected to the second device port **2994** and is drawing 22.66 W; and PAN 3 port **10096** indicates that a helmet system is connected to the third device port **2996** and is drawing 1.41 W.

[0163] A lightning bolt shaped icon indicates, e.g. **10085**, associated with a port tab, e.g. with EUD tab **10084**, indicates that a power device connected to a corresponding device port, e.g. to EUD port **2984**, is drawing power from a hub represented by a corresponding UI layout. A green battery shaped icon, e.g. **10065**, associated with a port tab, e.g. with MPP tab **10060**, indicates that a power source connected to a corresponding power port, e.g. **2960**, is providing power to a hub representing by a corresponding UI layout. A yellow battery shaped icon that includes a lightning bolt symbol, e.g. **10067**, associated with a port tab, e.g. with MPP port **10060** (see FIG. 10C), indicates that a rechargeable power supply, e.g. a conformal battery, is connected to a corresponding power port, e.g. to MPP port **2960**, and is drawing power from a hub to recharge the rechargeable power supply. A green lightning bolt shaped icon, e.g. **10020**, indicates that an auxiliary power source, for example an AC power source, is connected to the auxiliary power port, e.g. to APP **2920**, and is providing power to a hub corresponding to the UI layout.

[0164] A live power UI layout, e.g. **10001**, **10002**, **10003**, **10004**, **12001**, **12002**, **12003**, also includes a display of total system demand **10100**, **12100** and a display of system supply **10200**, **12200**. System demand, e.g. **10100** can include a current, live, value **10110**, **12110** and a peak value **10115** of power requested or consumed by all connected power loads, including power devices and the EUD represented by tabs **10084**, **10092**, **10094**, **10096**, **12084**, **12092**, and **12094**. The system demand display **10100**, **12100** includes a curved bar plot indicating the current, live, demand of connected power devices. System supply includes **10200**, **12200** a current, live, value of remaining energy (expressed in watt-hours Wh) and maximum energy available (i.e. a total energy capacity) corresponding to all connected power sources and a curved bar plot **10200**, **12200** indicating live energy available **10210**, **12210** as a function of maximum energy available. In an exemplary embodiment, a hub, e.g. **1001**, **2001**, communicates with a power supply that includes a smart battery, e.g. main power supply **2010** comprising a Galvion SharePack battery, via SMBus to determine actual capacity based on battery health and the UI, displays the actual battery capacity.

[0165] As shown in FIG. 10A, a first exemplary configuration **10001** of the first live power UI layout **10010** can include a Remaining Run value **10300**, i.e. a predicted remaining time that sufficient power to meet demand will be available based on current system demand **10110**, system energy supply **10210**, and historical power data regarding power performance of the connected power supplies and power devices.

[0166] As shown in FIG. 10B, a second exemplary configuration **10002** of the first live power UI layout **10010** can

include an alert **10400** which is displayed if a predicted live value of remaining run time falls below a pre-configured threshold value.

[0167] Referring now to FIGS. 12B and 10C, a third exemplary configuration **10003** of the first live power UI layout **10010** is displayed when a source of charging power, e.g., an auxiliary power supply **2010** comprising an AC power adapter, is connected to the auxiliary power port **2920**. APP tab **10020** indicates that the AC adapter is connected to the APP port **2920** and indicates and amount of power that is provided by the AC adapter. An icon **10025** associated with APP tab **10020** may include a specific color when power is being supplied to the APP port **2920**. In an embodiment the icon **10025** is colored green when a power source is connected to the APP port. It is noted that in FIG. 10A and 10B, a battery icon **10065** associated with MPP tab **10060** may also be colored green when a power supply is attached to a corresponding MPP port **2960** and is supplying power. Referring once again to FIG. 10C, a battery icon **10067** that includes a lightning bolt symbol may be displayed at MPP tab **10060** when a rechargeable power source, for example a conformal battery, is attached to corresponding MPP port **2960** and is being charge by power provided by a hub, e.g. by hub **2001** (see FIG. 2B). Referring to FIG. 10A, one or more of power device ports, e.g. **10084**, **10092**, **10094**, and **10096**, may each include a lightning-bolt shaped live power icon, e.g. **10085**, when a power device is drawing power from a corresponding hub device port, e.g. from EUD port **2984** (see FIG. 2B). In embodiments, the icon **10085** is colored blue.

[0168] Referring now to FIGS. 2B and 10D, as shown in FIG. 10D, an embodiment of an EUD **10500** includes the fourth exemplary configuration **10004** of the first live power UI layout **10010** shown on a display screen **10550** of the EUD. The fourth exemplary configuration **10004** of the first live power UI layout **10010** includes MPP tab **10060** and APP tab **10020** both popped up, indicating that a separate power source is connected to each of corresponding power ports **2960** and **2920**. For example, a main power supply **2060** comprising a Galvion SharePack battery is connected to Main PP port **2960** and is identified in MPP tab **10060** while an auxiliary power supply **2010** comprising a Galvion Solo Pack battery is connected to AUX PP **2920** and is identified in APP tab **10010**. As illustrated in FIG. 10D, the fourth exemplary configuration **10004** of the first power UI layout **10010** is shown displayed on a display screen **10550** of an exemplary EUD **10500**. Exemplary embodiments of EUD **10500** include a Samsung device, for example a Samsung S20 tactical edition smartphone or a Samsung S9 smartphone.

[0169] Referring now to FIG. 10E, multiple views of the EUD **10500** with display screen **10550** of FIG. 10D are shown. The exemplary embodiment of EUD **10500** is shown as a Samsung S9 smartphone although it is noted that one skilled in the art would recognize that one or more different smartphone or similar devices can be substituted for the Samsung S9 without departing from the disclosed technology. **10501** is a front view of the EUD **10500** with an illustrative embodiment of the first power layout **10010** displayed on the screen **10550**. FIG. 10D further includes a top rear view **10502**, left side view **10503**, right side view **10504**, top view **10505**, and bottom view **10506** of the EUD **10500**. It is noted that any of the first power UI layout **10010**, second power UI layout **10020**, power port pop-up window

11010, and port detail UI layout **13010** each may similarly be displayed on a display screen **10550** of the EUD **10500**.

[0170] Referring now to FIG. 11 a single port live power port pop-up window **11010** that can be opened by a user to display power load voltage **11100** at a selected port, e.g. at port **10060**, is shown. The user can open the single port live power pop-up window **11010** by tapping a power display tab, for example the MPP power display tab **10060**. In the illustrated example, voltage **11150** of a power signal provided by a power source connected to the main power port is shown. The single port power port pop-up window **11010** can show live power characteristics for the selected power port updated at a faster rate than the power data is saved to a data store, for example once every 100 ms. This is useful when a user desires to understand power characteristics of a particular power device on a more granular level. An exemplary implementation includes configuring a single live power UI layout **11010** to display live power corresponding to a power port, e.g. **2960** (see FIG. 2B) to which a solar blanket is attached. The user can observe changes in displayed voltage **11150** while positioning and repositioning the solar blanket to enable selection of a position of the solar blanket to receive a maximum amount of power from the solar blanket.

[0171] As shown in FIG. 12A, a first exemplary configuration **12001** of the second live power UI layout **12010** is shown. In this first configuration **12001**, no power devices are connected to power ports of a four port power hub represented by the power layout **12010**. All four power display tabs **12111**, **12084**, **12092**, and **12094** are shown in a down position and grayed out.

[0172] The second exemplary configuration **12002** of the second live power UI layout **12010** shown in FIG. 9B includes the BATT power display tab **12111** popped up, with an indication that a Galvion SharePack battery has been connected to a corresponding BATT power port (not shown). The BATT power display tab **12111** includes a display of live power provided by the battery to the hub. The EUD power display tab **12084** is also popped up, indicating that an EUD, in the illustrated example an EUD identified as Samsung S9, has been connected to an EUD port of a corresponding 4-port hub (not shown). The EUD power display tab **12084** includes a display of live power being drawn from the hub by the EUD. The System Supply bar graph **12200** has been updated to indicate an amount of energy available from the connected battery and a system demand value **12110** has been updated to reflect a power demand of the connected EUD. A value or Remaining Run **10300** has been updated to indicate an estimated runtime of the system generated by a power application based on characteristics of the connected battery and EUD.

[0173] The third exemplary configuration **12003** of the second live power UI layout **12010** shown in FIG. 12C includes the second power device port power display (PAN2) **12094** popped up with information displayed indicating that a head lamp has been attached to a corresponding hub (not shown) and that the head has a live power draw of 6.60 W. The displayed Remaining Time estimated value **12300** has been updated to reflect the addition of the head lamp power draw to the hub.

[0174] As shown in FIG. 13, a port detail UI layout **13010** corresponding to the battery port (BATT) **12111** of the four port hub represented by the second live power UI layout **12010** is shown. Referring to FIGS. 12C and 13, the port

detail UI layout **13010** can be accessed by a user by tapping the BATT power display tab **12111** on the second live power UI layout **12010**. The port detail UI layout **13010** includes multiple data values **13200** corresponding to a battery connected to the BATT port including: amount of the time the battery has been connected to the hub, live voltage, current, and power supplied by the battery, peak voltage and current values, temperature of the battery, health of the battery (i.e. a value representing battery state of health), and energy (in Wh) available from the battery. The port detail UI layout **13010** also includes a voltmeter and corresponding bar graph **13100** which displays an instantaneous voltage value **13150** corresponding to the BATT power display tab **12111**. Referring to FIGS. **11** and **13**, the voltmeter **13100** portion of the port detail UI layout **13010** is similar to the voltmeter **11100** of the power port pop-up window **11010**. In some embodiments, the power port pop-up window **11010** may be replaced by a port detail UI layout similar to **13010** or the port detail UI layout **13010** may be replaced by a power port pop-up window similar to **11010**.

[0175] In some embodiments, UI elements representing power demand and elements representing power supply may be shaded using different colors. For example, in an embodiment power demand display elements including a system demand bar graph **10100**, **12100**, text displaying individual and aggregated system demand values, and symbols representing live system demand (e.g. lightning bolt symbols **10085**) included blue coloring. In embodiments, power supply display elements, including a system supply bar graph **10200**, **12200**, text displaying individual and aggregated system supply values, and symbols representing live system supply (e.g. a battery symbol **10065** corresponding to a discharging battery or lightning bolt **10025** corresponding to an AC power supply) include green coloring. In embodiments, UI display elements representing a battery being charged, for example referring to FIG. **12C** battery symbol **10067** and power value text corresponding to a conformal battery being charged, include a yellow color. In embodiments, values representing remaining run **10300** may be shaded yellow and, referring to FIG. **10B**, may be indicated with different color, for example orange, when an alert **10400** is displayed. Referring to FIG. **11** and FIG. **13**, a voltmeter bar graph display **11100**, **13100** may include orange coloring in embodiment.

[0176] Referring now to FIG. **14**, a first exemplary power tracking method **14000** for generating, transmitting, receiving, and using information contained in live power data messages according to embodiments is shown. The first method **14000** includes generating and exchanging live power data messages, similar to those discussed in relation to FIG. **3**. As described herein, the first method **14000** is carried out using a hub processor of a smart hub, for example hub processor **2071** of hub **3001** shown in FIG. **3** or host processor **2071** of hub **4001** shown in FIG. **4**. It is recognized that the method **14000** may be carried out by any processing device that is capable of receiving live power data from one or more power devices, for example by the EUD **2110** connected to a non-smart hub **2001** as shown in FIG. **2B** or by a compute module, for example a helmet compute module **4179** as shown in FIG. **4**.

[0177] At step **14100** a hub processor collects live power data from one or more power devices, each of which is connected to a device port of the hub. The hub processor may probe a device port corresponding to each power device

to obtain the power data, for example, and referring to FIG. **2B**, power ports **2920** and **2960**, device ports **2992**, **2994**, and **2996**, and EUD port **2984**. In a particular embodiment, each power device, e.g. power supplies **2010**, **2060** and power loads **2101**, **2103**, **2105**, and **2110**, is a USB power device and the hub processor receives, from each USB power device, a data stream including live power data. In other embodiments, power devices may include power-only devices in which case the hub processor determines live power data based on measurements, as previously discussed. In still other embodiments, one or more power devices may communicate power data to the hub processor over a network protocol other than USB, for example over ISW or SMBus. The hub processor collects data at a fast rate determined by the system.

[0178] In some embodiments, one or more power devices each include a memory and are configured to store power readings for a period of time specified by the system, or overridden by a user, and to periodically aggregate live power data to generate an average and peak values over the period of time. The power devices may provide the average and peak power for the period when polled by the hub. In other embodiments, the hub receives non-aggregated data from, or corresponding to, one or more power devices at and generates average and peak power values.

[0179] The hub processor stores aggregated live power data to a datalog storage, for example to a database or storage file system, e.g. to hub power log **3307** (see FIG. **3**), by date on the hub for later retrieval. The hub processor may store addition information corresponding the live power data, for example a device type or device ID corresponding to live power data, a time of day, temperature, GPS location, or other metadata corresponding to the live power data. The hub processor may receive at least some of the metadata, for example a GPS location, from a power device that is communicatively connected to the hub, for example from a GPS system carried by a user.

[0180] At step **14120**, the hub processor assembles the live power data, which can include live power data from the multiple power devices, into a live power data message. The live power message may include live power data from each of the one or more power devices.

[0181] In some embodiments, the hub processor includes a host bridge service or bridge module and generates one or more universal bridge format messages that include data payload with the live power data, as previously discussed in relation to methods **7000**, **8000**, and **9000** (see FIGS. **7**, **8**, and **9**). In these embodiments, at step **14140**, the hub processor, using a uniform messaging platform, e.g. UMP **1001** (see, for example, FIG. **3**), generates one or more uniform bridge format messages that includes the live power data. In alternative embodiments, the hub processor communicates the live power data without including the power data in uniform bridge format messages. For example, the hub processor may receive a request for live power data, for example from an EUD connected to the hub, and, in response send the requested live power data.

[0182] At step **14160**, if the hub includes multiple transport protocols, the hub processor selects a soldier PAN transport protocol to use for communicating the live power data message as bridge message frames. In some embodiments of hubs, only USB is available and selection between protocols is not required. The hub processor can select an available transport modality based upon one or more crite-

ria. In an embodiment, the hub processor selects USB transport if available and falls back first to a soldier wireless network, e.g. an ISW network, and second to a BT network. The hub processor can use other criteria for selecting a transport modality, for example based on a measured or inferred speed or congestion of each of multiple networks, a user-selected preference, or a preferred network corresponding to parameters of a particular mission.

[0183] At step **14180**, the hub transmits the live power message to an endpoint device, for example to an EUD **2110** connected to hub EUD port **2984** (see FIG. 3). The hub processor transmits live power message, e.g. **3500**, for example uniform bridge formatted frames, over the selected transport modality.

[0184] At step **14200**, an endpoint device, for example the EUD, receives the live power data message that were transmitted by the hub processor in step **14180**. If the live power data includes uniform bridge formatted frames, then at step **14220**, an EUD uniform messaging platform (UMP), e.g. **1001**, operating on the EUD extracts the live power data payloads from the uniform bridge formatted frames received from the hub processor and assembles the payloads into the live power data message. The live power data is then transmitted to one or more applications operating on the EUD, including a power application **3304**, as discussed in relation to FIG. 3. In some embodiments, the EUD UMP **1001** generates native-format messages comprising the live power data, for example SMBus formatted messages, and communicates the native format messages to the one or more applications, for example using a broadcast message protocol. In some embodiments, each of the one or more applications may subscribe to particular types of data, for example to power data, and the EUD bridge service will only communicate, to the one or more applications, data of the types that the applications subscribe to.

[0185] At step **14240**, the power app generates, based on information included in the live power data message, at least one power information product that includes live power data corresponding to the multiple power devices and including in the live power message. The power information product can include, for example, power consumption and power provision per power device. The power information product can also include a statistical power information product, for example one or more statistical values generated based on the live power data, for example total power drawn by all powered devices, total power provided by one or more power sources, remaining power available from one or more power sources, peak and/or average power drawn by each of the power loads, and aggregated peak and/or average power drawn by all powered devices connected to the hub. For example, referring to FIGS. **10A** through **10E**, **11**, **12A** through **12C**, and **13**, the power app can generate one or more live power data layouts, for example any of live power data UI layouts **10010**, **11010**, **12010**, and **13010**. In an embodiment, the power app then causes the live power data layout to be displayed on the screen of the EUD, e.g. on display screen **10550** of EUD **10500** (see FIG. **10D**, **10E**) or on display **2012** of EUD **2110** (see FIG. **2B**).

[0186] The first method **14000** includes generating and transmitting live power data messages, which can include power data corresponding to multiple power devices. A method substantially similar to the first method can be used for uniform bridge formatted messages that include other types of data, for example digital audio data (e.g. USB audio

data) which can be received, by a hub processor, from multiple devices, each connected to a device port or otherwise in communication with a hub processor. In the example including digital audio data, an EUD or other device can receive one or more universal bridge formatted messages that each include digital audio data from multiple audio power devices and extract the audio data corresponding to each audio device from the bridge message payloads.

[0187] An example method similar to the first method **14000** can be used to transmit a datalog of historical power data from the hub to an EUD or other device. In this example method, the historical power data is too large to transmit within a single universal bridge formatted frame. The hub UMP **1001** service can generate multiple universal bridge message payloads that each include a portion of the historical power data. The hub UMP **1001** adds a bridge header to each bridge message payload to generate multiple universal bridge message frames. The universal bridge message frames are received by the EUD. At step **14220**, an EUD UMP **1001** operating on the EUD extracts live power data from the multiple universal bridge message frames received from the hub processor and assembles the payloads into the datalog of historical power data.

[0188] Referring now to FIG. **15** a second exemplary power tracking method **15000** that can be implemented by embodiments of the technology described herein is shown. The second method **15000** includes generating, at an EUD, e.g. at EUD **2110** (see FIG. 3), one or more live power data layouts, for example one or more of the live power data layouts illustrated in FIGS. **10** through **13** for display on a screen, e.g. on EUD display **2112** screen (see FIG. 3). In embodiments, and referring to FIGS. **14** and **15**, the second method **15000** may include an exemplary implementation of step **14240** of the first method **14000**.

[0189] Referring once again to FIG. **15**, at step **15100**, the EUD receives a live power message that includes live power data corresponding to multiple power devices that are operatively connected to an intra-soldier PAN, for example power devices connected to a hub. For example, and referring now to FIG. 3, the EUD **2110** receives a live power message **3500** from hub **3001** that includes live power data corresponding to power devices **3101** through **310n**. In an exemplary embodiment, the live power message is formatted as uniform bridge message and an EUD uniform messaging platform **1001** extracts the live power message from one or more bridge frames. Live power data included in the live power message is passed to a power application, for example Power App **3304**.

[0190] Referring now to FIGS. **3** and **15**, at step **15120**, the EUD displays live power data that includes power consumed by each power load on connected to the PAN and power provided by each power sources connected to the PAN. The EUD also displays aggregated power consumed by all of the power devices connected to the PAN and aggregated power provided by all power supplies connected to the PAN. In an exemplary embodiment, an analysis module, e.g. **2189**, of the power application generates live power statistics data based on live power data included in the live power data message, for example aggregated power consumed and aggregated power provided. A display module, e.g. **2187**, of the power application generates one or more live power data layouts, for example a live power data layout illustrated in any of FIGS. **10A** through **13**, that includes the live power

data corresponding to individual power devices and aggregated live power data (System Demand and System Supply).

[0191] At step 15140, the power application retrieves, from a historical power database, historical power data corresponding to each power device that is connected to the PAN. In some embodiments, a universal messaging service according to the disclosed technology performs a device discovery operation to determine an identity of each of the power devices and provides the device identity information to the power app. An example device discovery operation is described herein in relation to FIG. 7. In some embodiments, the historical power data includes data generated based on past performance of a particular power device, for example based on performance of a particular power supply that is connected to a power port of a hub. In some embodiments, historical power data include power data aggregated by device type. The power application can filter historical power data, for example by removing out-of-bounds values or by selecting historical power data corresponding to conditions expected for a particular mission, e.g. based on ambient temperature or time of day (e.g. data collected during night operations or during day operations).

[0192] At step 15160, the power application generates, based on the live power data and historical power data, a prediction of remaining run time of the power loads connected to the soldier PAN, or an amount of time remaining during which all power loads connected to the PAN can be powered by power sources connected to the PAN. In an exemplary embodiment, a modeler module of the power application uses one or more trained machine learning models or algorithms to generate a predicted run time. The power application also generates, at step 15160, one or more statistical power products, which can include, for example, one or more of an aggregated power draw of all power loads that make up the soldier PAN and an amount of power available from power sources connected to the soldier PAN. The power application may use one or more of the statistical power products to generate the prediction of remaining run time.

[0193] At step 15180, the power application displays the predicted runtime on the EUD display. For example, the display module of the power application may generate a UI similar to that depicted in FIG. 7, including predicted runtime (Remaining Run).

[0194] At step 15200, the power application compares a value of the predicted runtime to a threshold value, for example to a threshold value configured by a user, and determines whether the value of the predicted runtime is less than the threshold value. If not, the method returns to step 15100 to receive additional live power data. If the predicted runtime is less than the threshold value, the power application generates and displays an alert, for example an alert similar to the alert 10400 illustrated in FIG. 10B, and returns to step 15100 to receive additional live power data.

[0195] At step 15230, the power application optionally communicates information to one or more devices located outside of the soldier PAN, for example to one or more endpoint devices. The information may include, for example, one or more of an alert generated at step 15220, a statistical power product generated at step 15160, and all or a portion of live power data collected by the power application. For example, and referring now to FIGS. 3 and 4, the power application 3304 (e.g. APP A) may communicate the information to one or more of a central command processor

4509, a TCP/IP device 4510, a satellite endpoint 4512, or an other network endpoint 4514.

[0196] Referring now to FIG. 16A, and 17 through 28, UI layouts that can be displayed on a screen of an EUD are shown. At least some of the UIs shown in FIG. 16A, and 17 through 28 can be interacted with by a user to select data to be imported from a hub and to select data for use in analysis and modelling are shown. During a mission, live power data is automatically recorded on a hub, as previously discussed. In some embodiments, live power data can also be recorded on a EUD. The EUD can be connected to the hub, or left connected to the hub, post-mission to load data for analysis. The system, e.g. the EUD, sets date and time on the hub to organize power logs. For example, and referring now to FIG. 3, a hub processor 2071 can store live power data, and associated metadata, in a hub power log 3307 during a mission. An EUD 2110 can also store live power data to a local power log 3705. Following a mission, a user can download, onto the EUD power log 3705, live power data that was stored by the hub processor 2071 in the hub power log 3307, thereby consolidating live power data. A user can select some or all of the power data stored in power log 3705 and/or on the hub power log 3307 for inclusion in a configuration of the historical power database 3703. In an embodiment, selected power data is used as input for an analysis module 2188 and/or modeler module 2189 which generate one or more information products based on the selected power data.

[0197] The system compares logs and data on the EUD against logs on the hub and graphically shows the differences by date. Referring to FIG. 16A, a calendar display UI 16000 includes blocks 16010, each corresponding to a day, for example to a day of a selected month. Each of the blocks 16010 includes an indicator corresponding to data stored that corresponds to the block, including whether power data has been recorded, either on the hub, on the EUD, or both. FIG. 16B shows the meanings of the indicators shown in FIG. 16A. Indicator 16012 indicates that no data is available for a corresponding day. Indicator 16014 indicates data corresponding to a day is only stored on the EUD. Indicator 16016 indicates that data corresponding to a day is only stored on the hub. Indicator 16018 indicates that data corresponding to a data is stored both on the hub and the EUD. A user can interact with the calendar display UI 16000 to download any data stored on the hub that is not already stored on the EUD, and thereby concatenated EUD and hub power data, for example in the EUD power log 3705.

[0198] A user can select data stored on the hub for import to the EUD. For example, the user can tap on a date that is marked with an indication that data from that date is stored on the hub only to cause the data to be downloaded to the EUD.

[0199] Data downloaded to the EUD can be used for analysis and modelling, for example, and referring to FIG. 3, by loading the data into a historical power database. Data included in the historical data database can be used by an application, for example by a power app, for making predictions. The data in the historical database can also be shown on a user interface displayed on the EUD screen.

[0200] FIG. 16C shown an illustrative, exemplary, method 16500 for downloading concatenating power data and selecting data for use in one or more of analysis and modelling. At step 16510, a user selects data that is on hub for download to the EUD, for example data corresponding

to one or more days marked with indicator **16016** on the calendar display **16000**. The selected data is prepared and sent by the hub, in some embodiments as one or more uniform bridge formatted messages, and is received by the EUD at step **16520**. Referring to FIGS. **3** and **16C**, at step **16530**, the EUD processor **2114** combines the power data received from the hub with power data already stored on the EUD, if any, thereby generating concatenated power data which it stores in the EUD power log **3705**. At step **16540**, a user can select some or all of the power data for using in one or more analysis tasks, for example for use as input to analysis module **2188**. At step **61550**, a user can select some or all of the power data for use in one or more modeling tasks, for example for input for the modeler module **2189**. Data selected at steps **61540** or **61550** may be added to the historical power database **3703**. For example, a user may select, at step **16550**, data to be used as historical power data when the modeler module **2189** generates a predicted run time before or during a mission. The selected data is moved or copied into the historical database **3703** for retrieval and use by the modeler module **2189**.

[0201] The user can decide which data to import from the hub, which data to delete or retain, and which data to use for making predictions. For example, a user can select only known good data, according to one or more criteria, and can delete data that is not good. For example, a user can only import and retain data from dates on which actual missions occurred, as opposed to dates on which equipment was tested or used for some purpose other than one a mission. A user can also choose to export power data, for example to forward power data corresponding to a user's own PAN to a central location, e.g. to a squad leader, where it can be combined with power data from other users, for example power data from each squad member's PAN. In addition, the user can decide what devices profiles should be imported, exported, or deleted. FIG. **17** shows a data management UI layout **17001** that includes interfaces that a user can interact with to import, export, or delete power data and/or device profiles from one or more of the hub and EUD data stores.

[0202] Referring now to FIG. **18** a UI layout **18101** that a user can interact with to select data from particular missions to be included in analysis and modelling, for example to be included in a database of historical power data, is shown. A user is presented with a list of missions, which can be sorted by mission name, call sign, and/or date. A user can interact with a selection icon **18300** to select a particular mission for inclusion in an analysis or modelling task. The user can interact with an edit icon **18350** to edit data associated with a mission, for example a mission name or callsign, or to add information corresponding to the mission, for example a type of mission or one or more conditions corresponding to the mission. The user can interact with a delete icon **18360** to delete information corresponding to a mission.

[0203] The novel technology described herein simplified power analysis and enables a deeper understanding by automatically integrating power data with charts. The novel technology enables a user to analyze detailed power usage over time per port and device type and detailed power supply usage and charging over time.

[0204] Referring now to FIG. **19**, an exemplary UI layout **19003** for showing detailed power consumption data is shown. The UI layout **19003** includes plot **19200** that shows power consumption corresponding to each of multiple power devices (Samsung 9, head lamp, and Galvion Hub).

The power consumption includes mean, range, and quartile. The UI **19003** layout displays, as horizontal lines, first and third quartiles of power consumed by each power device as well as power draw maximum and minimum values, as vertical lines, for any power devices that fluctuated measurably in power draw over the mission, for example for the head lamp. The UI layout **19003** includes a checkbox **19300** that a user can interact with to indicate whether the displayed data should be included in data used by a modeler component of a power app, for example whether the data should be included in a historical power data database.

[0205] Referring now to FIG. **20**, an exemplary UI layout **20003** for showing power consumed by power devices during a particular mission is shown. The UI layout includes power consumption data corresponding to a particular mission (**444**) and to a particular soldier who participated in the mission, identified by callsign Hammer. The UI layout includes a checkbox **20300** for selecting, by a user, whether the displayed data should be included in data used by a modeler component of a power app, for example whether the data should be included in a historical power data database. The UI layout includes a table **20200** that includes display of, for each device port (Pan1, Pan2, and Pan3) and for the EUD port (EUD) what device (Radio, MBITR, Helmet, and USB Device) was operatively connected to the port during the mission. The table **20200** includes an amount of power consumed by each device as well as an amount of time that the device drew power during the mission. The UI **20003** layout includes a circular bar graph **20100** with an individually shaded segment indicating power usage of each power device as a portion of total power consumed.

[0206] Referring now to FIG. **21** a UI layout **21003** for showing power consumed by a particular power device (e.g. MBITR), as indicated in selection box **21350**, over the course of a mission (mission **444**) is shown. FIG. **21**, includes a power draw plot **21200** that illustrates power drawn from a hub over time. A user can interact with selection box **21350** to select a device for which power data should be displayed in the UI layout **21003**. A solid line **21250** indicates power consumed by the selected power device, i.e. by the MBITR, over the course of a mission (**444**) and a dashed line **21230** indicates total power consumed by all power devices that were connected to the hub with the MBITR during the mission. The UI layout **21003** is advantageous in that it can show how a particular power device affects network power usage and how changes in the power device usage affect changes in network power usage.

[0207] Referring now to FIG. **22** a UI layout **22003** for showing power provided by a particular power source (SharePack) shown in selection box **22350** over the course of the mission is shown. A user can interact with the selection box **22350** to select a power device for which power data should be displayed on the UI layout **22003**. The UI layout **22003** includes a power supply plot **22200** that indicates power supplied to a hub over time. The solid line **22250** indicates power provided by the SharePack over time and the dashed line **22230** indicate total power capacity over the course of the mission.

[0208] The UI layouts **21003** and **22003** depicted in FIG. **21** and FIG. **22** allow a user to select, by interacting with selection icons **21300** and **22300** whether the data corresponding to individual power devices should be used by a modeler, e.g. by modeler module **2189** of power app **3304** (see FIG. **3**), to make predictions.

[0209] Referring now to FIG. 23 a UI layout 23104 for selecting and displaying historical power data corresponding to a particular power device is shown. The UI layout 23104 includes a table 23200 of indicator boxes 23210, each corresponding to a power device that is currently connected to a hub represented by UI layout 23104 and other known power devices. An indicator box 23210 includes a lightning bolt icon 23215 when a power device represented by the indicator box is connected. A system according to the technology described herein detects when a particular power device is connected to a particular device port of a hub represented by UI layout 23104, by device ID, and displays a lightning bolt shaped icon 21215 next to the name of the connected device. When a user selects one of the displayed devices, for example by clicking on a corresponding indicator box 23210, the system, e.g. EUD 2110, displays, in display window 23300, historical power data, e.g. an average historical power consumed by the power device, with standard deviation, as well as the number of missions from which data was included to generate the average. The UI layout 12104 allows a user to enter a custom device name for any of the displayed devices included in table 23200.

[0210] Referring now to FIG. 24 a UI layout 24104 for displaying power usage data of a selected device, e.g. by MBITR, over a number of missions is shown. The UI layout 24104 includes a plot 24200 that shows power consumed by the power device on each of multiple missions 24202, 24204, 24206, and 24208 and a line 24210 indicating historical average power. The UI layout 24104 includes a pull-down menu 24350 for filtering data by callsign. A user, e.g. a team leader, can load data from multiple users and select whether to display data for all call signs of for particular users.

[0211] In additional embodiments, not shown, a user can select data for analysis and use by a modeler, e.g. by modeler module 2189 of power app 3304 (see FIG. 3), based on additional or alternative criteria. For example, in an embodiment historical power data is stored with tags including temperature during a mission and a user can filter data for use in a modeler based on temperature. This is useful when temperature can affect power consumption or power capacity in that the user can cause the modeler to use only power data gathered when a temperature was similar to temperature expected during a planned mission. Similarly, a user can filter data by time of day (e.g. night vs. daylight), or by one or more other criteria that may be relevant to predicting power consumption or power provision of particular power devices.

[0212] Referring now to FIG. 25 and FIG. 26, UI layouts 25202 and 26202, respectively, that include example modeler outputs 25400 and 26300, e.g. outputs generated by modeler module 2189 of power app 3304 (see FIG. 3), are shown. The modeler outputs can be generated by the modeler module using, as inputs, personal power usage data collected over time and selected for inclusion in a historical power data database. The personal power usage data includes data corresponding to a soldier's own equipment or imported from other known good data. The system, for example a power application, creates power profiles of devices based on a user's real usage. The power profiles are used to predict future power needs.

[0213] FIG. 25 depicts a UI layout 25202 for generating and displaying predicted power required for a mission and suggested power supply to power the mission. A user enters,

in device selection interface 25100, information regarding power devices to be used on a particular mission, for example power loads (e.g. a first radio 25110, a second radio 25112, and MBITR 25114). The user enters, in runtime selection interface 25300, length, in time, of the mission. Referring to FIGS. 3 and 25, the modeler module 2189, which in some embodiments is a component of a power app 3304, retrieves, from the historical power data base 3703, historical power data corresponding to each of the selected power devices and historical power data corresponding to one or more available power supplies. The modeler module 2189 generates an estimated power draw 25120 corresponding to the selected power devices 25110, 25112, and 25114. The modeler module 2189 generates, based on the historical power device data and length of the mission, an estimated value of total power capacity needed from one or more power supplies to complete the mission. The modeler displays, in a suggested power supply window 25400, at least one suggested power supply 25410 for the mission. The modeler displays, at 25420 the power capacity of the selected power supply and the estimated required total power capacity to complete the mission.

[0214] FIG. 26 depicts a UI layout 25202 for generating and displaying an estimated powered run time based on a set of power loads and power sources selected for a particular mission. A user enters, in power device selection interface 26100 selected power loads (e.g. a first radio 26110, a second radio 26112, and MBITR 26114). The user enters, in power supply selection interface 26400 one or more selected power supplies (e.g. a SharePack 26410). In some embodiments, the power supply selection window is populated, either automatically or by a user, with the one or more power supplies suggested by the modeler. Referring to FIGS. 3 and 26, the modeler module 2189, retrieves, from the historical power data base 3703, historical power data corresponding to each of the selected power devices and power supplies. Modeler module 2189 displays the historical power data corresponding to each selected power source and power load, for example average power consumed by each power load, with standard deviation, in the power device selection interface 26100 and power capacity of the selected power supply in the power supply selection interface 26400. The modeler module 2189 generates, based on the historical power data, an estimated runtime of a power network that includes the selected power loads and selected power source in a runtime display interface 26300. The modeler module 2189 can display a typical estimated runtime, based on average values of power data, and high and low estimated runtimes, for example based on average values and standard deviations of the power data. A user can build a power plan by power loads and power supply used by the system. In this manner, a user can build out a mission and know how much power will be needed. It is noted that, in the preceding description, when the modeler module 2189 is recited as displaying information, the display of information may include interaction of the modeler module with the display module 2187. For example, the modeler module 2189 may communicate information, for example power data, to be displayed on an EUD display device 2012 to the display module 2187, and the display module 2187 may generate one or more instructions for the display device which cause the display device 2012 to present a UI layout that includes the information.

[0215] Referring now to FIG. 27 and FIG. 28, UI layouts 27301 and 28301 for selecting, by a user, one or more settings are shown. Using the UI 27301 depicted in FIG. 27, the user can be identified by an ATAK callsign or by a custom callsign, for example a custom callsign entered by the user. For example, the user may be identified by an ATAK callsign 27110 and can enter a custom callsign 27130 to be used in place of the ATAK callsign by interacting with selection icon 27120 and entering the custom callsign 27130. The user can use the UI 28301 depicted in FIG. 28 to allow power data corresponding to imported devices, e.g. power data corresponding to one or more other users, to be used by the modeler, e.g. by modeler module 2189 (see FIG. 3). A user may set a low power alert threshold (e.g. a threshold value for time remaining) by interacting with a time remaining alert interface 28110, and may set a frequency for live data transfer from a hub to an EUD by interacting with the update frequency interface 28120.

[0216] As shown in FIG. 29, UI layout 29110 according to the technology herein can be shown on a display device 29100 or 29200 of a non-EUD device, for example a laptop or desktop computer display or a tablet display. One or more UI layouts 29110 that are typically displayed on a EUD screen can be ported to a PC for display on a larger monitor 29100 using, for example, Samsung DeX. In addition, analysis data can be shown on a larger display 19200, for example power provision and power draw 29210, power draw per device 29220, individual power source provision 29230, individual power device draw 29340, and data transfer indicators 29250. In this manner, information corresponding to multiple UI layouts, e.g. to two or more of the UI layouts depicted in FIGS. 19 through 24, can be displayed together on one screen, which may facilitate analysis and understanding of characteristics of one or more power networks.

[0217] Referring now to FIGS. 30 through 33, exemplary improved live power user interface (UI) layouts 30001, 30002, 30003, and 30004 for display on a watch face, for example on the face of a smart watch, according to the technology disclosed herein are shown. A smart watch 30010 includes a watch band 30015 for wearing the watch and function buttons 30014, 30016 for interacting with and controlling operation of the watch 30010. The watch 30010 includes a display screen 30012 for presenting information to a user.

[0218] FIG. 30 depicts a first watch live power UI layout 30001 displayed on the display screen 30012 of the smart watch 30010. The first watch live power UI layout 30001 includes a curved power demand bar graph 30100 including a representation system power demand 30110 displayed near a perimeter of a first, e.g. top, half of the display screen. The UI layout 30001 includes a power provision curved bar graph 30100 including a representation of available system power supply 30210 displayed near a perimeter of a second, e.g. bottom, half of the display screen. The first watch live power UI 30001 layout In some embodiments, the system demand bar graph 30100 and a corresponding power demand indicator 30150 and a system demand symbol 30155 include blue shading while the system supply bar graph 30200 and corresponding power supply indicator 30250 and a system supply symbol 30255 include green shading. As shown, the smart watch display includes a

selected portion of the information that may be displayed on an EUD UI layout, for example on UI layout 1004 shown in FIG. 10D.

[0219] As shown in FIG. 31 through FIG. 32, a user can select one or more second watch live power UI layouts 30002, 30003 respectively, to be shown on the smart watch display 30012. A user can select, for example by interacting with one or more of the function buttons 30014, 3000016, one of the first 30002 and second 30003 configurations of the second watch live power UI layout. The first configuration 30002 includes a Pan1 display region 30092, a Pan2 display region 30094, and a Pan3 display region 30096, each of which display power consumption data for a corresponding device port, e.g., referring to FIG. 2B and 3, for ports 2992, 2994, and 2996, respectively. Referring to FIG. 2B, 3, and 32, the second configuration 30003 of the second UI layout includes a MPP display region 30060, an APP display region 30020, and an EUD display region 30084, each of which display power provision and consumption data for a corresponding power or EUD port, e.g., referring to FIG. 2B and 3, for ports 2920, 2960, and 2984, respectively.

[0220] Referring now to FIG. 33, a user can select a watch live power port detail UI layout 30004 that includes details corresponding to a selected port, for example details of a power device connected to a PAN2 port of a hub. The port detail UI layout 30004 may include a port name indicator 30294 (e.g., Pan2), a connected power device ID indicator 30296 (e.g., Radio), and live (i.e. near real time) power demand indicator 30297 that display power demand (e.g. -22.64 W) of the connected power device.

[0221] The first and second watch live power UI layouts 30001, 30002, and 30004 and the watch live power port detail UI layout 30004 advantageously allow a user to view system demand, supply, and remaining run time, and details corresponding to individual ports in an easy to comprehend format without needing to operate the EUD.

[0222] It will also be recognized by those skilled in the art that, while the subject technology has been described above in terms of preferred embodiments, it is not limited thereto. Various features and aspects of the above-described subject technology may be used individually or

[0223] jointly. Further, although the subject technology has been described in the context of its implementation in a particular environment, and for particular applications (e.g. for sharing live power data generated by multiple power devices in a solidier power network), those skilled in the art will recognize that its usefulness is not limited thereto and that the present subject technology can be beneficially utilized in any number of environments and implementations where it is desirable to share live data from multiple devices over a communication link that is known to be usable to share live data from only a single device. Accordingly, the claims set forth below should be construed in view of the full breadth and spirit of the subject technology as disclosed herein.

What is claimed is:

1. A system for providing a uniform messaging platform in a heterogeneous environment, the system comprising:
 - at least one end-point device;
 - at least one network;
 - a hub for routing messages from a source to a destination;

a uniform messaging service comprising:

- a device interface that is communicatively coupled to the at least one end-point device and the at least one network;
- at least one processor;
- at least one memory having stored thereon a uniform messaging application including a plurality of machine executable instructions that when executed by the at least one processor, cause the processor to:
 - receive at least one message from a source connected to the at least one network for transmission to a destination;
 - determine whether the destination of the at least one message is a bridge-compatible device;
 - convert the at least one message into a uniform bridge message when the destination is determined to be a bridge-compatible device;
 - convert the at least one message into a native destination-format message when the destination is determined to be a non-bridge compatible device; and
 - output the converted message to the at least one network for transmission to the destination.

2. The system of claim 1 wherein the hub further comprises a universal serial bus (USB) interface, a Bluetooth interface, an ultrawideband (UWB) interface, serial data interface and a TCP/IP interface.

3. The system of claim 1 wherein the uniform messaging service further comprises a multi-protocol routing application in the at least one memory having a plurality of machine executable instructions that when executed by the at least one processor, cause the processor to:

- select at least one interface from a plurality of interfaces to route the converted message to the destination based on at least one selected transport protocol and availability status of the plurality of interfaces; and

- route the converted message to the destination using the at least one selected interface.

4. The system of claim 1 wherein the instructions of the uniform messaging application further cause the processor to:

- select at least one transport protocol for transmission of the converted message to the destination; and
- output the at least one selected transport protocol along with the converted message.

5. The system of claim 4 wherein the at least one transport protocol is selected from a group comprising Bluetooth, USB, UWB, Wi-Fi and SMBus.

6. The system of claim 1 wherein the instructions of the uniform messaging application further cause the processor to:

- detect an end-point device connected to the at least one network;

- create a profile for the detected end-point device; and
- store the created profile in a profile database.

7. The system of claim 6 wherein a source is an end-point device connected to the at least one network and the destination is an end-point device connected to the at least one network.

8. The system of claim 7 wherein the instructions of the uniform messaging application further cause the processor to:

- examine the profile of the destination to determine whether the destination is a bridge compliant device.

9. The system of claim 7 wherein a native destination-format message is a message formatted consistent with the operating platform of the corresponding destination end-point device.

10. The systems for providing a uniform messaging platform in a heterogeneous environment, the system as shown and described herein.

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